BRIDGE CONTRACT No. 692

INDEX

SHEET NUMBER	SECTION	STRUCTURE NO.	STRUCTURE TYPE	STRUCTURE SPAN	STRUCTURE OVER	STRUCTURE STATION	DRAWING NUMBER	SUBJECT	BRIDGE CONTRACT NO.
1								Index and Title Sheet	
2	O	230	R.C.Arch	4@72'-5"&15"	Wabash River	66+10	C1	Layout	
3							C2	General Plan	
4							C3	Abutment No.1 Details	
5							C4	Abutment No.5 Details	
6							C5	Footing Plan & Bracket Details-Abuts. No.1&5	
7							C6	Piers No. 2, 3, & 4 Details	
8							C7	Sections, Abuts, Spans, & Piers-Miscell. Detls.	
9							C8	Elevation, Plan, & Sidewalk Section- Spans	
10							C9	Span Counter-fort Details	692
11							C10	Miscellaneous Details	
12							C11	Bending Diags., Splice Bars, & Summary	
13							C12	Bill of Materials	
14							C13	End Gutter Drain, Spec. Headwall, Wing Wall, Cover Plate	
15								Road Plan & Profile for Temporary Bridge	
16					Road Dept. Sheet No.		0	Road Plan & Profile - Rd. Proj. 377-B	
17							Std.	Plan Culverts & Headwalls	
18-19							Stds.	Miscell. Road Stds.- Sheets J, & B&C	
20							Std.	Std. Detour, Barricades, Etc.	
21					Road Dept. Sheet No.		5A	Typical Cross Sections-Rd. Proj. 377-B	
22							Std.	Std. 2C-8-6 Paving Section	

STATE OF INDIANA
STATE HIGHWAY COMMISSION

BRIDGES OVER 20 FT. SPAN

REG. ROAD PROJ. NO.	STATE	DIST. NO.	FEDERAL PROJ. NO.	DEPOSIT NO.	TOTAL SHEETS
7	IND.	1	ROA	1	22

SECTION - O

PLANS FOR BRIDGES OF SPANS OVER 20 FEET FOR PROPOSED

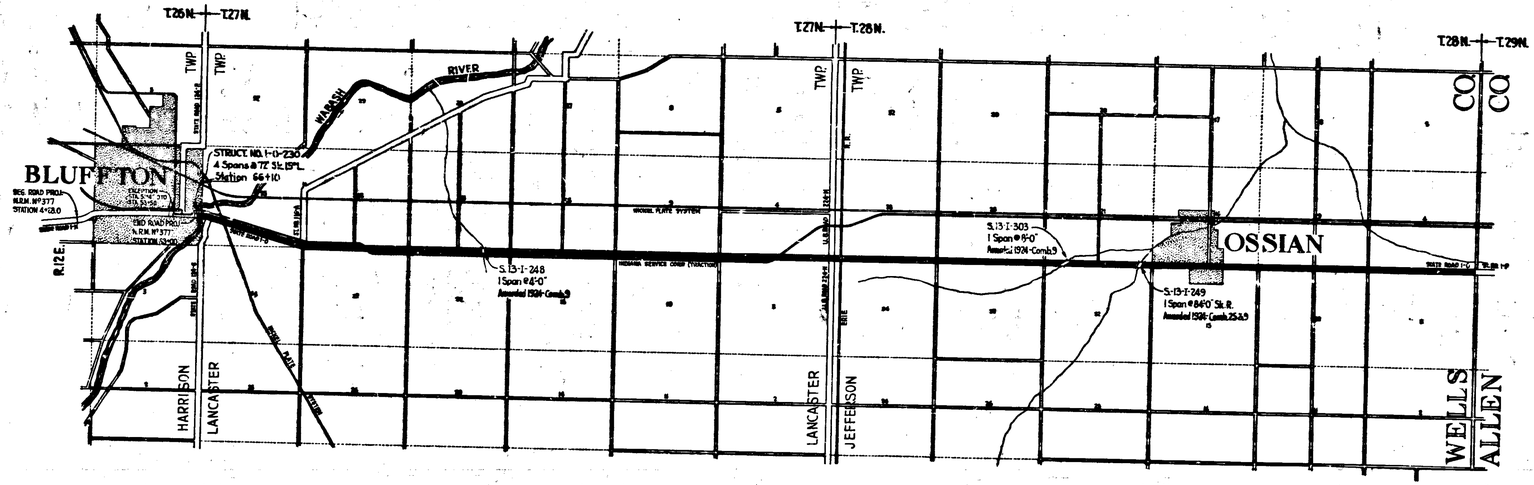
STATE HIGHWAY

P.W.A. PROJECT NO. 377

STATE ROAD NO. 1 SECTION O

LOCATION OF STRUCTURE NO. 1-0-230 ON STATE ROAD NO. 1 OVER THE WABASH RIVER AT THE NORTH EDGE OF BLUFFTON IN WELLS COUNTY.

Scale 1" = 1,000'



INDIANA STATE HIGHWAY STANDARD BRIDGE SPECIFICATIONS 1922 TO BE USED WITH THESE PLANS.

APPROVED AND ADOPTED DECEMBER 15, 1933
BY STATE HIGHWAY COMMISSION OF INDIANA

James D. Adams
CHAIRMAN, STATE HIGHWAY COMMISSION OF INDIANA

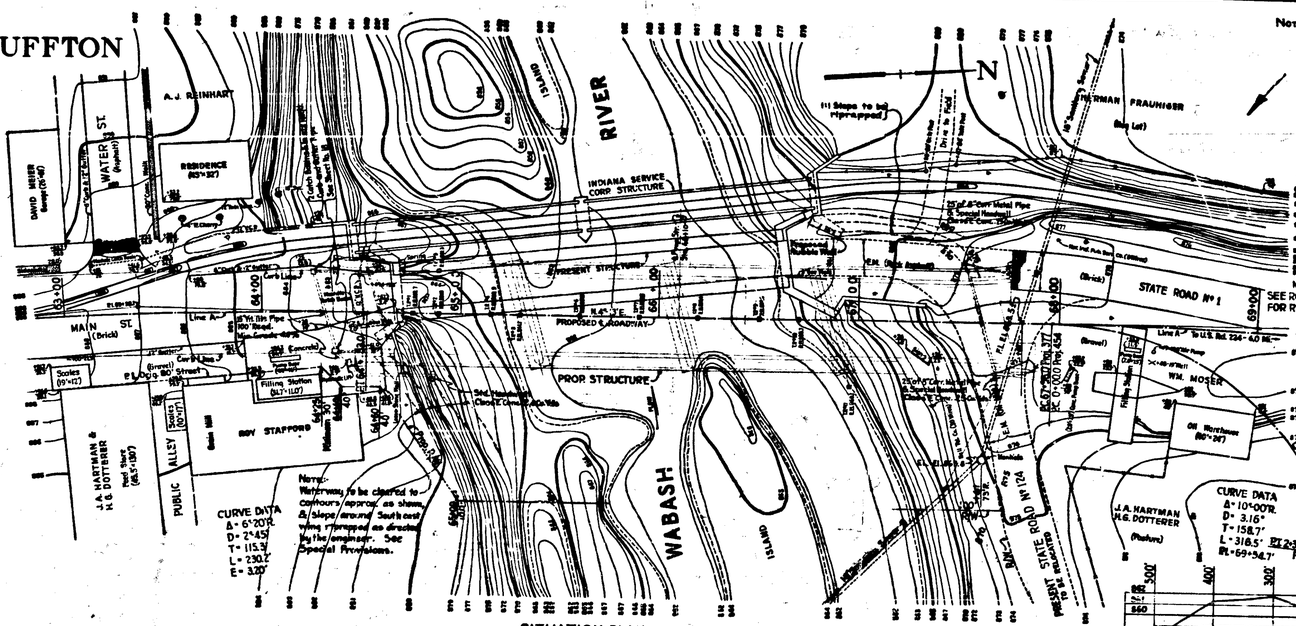
APPROVED DECEMBER 15, 1933

CHIEF ENGINEER, STATE HIGHWAY COMMISSION

1-0-230

Revised 2-8-34 Changed from P.W.A. to State Project & Add Sheets 21, 22 & Misc. 1941.

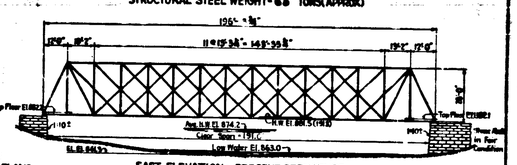
BLUFFTON



SITUATION PLAN
Contour Interval-1 Ft.
Scale-1" = 30'-0"

DRAINAGE AREA- Approximately 554 Sq. Mi. of Lightly Rolling, Sandy Clay Soil.

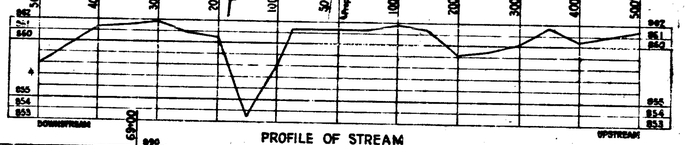
STRUCTURAL STEEL WEIGHT- 68 TONS (APPROX.)



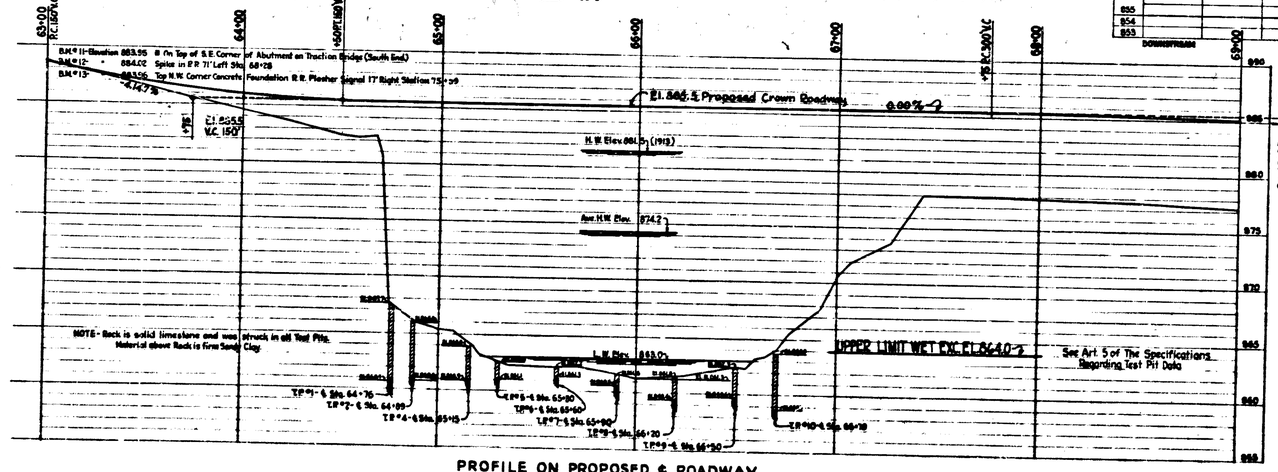
EAST ELEVATION - PRESENT STRUCTURE
Scale-1"-30'-0"

NOTE
General Condition of Structure is fair. Five Verticals in East Truss have been dismissed and repaired by I.S.H.C. 5-29-32 as Structure 1-0-230.

Clear Roadway = 17'-11"
13 Floor Beams- Built Up B
6 @ 3'-4.69" and 24" R = 2864"
15 @ 9'-1.5" = 136'-6"
7 Stringers
42 Tr. 9" 21.8" = 26'-6"
Sidewalk:- Floor 3"-8" Plank Transv.
2-3"-8" Plank Filler Under Sidewalk Stringers.
3-3"-12" Plank Stringers Under Each Sidewalk.
Floor:- 1 Layer 3"-8" Plank Transv.
1 Layer Wood Block (3"-3"-8") Coarse Run Longitudinal
Abutments- Plane, Ave. Size 35'-0" x 6'



PROFILE OF STREAM
Scales- Hor. 1"=100'-0" Vert. 1"=10'



PROFILE ON PROPOSED ℄ ROADWAY
Scales- Hor. 1"=30'-0" Vert. 1"=5'-0"

THIS IS STRUCTURE NO. 1 ON ROAD PROJ. NO. 377 SEE SHEET OF ROAD DEPT. PLANS FOR GRADE LINE, BENCH MARKS, AND REFERENCES.

LAYOUT
REINFORCED CONCRETE BRIDGE
4 SPANS @ 72' SKEW 15° L. 3' WALLS 44'-0" ROADWAY
OVER WABASH RIVER ON STATE ROAD.-1-0

INDIANA STATE HIGHWAY COMMISSION
WELLS COUNTY

SCALE:- AS NOTED DECEMBER 15, 1933

RECOMMENDED FOR APPROVAL:- Clifford Hardy ENGINEER OF BRIDGES

PROJECT:- 1
SECTION:- 0
STATION:- 66+10

DRAWING:- C1 OF 15

STRUCTURE NO. 230

BRIDGE CONTRACT NO. 692

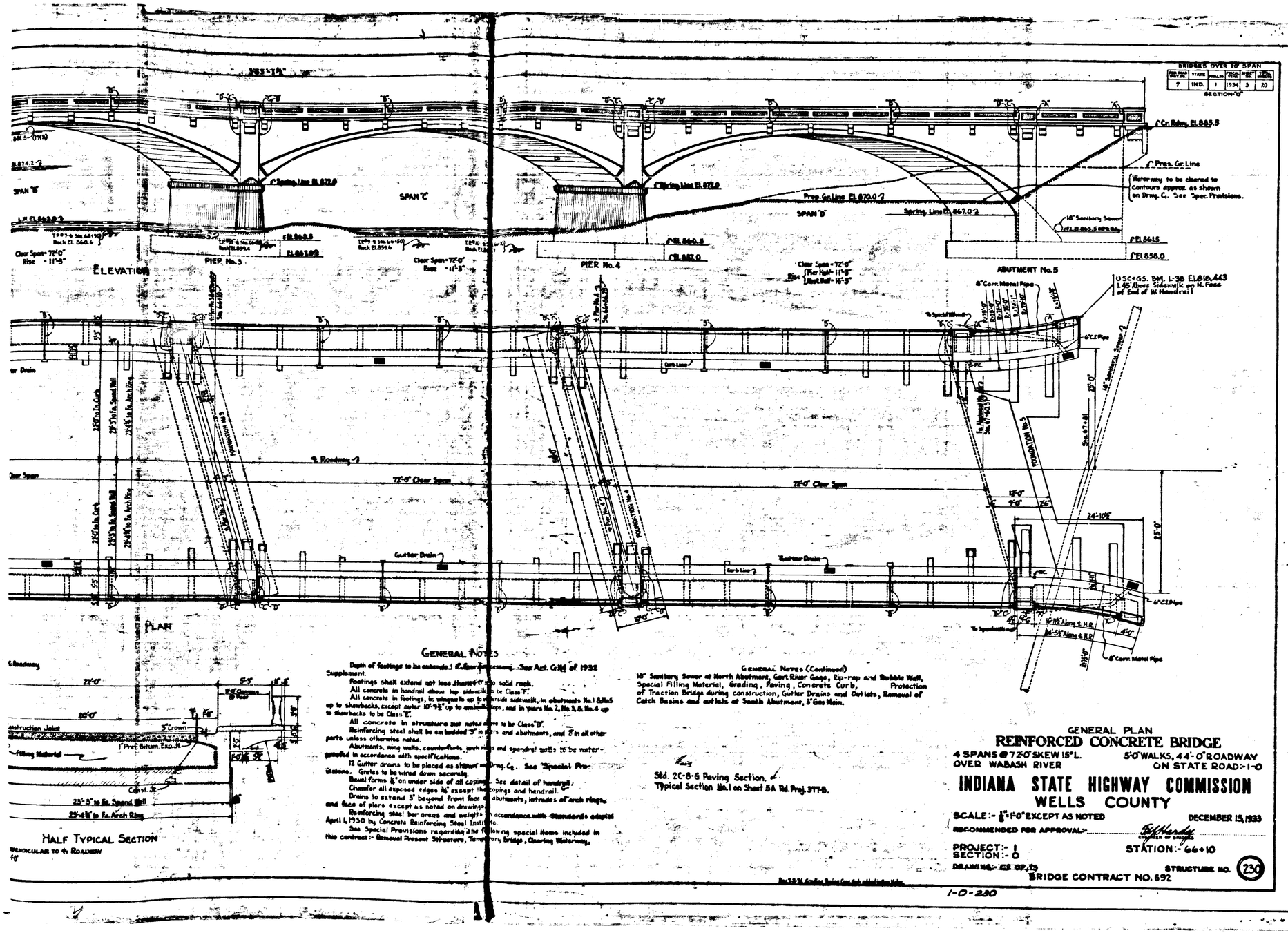
FIELD NOTES- Book Nº 885- pp. 28-70

1-0-230

BRIDGES OVER 10' SPAN

FED. PROJ. OR SEC. NO.	STATE	ROUTE NO.	PROJ. OR SEC. NO.	SHEET NO.	TOTAL SHEETS
7	IND.	1	1534	3	20

SECTION-O



HALF TYPICAL SECTION
PERPENDICULAR TO ℄ ROADWAY

GENERAL NOTES

Depth of footings to be extended if necessary. See Act. G.Rd. of 1932 Supplement.

Footings shall extend not less than 4'-0" into solid rock.

All concrete in handrail above top sidewalk to be Class "F".

All concrete in footings, in wingwalls up to riverside sidewalk, in abutments No. 1 & No. 5 up to shoulders, except outer 10'-9½" up to coping tops, and in piers No. 2, No. 3, & No. 4 up to shoulders to be Class "E".

All concrete in structure not noted above to be Class "D".

Reinforcing steel shall be embedded 3" in piers and abutments, and 2" in all other parts unless otherwise noted.

Abutments, wing walls, counterforts, arch rings and spandrel walls to be water-proofed in accordance with specifications.

12 Gutter drains to be placed as shown on Drwg. C. See "Special Provisions." Grates to be wired down securely.

Bevel forms ¾" on under side of all copings. See detail of handrail.

Chamfer all exposed edges ¾" except the copings and handrail.

Drains to extend 3" beyond front face of abutments, intrados of arch rings, and face of piers except as noted on drawings.

Reinforcing steel bar areas and weights in accordance with "Standards" adopted April 1, 1930 by Concrete Reinforcing Steel Institute.

See Special Provisions regarding the following special items included in this contract:- Removal Present Structure, Temporary Bridge, Clearing Waterway.

GENERAL NOTES (Continued)

18" Sanitary Sewer at North Abutment, Cont. River Gage, Rip-rap and Rubble Wall, Special Filling Material, Grading, Paving, Concrete Curb, Protection of Traction Bridge during construction, Gutter Drains and Outlets, Removal of Catch Basins and outlets at South Abutment, 3" Gas Main.

Std. 2C-8-6 Paving Section.
Typical Section No. 1 on Sheet 5A Rd. Proj. 377-B.

GENERAL PLAN
REINFORCED CONCRETE BRIDGE

4 SPANS @ 72'-0" SKEW 15° L. 5'-0" WALKS, 44'-0" ROADWAY
OVER WABASH RIVER ON STATE ROAD :- 1-O

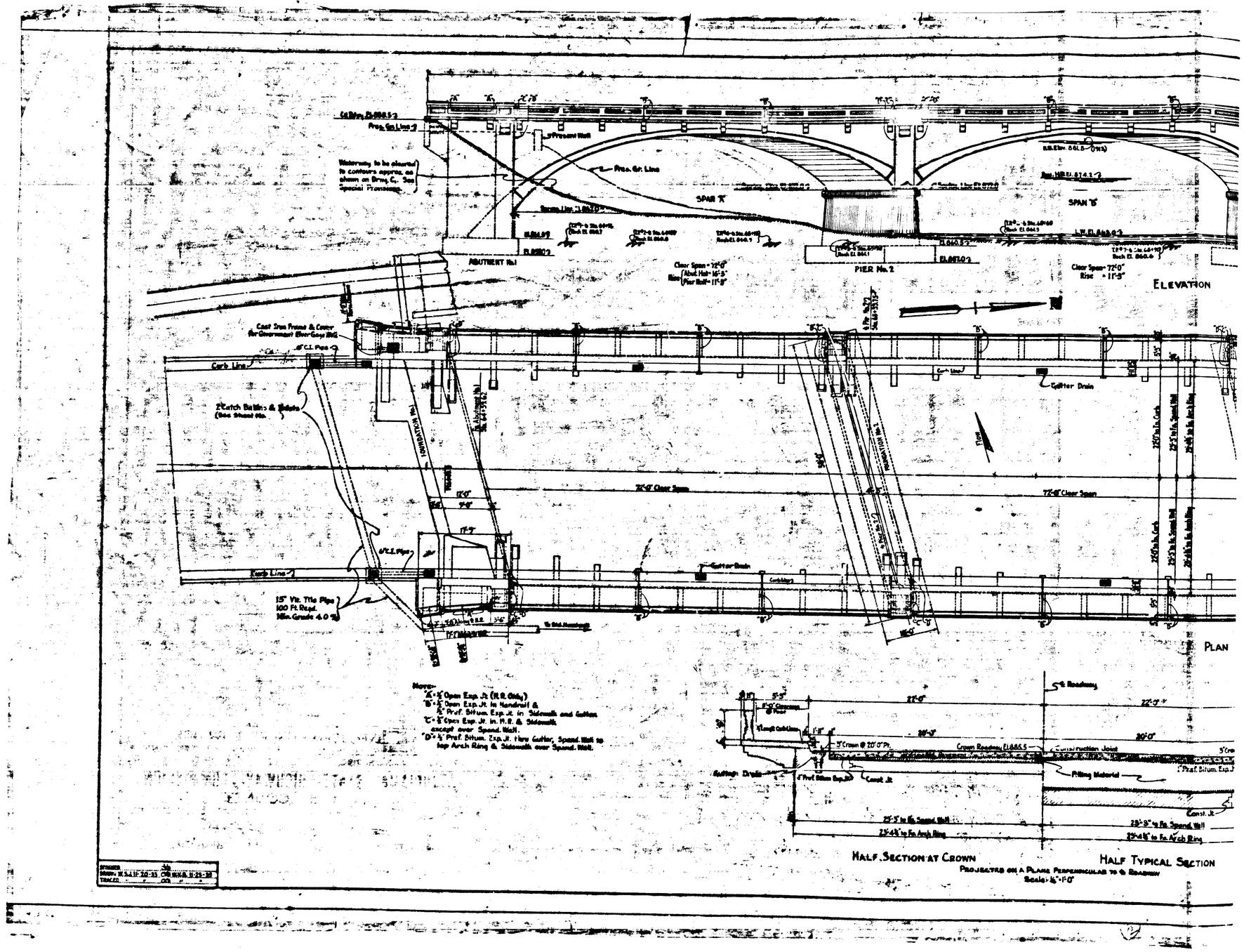
INDIANA STATE HIGHWAY COMMISSION
WELLS COUNTY

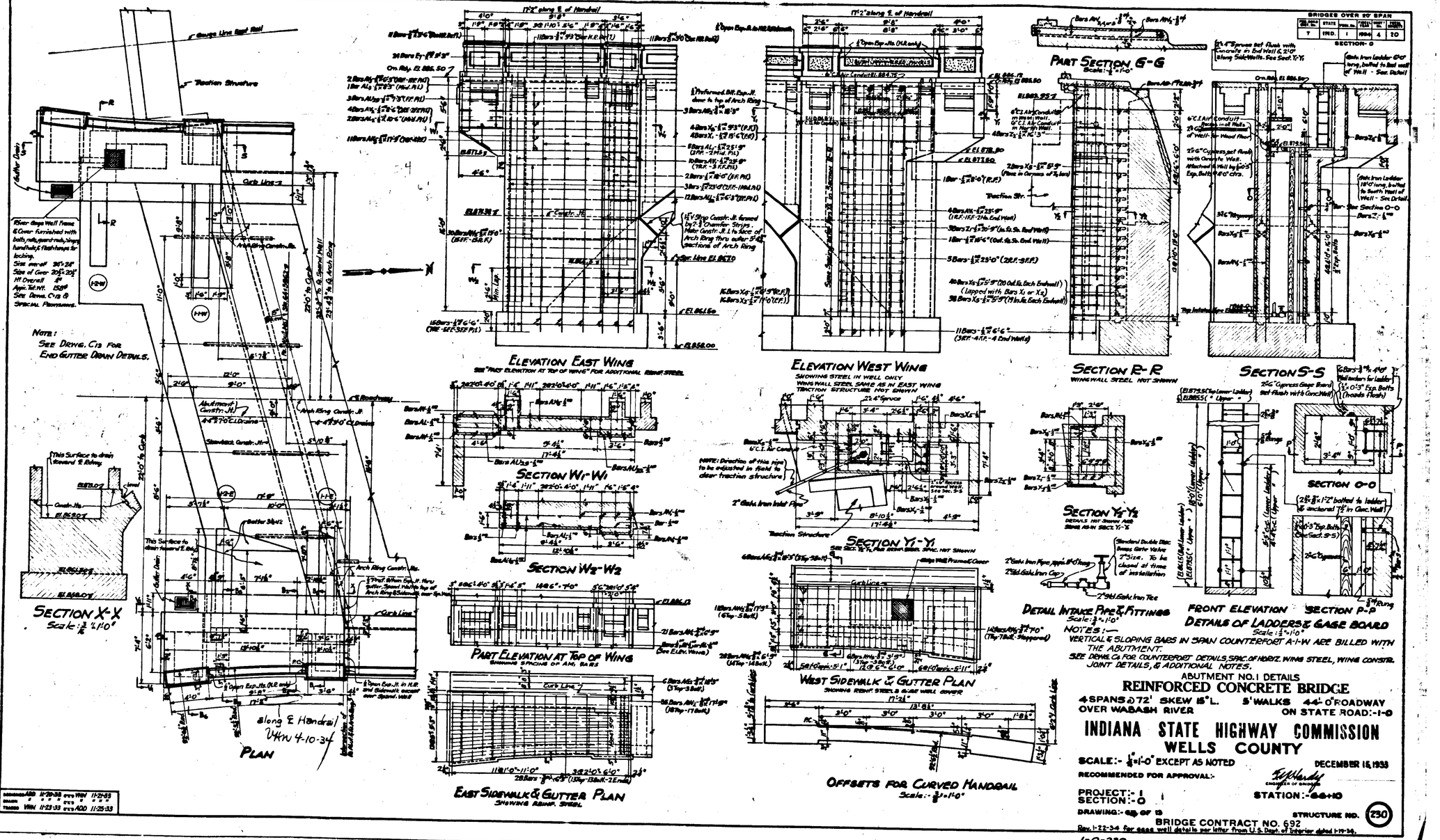
SCALE :- 1" = 10'-0" EXCEPT AS NOTED DECEMBER 15, 1933

RECOMMENDED FOR APPROVAL :- [signature] ENGINEER OF BRIDGES

PROJECT :- 1 STATION :- 66+10
SECTION :- O
DRAWING :- CE 132, 13 STRUCTURE NO. 230
BRIDGE CONTRACT NO. 692

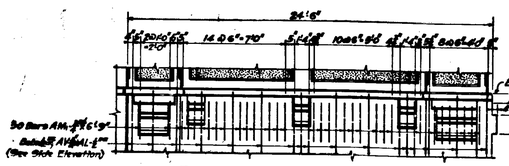
1-O-230


Waterway to be excavated to contours approx. as shown on Drwg. C. See Special Provisions
Prop. Gr. Line
Present Wall
SPAN 'A'
SPAN 'B'
ABUTMENT No.1
PIER No. 2
Clear Span = 72'0"
Rise = 11'9"
ELEVATION
Cast Iron Frame & Cover for Government River Gage Well
6" C.I. Pipe
Curb Line
2" Catch Basins & Inlets (See Sheet No.)
FOUNDATION No.1
FOUNDATION No.2
Gutter Drain
Flow
72'-8" Clear Span
15" Vit. Tile Pipe 100 Ft. Rad. Min. Grade 4.0 %
PLAN
Notes:
'A' = ½" Open Exp. Jt. (R.R. Obs.)
'B' = ½" Open Exp. Jt. in Handrail & ½" Pref. Bitum. Exp. Jt. in Sidewalk and Gutter.
'C' = ½" Open Exp. Jt. in R.R. & Sidewalk, except over Spand. Wall.
'D' = ½" Pref. Bitum. Exp. Jt. thru Gutter, Spand. Wall to top Arch Ring & Sidewalk over Spand. Wall.
5" Roadway
22'-0"
20'-0"
5" Crown @ 20'-0" Pt.
Crown Roadway El. 886.5
Construction Joint
Roadway Drain
Filling Material
Const. Jt.
25'-5" to Fc. Spand. Wall
25'-4.5" to Fc. Arch Ring
HALF SECTION AT CROWN
HALF TYPICAL SECTION
PROJECTED ON A PLANE PERPENDICULAR TO ℄ ROADWAY
Scale: ½"=1'-0"

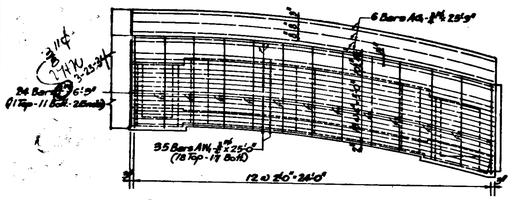

Section X-X
Scale: 3/8"=1'-0"
Plan
Elevation East Wing
Section W1-W1
Section W2-W2
Part Elevation at Top of Wing
East Sidewalk & Gutter Plan
Elevation West Wing
Section Y1-Y1
West Sidewalk & Gutter Plan
Offsets for Curved Handrail
Part Section G-G
Section R-R
Wingwall Steel Not Shown
Section S-S
Section O-O
Detail Intake Pipe & Fittings
Front Elevation
Section P-P
Details of Ladders & Gage Board
Bridges over 20' Span
NOTE: See Drwg. C13 for End Gutter Drain Details.
NOTES:—
Vertical & sloping bars in span counterfort A-1-W are bilked with the abutment.
See Drwg. C4 for counterfort details, spec. of horiz. wing steel, wing constr. joint details, & additional notes.
Abutment No. 1 Details
Reinforced Concrete Bridge
4 Spans @ 72' Skew 15° L. 5' Walks 44'-0" Roadway
over Wabash River on State Road:-1-0
Indiana State Highway Commission
Wells County
Scale:- 1/4"=1'-0" except as noted
December 15, 1933
Recommended for Approval:-
Engineer of Bridges
Project:- 1
Section:- D
Station:- 66+10
Drawing:- 6th of 13
Structure No. 230
Bridge Contract No. 692
Rev. 1-22-34 for 6666 wall details per letter from U.S. Dept. of Interior dated 1-19-34.
1-0-230

BRIDGES OVER 80' SPAN					
FED. ROAD DIST. NO.	STATE	FED. NO.	SHEET NO.	SHEETS IN SET	SHEET IDENTIFICATION
7	IND.	1	1964	5	70

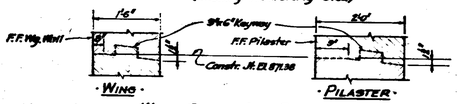
SECTION-C



· PART · ELEVATION · AT · TOP · OF · WING ·
(Showing Spacing of AM₁ Bars)

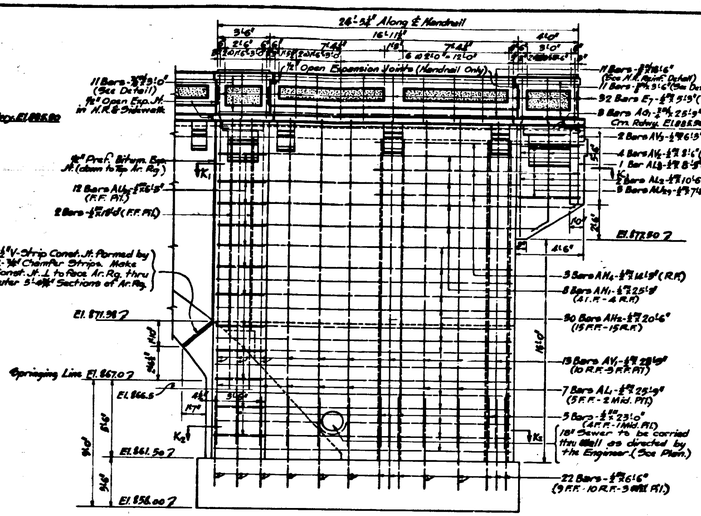


· SIDEWALK · & · GUTTER · PLAN ·
(Showing Reinforcing Steel)

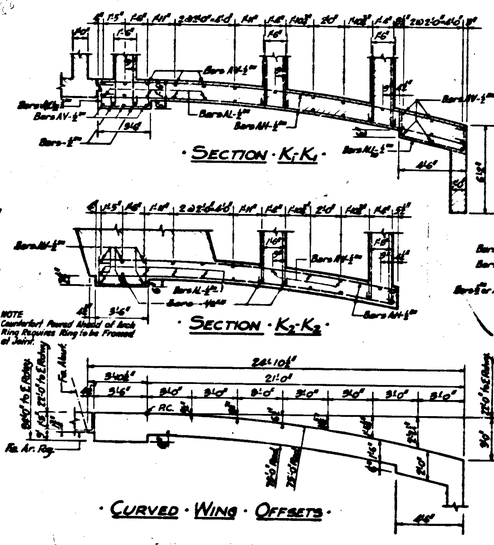


· WING · · PILASTER ·

· WING · CONST · JTS · DETAIL ·
Scale: ¾"=1'-0"



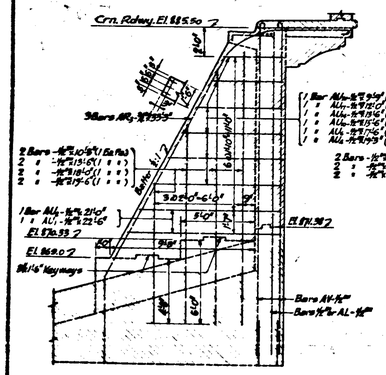
· SIDE · ELEVATION ·
(See "Part Elevation At Top Wing" For Additional Reinforcing Steel)



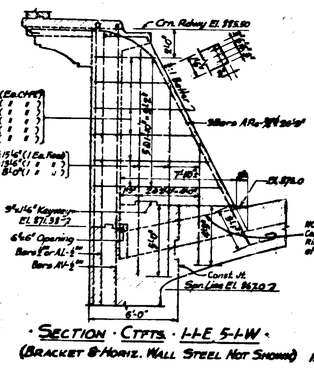
· SECTION · K₁-K₁ ·

· SECTION · K₂-K₂ ·

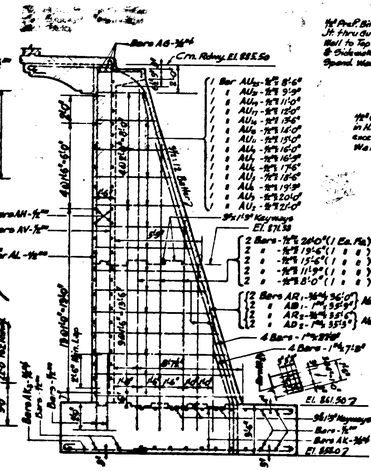
· CURVED · WING · OFFSETS ·



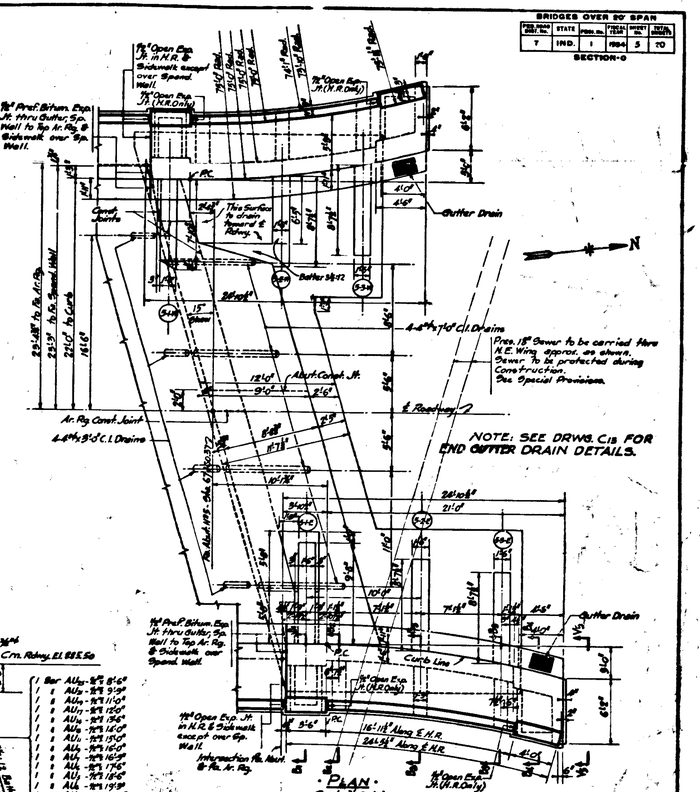
· SECTION · CTFTS · 1-1-E, 5-1-E ·
(BRACKET & HORIZ. WALL STEEL NOT SHOWN)



· SECTION · CTFTS · 1-1-E 5-1-W ·
(BRACKET & HORIZ. WALL STEEL NOT SHOWN)



· SECTION · CTFTS · 1-2-E, 1-2-W, 5-2-E, 5-3-E, 5-2-W, 5-3-W ·
(FOOTING STEEL NOT TYPICAL ALL CTFTS. - SEE PIER PLAN)
(BRACKET STEEL NOT SHOWN)



· PLAN ·
Scale: ¾"=1'-0"

NOTES: FOR DETAILS OF HANDRAIL, HR. POST, BRACKETS, CURB, GUTTER, HALF SECT. THRU STRUCT, SIDEWALK, WALK LINERS, EXP. JOINTS, AND GUTTER DRAINS, SEE DRWGS. C7 & C10.
FOR SPAN CTFT. DIMENSIONS, REINF. STEEL, AND DETAILS; AND SPAN BRACKET STEEL, SEE DRWG. C-9.
VERTICAL & SLOPING BARS IN SPAN COUNTERFORT D-10-E ARE BILLED WITH ABUTMENT.
BRACKET SPACINGS AND DIMENSIONS OF UMBRELLA TOP CONST. JOINTS ARE MEASURED AT FACE OF ARCH RING. SEE DRWG. C-5 FOR BRACKET DETAILS & FOOTING PLAN.
ALL REINF. BARS WITHOUT A LETTER & NUMBER ARE STRAIGHT.
ALL REINF. BARS WITH A LETTER & NUMBER ARE BENT.

ABUTMENT NO. 5 DETAILS
REINFORCED CONCRETE BRIDGE
4 SPANS @ 72' SKEW 15° L. 5' WALKS 44'-0" ROADWAY
OVER WABASH RIVER ON STATE ROAD: I-69
INDIANA STATE HIGHWAY COMMISSION
WELLS COUNTY

SCALE: - ½"-1'-0" (EXCEPT AS NOTED) DECEMBER 15, 1933
RECOMMENDED FOR APPROVAL: F. H. Hardy, Engineer of Bridges
PROJECT: - I
SECTION: - D
STATION: - 66+10
DRAWING: - C4 OF 15
STRUCTURE NO. 230
BRIDGE CONTRACT NO. 692

DESIGNED H.N.H. - 11/14/33 DET. A.B.D. - 11/17/33
DRAWN A. - DET. A.
TRACED L.Z.S. - 11/24/33 CHK. H.N.H. 11/24/33

I-D-230

BRIDGES OVER 20' SPAN

DIST. NO.	STATE	ROUTE NO.	YEAR	SHEET NO.	TOTAL SHEETS
7	IND.	1	1934	6	20

SECTION-B

Traction Structure

Skew & Parbus

Same Spacing as in N.E. Wing

See Coundwall Detail on Dwg. C2 for Bars and embedment

℄ Roadway

Abutment No. 1

Abutment No. 2

Foundation No. 1

Foundation No. 2

FLOW

N

FOOTING PLAN

SHOWING DIMENSIONS, HEAT LINES, & VERTICAL STEEL EXTENDING INTO FOOTING

Scale: $\frac{1}{8}$"=1'-0"

FOOTING PLAN

SHOWING FOOTING STEEL

Scale: $\frac{1}{8}$"=1'-0"

SECTION B-B₁ (See Dwg. C3) (See Dwg. C4)

ELEVATION C-C

ELEVATION V₁-V₁ (See Dwg. C3)

SECTION T₁-T₁

SECTION B₂-B₂ (See Dwg. C3)

SECTION B₃-B₃ (See Dwg. C3)

SECTION U-U

SECTION Z-Z

SECTION B₃-B₃ (See Dwg. C4)

Scale: $\frac{1}{2}$"=1'-0"

ELEVATION V₂-V₂ (See Dwg. C4)

SECTION T₂-T₂

ELEVATION D-D

SECTION B₄-B₄ (See Dwg. C4)

SECTION Q-Q

SECTION B₂-B₂ (See Dwg. C3 & C4)

FOOTING PLANS & BRACKET DET.-ABUTS. NO. 1 & 5

REINFORCED CONCRETE BRIDGE

4 SPANS @ 72' SKEW 15° L. 5' WALKS 44'-0" ROADWAY

OVER WABASH RIVER ON STATE ROAD:-1-O

INDIANA STATE HIGHWAY COMMISSION

WELLS COUNTY

SCALE:-$\frac{1}{8}$"=1'-0" EXCEPT AS NOTED DECEMBER 15, 1933

RECOMMENDED FOR APPROVAL:- E. H. Hardy

ENGINEER OF BRIDGES

PROJECT:-1

SECTION:-O STATION:-66+10

DRAWING:- C3 OF 13 STRUCTURE NO. 230

BRIDGE CONTRACT NO. 692

1-0-230

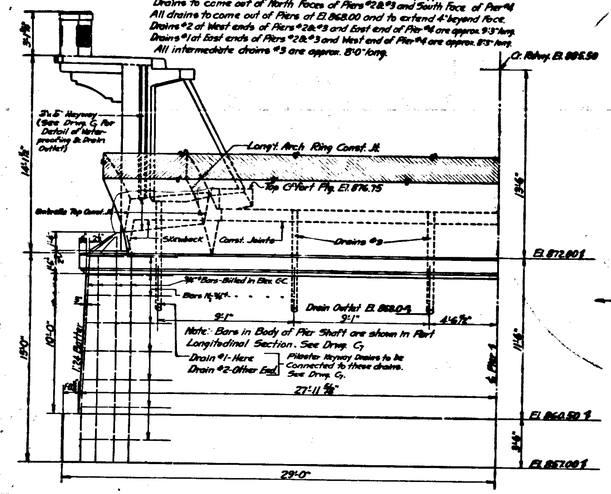
BRIDGES OVER 20' SPAN

FED. PROJ. NO.	STATE	PROJ. NO.	SHEET NO.	NO. OF SHEETS
Y	IND.	1984	7	20

SECTION-C

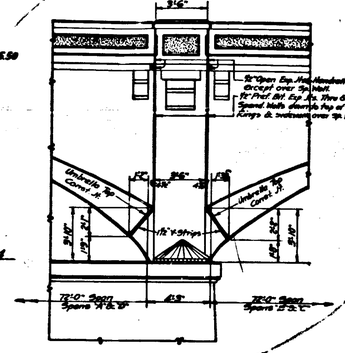
DRAIN NOTES.

Drains Shown, to be 4"ø C.I. Drains or 4"ø Steel Tubing Approx. 1/8" Thick.
Drains to come out of North Faces of Piers #2 & #3 and South Face of Pier #4
All drains to come out of El. 868.00 and to extend 4" beyond Face.
Drains #2 at West ends of Piers #2 & #3 and East end of Pier #4 are approx. 9'3" long.
Drains #1 at East ends of Piers #2 & #3 and West end of Pier #4 are approx. 8'3" long.
All intermediate drains #3 are approx. 8'0" long.

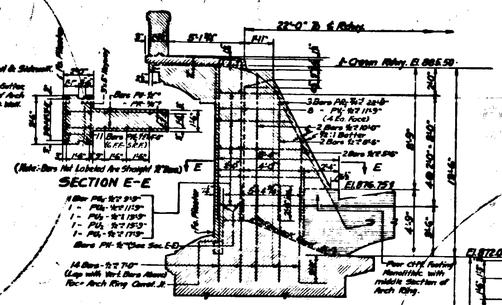


HALF SIDE ELEVATION

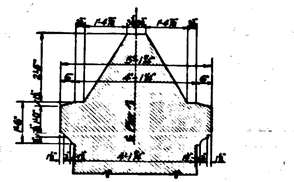
NOTE:- Umbrella Top Const. Jt. Dimensions Apply at Face of Arch Ring Only. Make Const. Jt. ⊥ to Face of Arch Ring Thru Outer 5'-4 7/8" Section of Arch Ring.



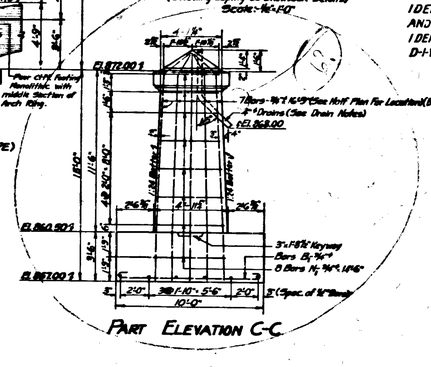
PART ELEVATION A-A



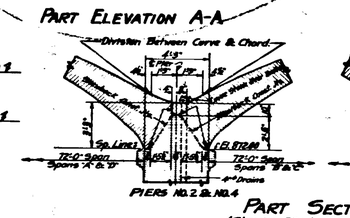
PART SECTION D-D
(Showing Steel in Counterforts PW and PE)



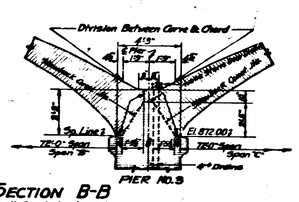
NOTE:- SLOPING BARS IN SPAN COUNTERFORTS A-10-E AND B-1-W ARE BILLED WITH PIER NO. 2 IDENTICAL BARS FOR COUNTERFORTS B-10-E AND C-1-W ARE BILLED WITH PIER NO. 3. IDENTICAL BARS FOR COUNTERFORTS C-10-E AND D-1-W ARE BILLED WITH PIER NO. 4.



PART ELEVATION C-C

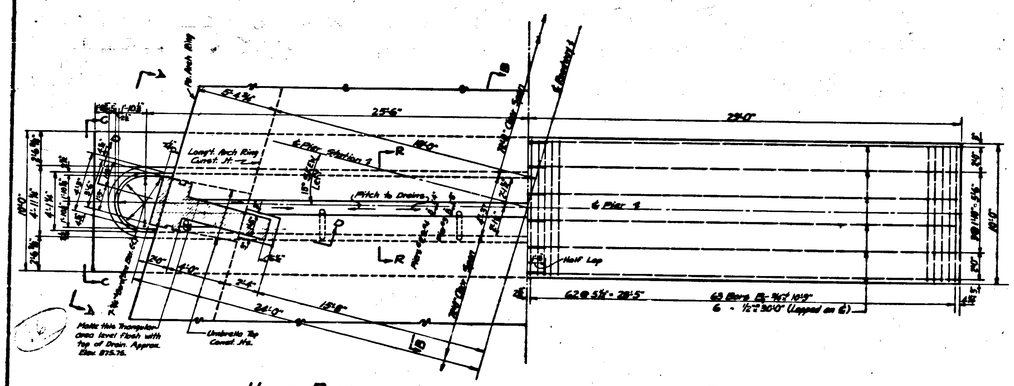


PIERS NO. 2 & NO. 4



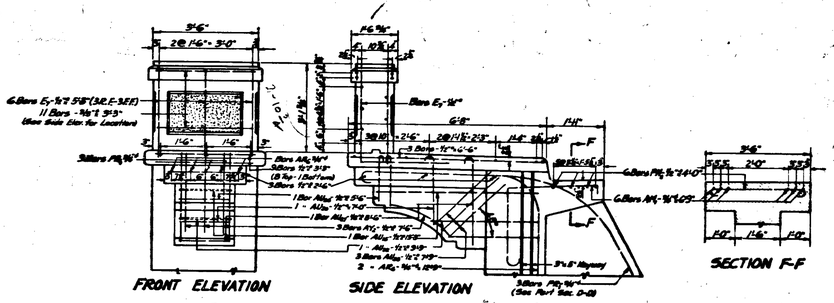
PIER NO. 3

PART SECTION B-B
(Showing Stirrback Const. Joints Between Pier Shaft and Arch Ring.)



HALF PLAN
(All Details above Top of Counterfort Footing and Pier Coping omitted)

HALF PLAN OF FOOTING



SIDEWALK, BRACKET AND HANDRAIL POST
Showing Steel Only. See Dwg. C10 For Dimensions.
Scale:- 1/2"=1'-0"

NOTE:-
FOR DETAILS OF H.R. POST, BRACKETS, CURB, GUTTER, HALF SECTION THRU STRUCTURE, SIDE WALK, WALK LINERS & EXPANSION JOINTS, SEE DRWGS. C9 & C10.
FOR SPAN COUNTERFORT DIMENSIONS, REINFORCING STEEL & DETAILS, SEE DRWG. C7.
ALL REINFORCING BARS WITHOUT A LETTER & NUMBER ARE STRAIGHT BARS.
ALL REINFORCING BARS WITH A LETTER & NUMBER ARE BENT BARS.

PIERS NO. 2, 3 & 4 - DETAILS

REINFORCED CONCRETE BRIDGE

4 SPANS @ 72' SKEW 15° L. 5' WALKS 44'-0" ROADWAY
OVER WABASH RIVER ON STATE ROAD:- I-0

INDIANA STATE HIGHWAY COMMISSION
WELLS COUNTY

SCALE:- 1/4"=1'-0" EXCEPT AS NOTED DECEMBER 15, 1933

RECOMMENDED FOR APPROVAL:- R. H. Hardy
ENGINEER OF BRIDGES

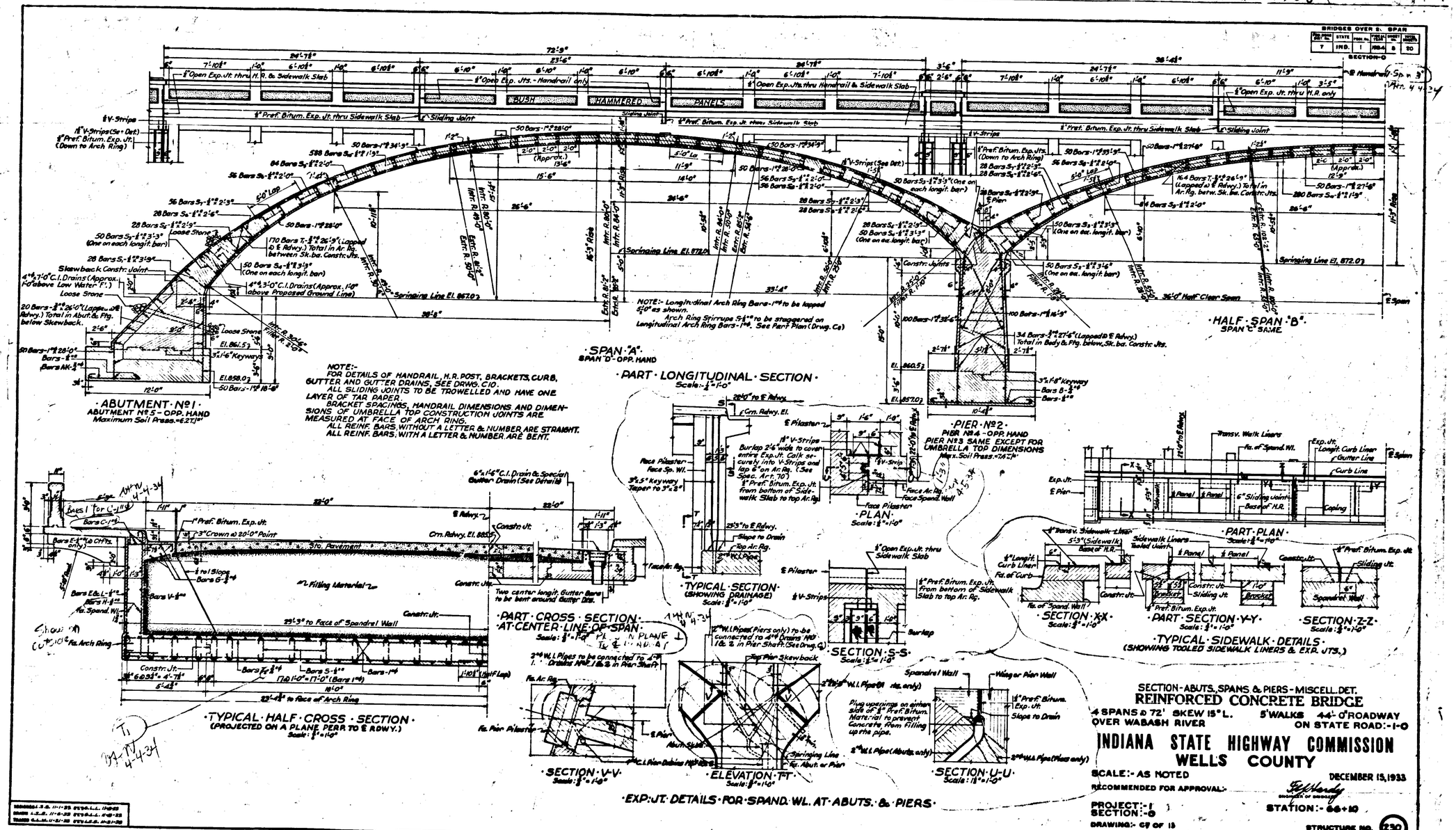
PROJECT:- I
SECTION:- C
STATION:- 66+10

DRAWING:- C6 OF 13
STRUCTURE NO. 230

BRIDGE CONTRACT NO. 692

Plans Rec'd. 11-8-33 Ctr's Rec'd. 11-9-33
Road 11-14-33 Ctr's A.E.M. 11-18-33
W.P.S. 11-29-33 Ctr's Rec'd 11-24-33

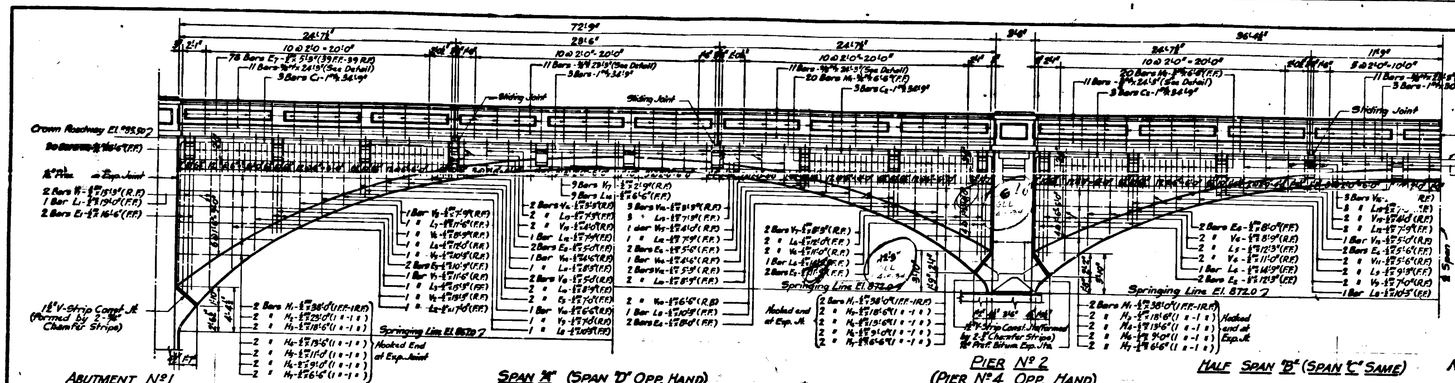
1-0-230


·ABUTMENT·Nº1·
ABUTMENT Nº5 - OPP. HAND
Maximum Soil Press.=6.27 T/Sq.Ft.
·SPAN·"A"·
SPAN "D" - OPP. HAND
·PART·LONGITUDINAL·SECTION·
Scale:- 1/4"=1'-0"
·HALF·SPAN·"B"·
SPAN "C" SAME
·PIER·Nº2·
PIER Nº3 SAME EXCEPT FOR UMBRELLA TOP DIMENSIONS
PIER Nº4 - OPP. HAND
Max. Soil Press.=7.47 T/Sq.Ft.
·PLAN·
·TYPICAL·HALF·CROSS·SECTION·
(PROJECTED ON A PLANE PERP. TO ℄ RDWY.)
·PART·CROSS·SECTION·
AT·CENTER·LINE·OF·SPAN·
·TYPICAL·SECTION·
(SHOWING DRAINAGE)
·SECTION·S-S·
·SECTION·V-V·
·ELEVATION·T-T·
·SECTION·U-U·
·PART·PLAN·
·SECTION·X-X·
·PART·SECTION·Y-Y·
·SECTION·Z-Z·
·TYPICAL·SIDEWALK·DETAILS·
(SHOWING TOOLED SIDEWALK LINERS & EXP. JTS.)
·EXP.JT. DETAILS·FOR·SPAND. WL. AT·ABUTS.·& ·PIERS·
NOTE:-
FOR DETAILS OF HANDRAIL, H.R. POST, BRACKETS, CURB, GUTTER AND GUTTER DRAINS, SEE DRWG. C10.
ALL SLIDING JOINTS TO BE TROWELLED AND HAVE ONE LAYER OF TAR PAPER.
BRACKET SPACINGS, HANDRAIL DIMENSIONS AND DIMENSIONS OF UMBRELLA TOP CONSTRUCTION JOINTS ARE MEASURED AT FACE OF ARCH RING.
ALL REINF. BARS, WITHOUT A LETTER & NUMBER ARE STRAIGHT.
ALL REINF. BARS, WITH A LETTER & NUMBER ARE BENT.
NOTE:- Longitudinal Arch Ring Bars-1"ø to be lapped 5'-0" as shown.
Arch Ring Stirrups-5/8"ø to be staggered on Longitudinal Arch Ring Bars-1"ø. See Part Plan (Drwg. C8)
BRIDGES OVER E. SPAN
STATE: IND. | ROAD NO.: I | ROAD OR STREET: 136-A | SHEET NO.: 8 | SHEETS: 20
SECTION-Q
SECTION-ABUTS., SPANS & PIERS-MISCELL. DET.
REINFORCED CONCRETE BRIDGE
4 SPANS @ 72' SKEW 15°L. 5'WALKS 44'-0" ROADWAY
OVER WABASH RIVER ON STATE ROAD:-I-0
INDIANA STATE HIGHWAY COMMISSION
WELLS COUNTY
SCALE:- AS NOTED
DECEMBER 15, 1933
RECOMMENDED FOR APPROVAL:-
ENGINEER OF BRIDGES
PROJECT:- I
SECTION:- B
STATION:- 66+10
DRAWING:- C7 OF 15
STRUCTURE NO. 230
BRIDGE CONTRACT NO. 692

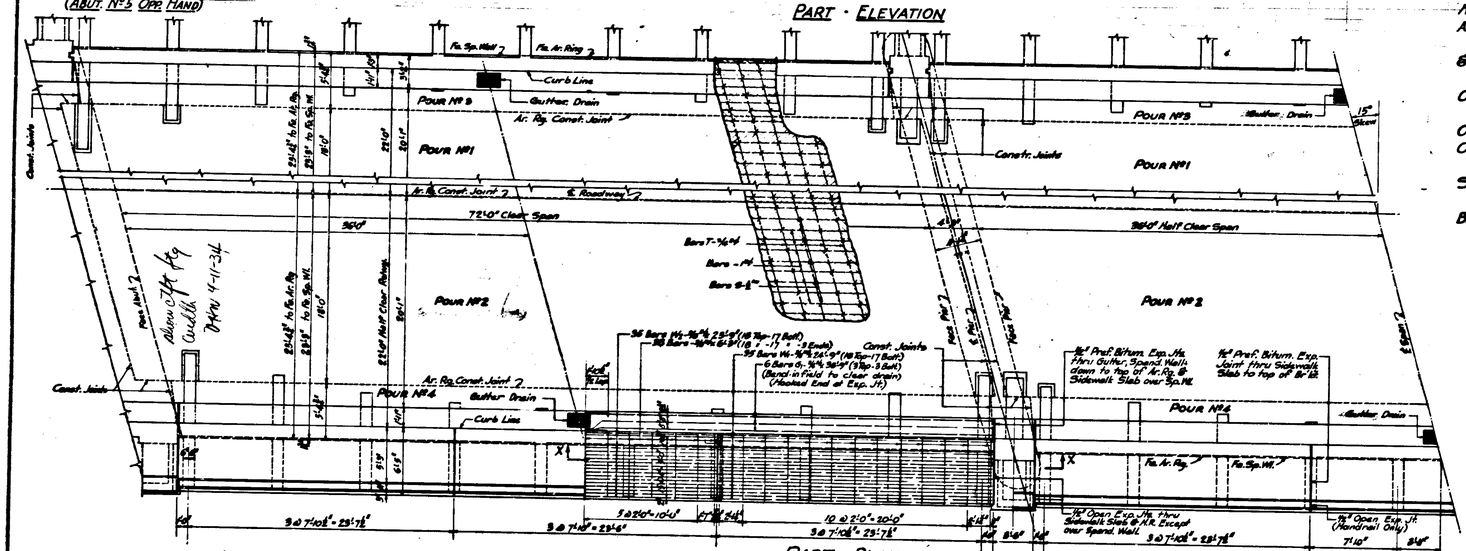
1-0-230

BRIDGES OVER 20' SPAN					
FED. AID PROJ. NO.	STATE	FED. NO.	YEAR	SHEET NO.	SHEETS
7	IND.	1	1934	9	20

SECTION-D



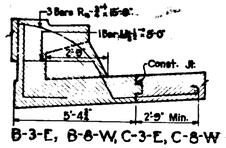
PART · ELEVATION


Pour No. 3
Pour No. 1
Pour No. 2
Pour No. 4
Curb Line
Gutter Drain
Const. Joints
Ar. Rg. Const. Joint
Roadway
72'-0" Clear Span
360° Half Clear Span
Pour No. 2
Pour No. 4
Pour No. 3
Pour No. 1
Pour No. 2
Pour No. 4
Curb Line
Gutter Drain
Const. Joints
Ar. Rg. Const. Joint
Roadway
Fa. Sp. Wall
Fa. Ar. Ring
Fa. Ar. Rg.
Fa. Sp. Wall
Sliding Joint
Sliding Joint
Sliding Joint
Const. Joints
Const. Joints
Const. Joints

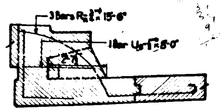
BRIDGES OVER 20' SPAN

DIST. NO.	STATE	PROJ. NO.	HIGHWAY NO.	SHEET NO.
7	IND.	1	1934	10	20

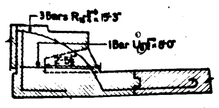
SECTION-B



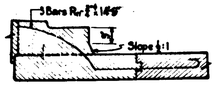
B-3-E, B-8-W, C-3-E, C-8-W



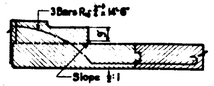
A-8-W, D-3-E



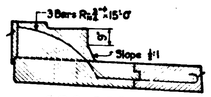
A-4-E, D-7-W



B-4-E, B-7-W, C-4-E, C-7-W
B-4-W, A-7-W, D-4-E



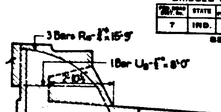
A-5-E, A-6-E, A-5-W, A-6-W
B-5-E, B-6-E, B-5-W, B-6-W
C-5-E, C-6-E, C-5-W, C-6-W
D-5-E, D-6-E, D-5-W, D-6-W



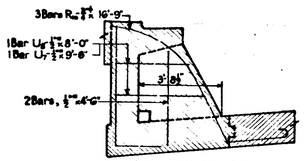
A-7-E, B-7-E, C-7-E
B-4-W, C-4-W, D-4-W



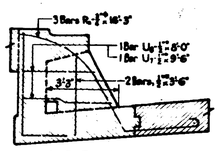
A-4-W, D-7-E



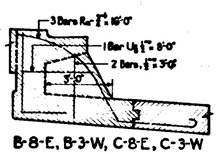
A-8-E, D-3-W



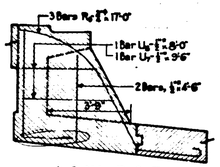
A-9-W, B-9-W, C-9-W
B-2-E, C-2-E, D-2-E



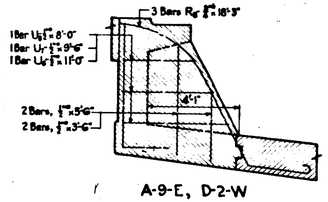
A-3-E, B-8-W



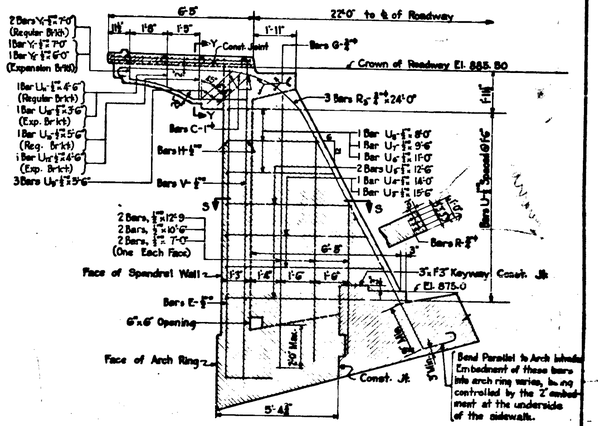
B-8-E, B-3-W, C-8-E, C-3-W



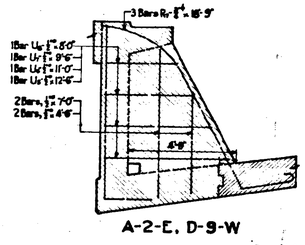
A-3-W, D-8-E



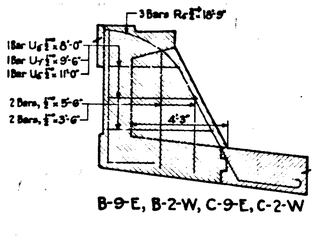
A-9-E, D-2-W



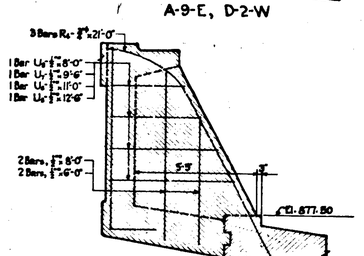
A-1-E, D-10-W



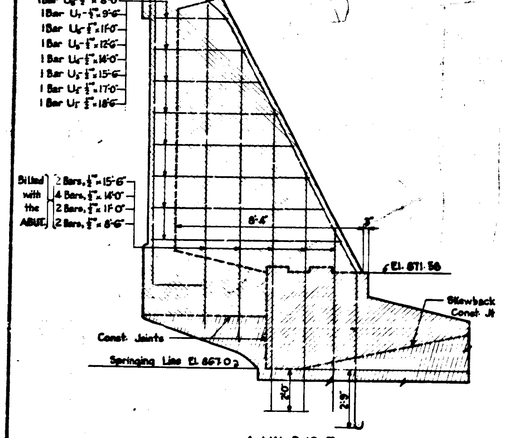
A-2-E, D-9-W



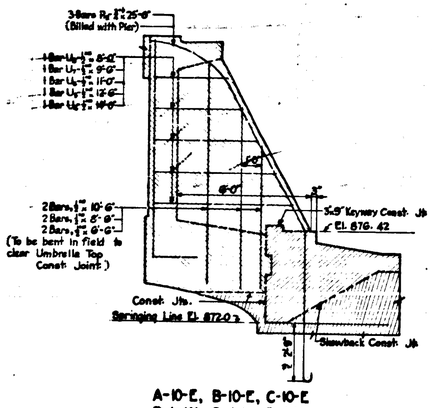
B-9-E, B-2-W, C-9-E, C-2-W



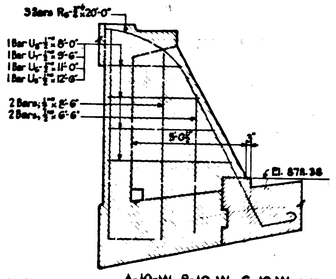
A-2-W, D-9-E



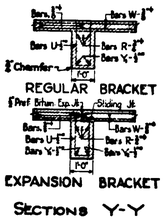
A-1-W, D-10-E



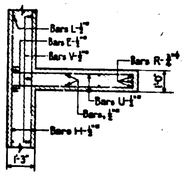
A-10-E, B-10-E, C-10-E
B-1-E, C-1-E, D-1-E



A-10-W, B-10-W, C-10-W
B-1-E, C-1-E, D-1-E



SECTIONS Y-Y



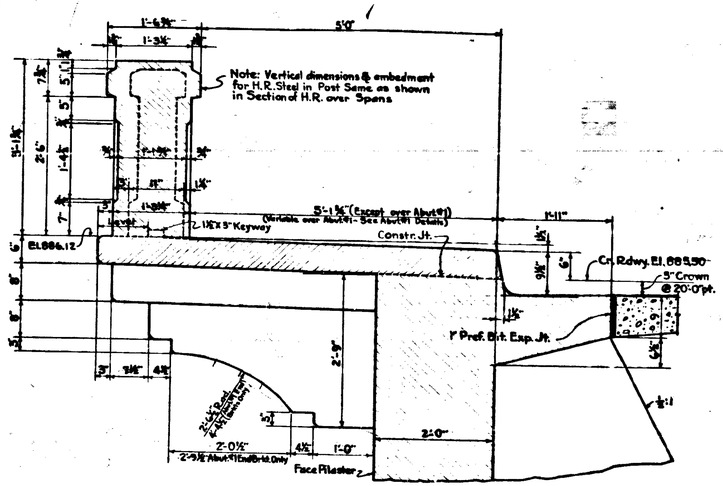
TYPICAL SECTION S-S

NOTE:-
ALL COUNTERFORT DETAILS NOT SHOWN ARE SAME AS SHOWN ON COUNTERFORT A-1-E, D-10-W.
ALL REINFORCING BARS WITHOUT A LETTER AND NUMBER ARE STRAIGHT.
ALL REINFORCING BARS WITH A LETTER AND NUMBER ARE BENT.

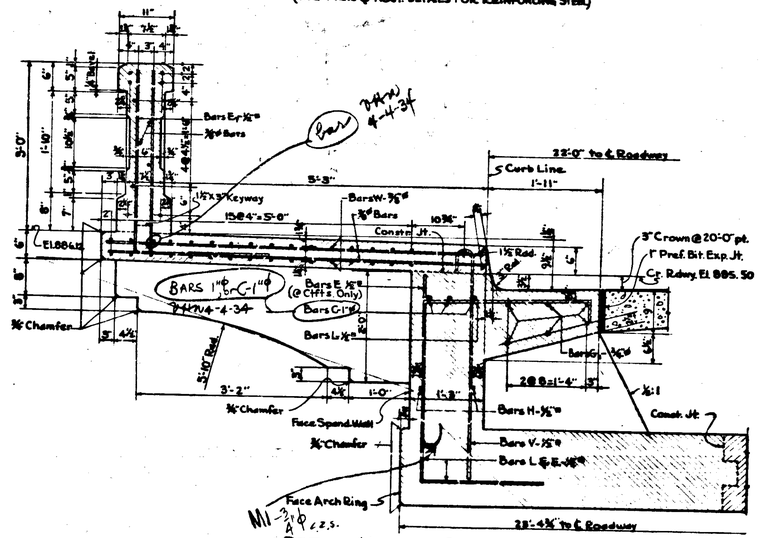
SPAN COUNTERFORT DETAILS
REINFORCED CONCRETE BRIDGE
4 SPANS @ 72' SKEW 15° L. 5' WALKS 44'-0" ROADWAY
OVER WABASH RIVER ON STATE ROAD:-1-0
INDIANA STATE HIGHWAY COMMISSION
WELLS COUNTY

SCALE:- ⅜"=1'-0" DECEMBER 15, 1933
RECOMMENDED FOR APPROVAL:- Hufford ENGINEER OF BRIDGES
PROJECT:- 1
SECTION:- B STATION:- 66+10
DRAWING:- CB OF 13 STRUCTURE NO. 230
BRIDGE CONTRACT NO. 692

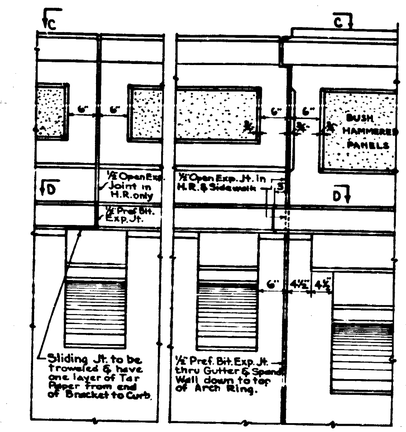
1-0-230



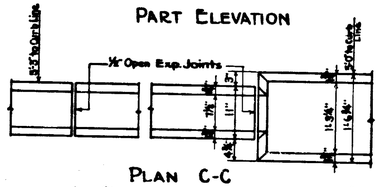
DETAIL H.R. POST, SIDEWALK, GUTTER & BRACKET AT PILASTERS
(SEE PIER & ABUT. DETAILS FOR REINFORCING STEEL)



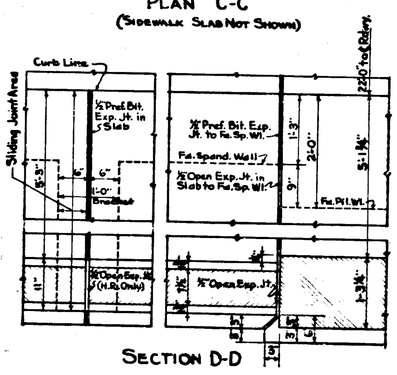
DETAIL- HANDRAIL, SIDEWALK, GUTTER & BRACKET OVER SPANS
(SEE DRWG. C9 FOR BRACKET STEEL)



PART ELEVATION

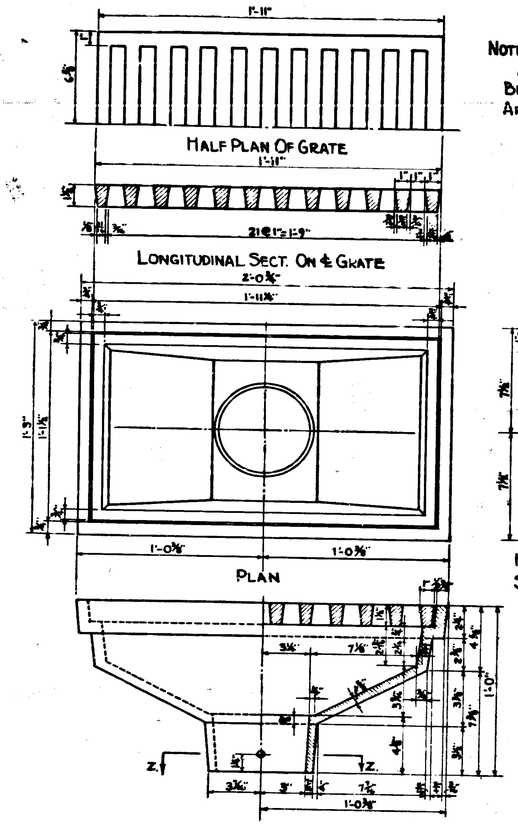


PLAN C-C
(SIDEWALK SLAB NOT SHOWN)



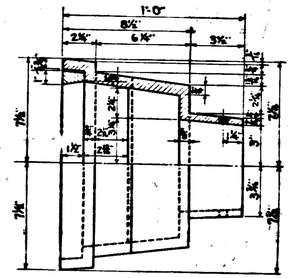
SECTION D-D

HANDRAIL & SIDEWALK EXP. JOINT DETAILS

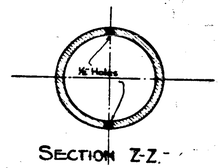


HALF SIDE ELEVATION
AND HALF SECT. ON LONGITUDINAL ℄

NOTE:-
ALL DIMENSIONS SHOWN ON DETAILS AND BENDING DIAGRAMS FOR REINFORCING BARS ARE MEASURED ON ℄ OF BAR.



HALF END ELEVATION AND HALF SECTION ON TRANSVERSE ℄



SECTION Z-Z

GUTTER DRAIN DETAILS
Scale 3"=1'-0"
Material:- Gray Iron in accordance with Art. 236-244 I.S.H.C. ... (1933 Specifications)
Approx. W.T. of Grate = 89 #
" " Box = 148 #
Total = 237 #

BRIDGES OVER 20' SPAN

DIST. NO.	STATE	ROAD NO.	YEAR	SHEET NO.	SHEETS
7	IND.	1	1933	11	29

SECTION-O

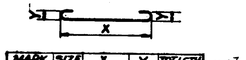
MISCELLANEOUS DETAILS
REINFORCED CONCRETE BRIDGE
4 SPANS @ 72' SKEW 15° R. 5' WALKS 44'-0" ROADWAY
OVER WABASH RIVER ON STATE ROAD:-1-O
INDIANA STATE HIGHWAY COMMISSION
WELLS COUNTY
SCALE:-1"=1'-0" (EXCEPT AS NOTED) DECEMBER 15, 1933
RECOMMENDED FOR APPROVAL:- H. Hardy
PROJECT:-1 STATION:-66+10
SECTION:-O
DRAWING:- C8 OF 13 STRUCTURE NO. 230
BRIDGE CONTRACT NO. 692

1-O-230

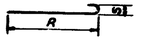
BRIDGES OVER 20' SPAN

FED. PROJ. NO.	STATE	PROJ. NO.	SHEET NO.	TOTAL SHEETS
7	IND.	I 1924	12	20

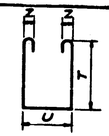
SECTION-O



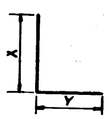
MARK	SIZE	X	Y	TOT. LGTH.
AG1	#6	24'-4"	48°	23'-9"
AG2	"	16'-11"	"	18'-9"
B1	"	9'-8"	"	10'-9"
AH1	#10	24'-"	9"	25'-9"
AH2	"	19'-8"	"	20'-6"
AH3	"	16'-10"	"	17'-9"
AH4	"	15'-2"	"	15'-9"
AH5	"	13'-2"	"	13'-9"
B2	"	3'-2"	"	4'-0"
S1	"	21'-11"	"	31'-9"
S2	"	2'-8"	"	3'-6"
S3	"	2'-5"	"	3'-3"
S4	"	2'-6"	"	3'-3"
S5	"	1'-8"	"	2'-6"
S6	"	1'-5"	"	2'-3"
S7	"	1'-10"	"	2'-7"
S8	"	1'-5"	"	2'-0"
S9	"	14.1"	"	2'-0"
S10	"	1'-0"	"	1'-9"
X3	"	8'-6"	"	9'-3"
X4	"	5'-11"	"	6'-9"
X5	"	4'-10"	"	5'-9"
AW1	#9	24'-6"	8"	25'-0"
AW2	"	17'-1"	"	17'-9"
AW3	"	6'-5"	"	7'-0"
AW4	"	6'-2"	"	6'-9"
AW5	"	3'-2"	"	3'-9"
W1	"	24'-2"	"	24'-9"
W2	"	23'-2"	"	23'-9"



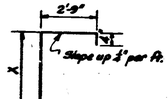
MARK	SIZE	R	S	TOT. LGTH.
AK1	#4	10'-2"	45°	11'-0"
AK2	"	6'-7"	"	7'-3"
AK3	"	5'-7"	"	6'-3"
AK4	"	4'-7"	"	5'-3"
B1	"	36'-1"	"	38'-3"
T1	"	26'-0"	"	26'-9"
AN4	#10	13'-10"	3"	14'-3"
AV1	"	23'-4"	"	23'-9"
AV2	"	8'-1"	"	8'-6"
AV3	"	5'-10"	"	6'-3"
H1	"	37'-6"	"	38'-0"
H2	"	22'-7"	"	23'-0"
H3	"	18'-1"	"	18'-6"
H4	"	18'-1"	"	18'-6"
H5	"	10'-7"	"	11'-0"
H6	"	8'-7"	"	9'-0"
H7	"	6'-1"	"	6'-6"
V1	"	14'-10"	"	15'-3"
V2	"	12'-10"	"	13'-3"
V3	"	11'-1"	"	11'-6"
V4	"	10'-7"	"	11'-0"
V5	"	9'-10"	"	10'-3"
V6	"	8'-3"	"	8'-9"
V7	"	7'-10"	"	8'-3"
V8	"	7'-4"	"	7'-9"
V9	"	6'-7"	"	7'-0"
V10	"	6'-1"	"	6'-6"
V11	"	5'-7"	"	5'-9"
V12	"	4'-10"	"	5'-3"
V13	"	4'-7"	"	5'-0"
V14	"	4'-1"	"	4'-6"
V15	"	3'-7"	"	4'-0"
V16	"	2'-10"	"	3'-3"
V17	"	2'-4"	"	2'-9"
AV1	"	11'-5"	"	11'-9"



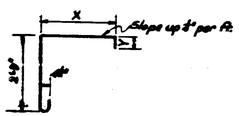
MARK	SIZE	U	T	Z	TOT. LGTH.
AU36	#8	2'-1"	2'-9"	3"	3'-3"
AU38	"	1'-5"	2'-1"	"	2'-5"
AU39	"	0'-10"	2'-5"	"	2'-9"
PU1	"	1'-1"	7'-11"	"	8'-3"
PU2	"	"	6'-11"	"	7'-3"
PU3	"	"	5'-11"	"	6'-3"
PU4	"	"	4'-11"	"	5'-3"
PU5	"	"	3'-11"	"	4'-3"
PU6	"	"	3'-11"	"	4'-3"
U3	"	0'-7"	2'-0"	"	2'-3"
Z2	"	1'-0"	7'-3"	"	8'-3"



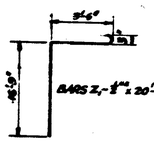
MARK	SIZE	X	Y	TOT. LGTH.
C1	#4	27'-7"	13'-3"	32'-9"
C2	"	26'-1"	6'-8"	34'-9"
E1	#6	14'-0"	2'-0"	16'-6"
E2	"	10'-3"	"	12'-3"
E3	"	8'-9"	"	10'-9"
E4	"	6'-0"	"	8'-0"
E5	"	5'-0"	"	7'-0"
E6	"	3'-9"	"	5'-6"
E7	"	2'-2"	2'-6"	4'-3"
E8	"	3'-0"	2'-0"	5'-0"

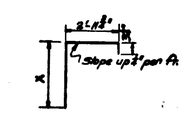


MARK	SIZE	X	TOT. LGTH.
L1	1/2"ø	13'-11"	19'-0"
L2	"	11'-11"	17'-0"
L3	"	10'-2"	15'-3"
L4	"	9'-2"	14'-3"
L5	"	7'-2"	12'-3"
L6	"	6'-11"	12'-0"
L7	"	6'-3"	11'-6"
L8	"	5'-2"	10'-3"
L9	"	4'-2"	9'-3"
L10	"	3'-8"	8'-3"
L11	"	3'-2"	8'-3"
L12	"	2'-8"	7'-9"
L13	"	2'-2"	7'-3"
L14	"	1'-8"	6'-8"



MARK	SIZE	X	Y	TOT. LGTH.
AM1	#4	2'-11"	3'-3"	6'-3"
AN1	"	3'-3"	3'-3"	6'-9"

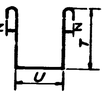




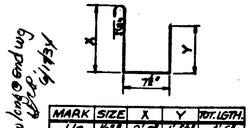
MARK	SIZE	X	TOT. LGTH.
AU35	#8	22'-6"	23'-3"
AU1	"	7'-3"	10'-6"
AU2	"	3'-0"	8'-3"



MARK	SIZE	G	N	TOT. LGTH.
AU30	#8	4'-11"	0'-7"	10'-6"
AU31	"	1'-7"	4'-7"	7'-3"
AU32	"	1'-7"	3'-0"	6'-3"
X1	"	2'-6"	8'-3"	13'-6"
X2	"	2'-7"	3'-0"	11'-0"



MARK	SIZE	U	T	Z	TOT. LGTH.
AU1	#8	1'-7"	10'-2"	3"	22'-6"
AU2	"	"	9'-2"	"	21'-0"
AU3	"	"	9'-2"	"	20'-0"
AU4	"	"	8'-2"	"	19'-3"
AU5	"	"	8'-2"	"	18'-3"
AU6	"	"	7'-2"	"	17'-6"
AU7	"	"	7'-2"	"	16'-9"
AU8	"	3'-7"	6'-3"	"	16'-6"
AU9	"	1'-8"	7'-2"	"	16'-9"
AU10	"	"	6'-2"	"	15'-6"
AU11	"	"	6'-6"	"	14'-0"
AU12	"	3'-1"	3'-6"	"	13'-6"
AU13	"	1'-11"	6'-1"	"	14'-0"
AU14	"	"	5'-9"	"	13'-6"
AU15	"	1'-8"	4'-5"	"	12'-3"
AU16	"	3'-1"	4'-6"	"	13'-0"
AU17	"	1'-11"	5'-0"	"	12'-0"
AU18	"	3'-1"	3'-6"	"	11'-6"
AU19	"	1'-11"	4'-6"	"	11'-0"
AU20	"	1'-8"	3'-9"	"	9'-3"
AU21	"	1'-11"	3'-6"	"	9'-3"
AU22	"	"	3'-1"	"	8'-3"
AU23	"	1'-8"	3'-1"	"	8'-0"
U4	"	0'-7"	2'-5"	"	3'-6"
U5	"	"	2'-0"	"	3'-0"
U6	"	"	3'-0"	"	4'-0"
U7	"	"	4'-0"	"	5'-0"
U8	"	"	3'-5"	"	8'-0"



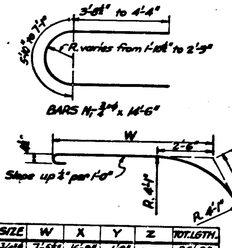
MARK	SIZE	X	TOT. LGTH.
AL1	#8	22'-6"	23'-3"
AL2	"	7'-3"	10'-6"
AL3	"	3'-0"	8'-3"

MARK	SIZE	X	Y	TOT. LGTH.
U2	1/2"ø	2'-0"	1'-3"	4'-6"
U3	"	1'-6"	0'-11"	3'-6"

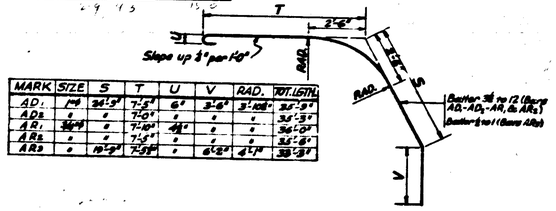
MARK	SIZE	K	TOT. LGTH.
AR2	#4	6'-8"	13'-6"
AR6	"	5'-11"	12'-9"
AR7	"	5'-8"	12'-6"



MARK	SIZE	R	S	T	U	RAD.	TOT. LGTH.
AV1	#6	1'-0"	1'-0"	2'-8"	3'-0"	4'-6"	8'-3"
AV2	"	"	0'-11"	1'-11"	"	2'-9"	7'-6"
AV3	"	1'-11"	"	3'-3"	2'-5"	7'-6"	7'-3"
AV4	"	"	"	2'-6"	"	6'-0"	7'-0"
V1	"	"	"	2'-1"	2'-3"	"	7'-0"
V2	"	0'-7"	"	"	"	"	6'-0"

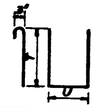


MARK	SIZE	W	X	Y	Z	TOT. LGTH.
AR4	3/4"ø	7'-5"	16'-9"	1'-9"		26'-9"
PR1	"	"	2'-9"	"		22'-9"
R1	"	7'-2"	15'-3"		7'-3"	30'-6"
R2	"	"	10'-1"		7'-1"	23'-0"
R3	"	"	13'-7"	1'-6"		24'-0"
R4	"	"	11'-2"	1'-6"		21'-0"
R5	"	"	10'-6"	1'-6"		20'-0"
R6	"	"	8'-0"	2'-3"		18'-9"
R7	"	"	8'-0"	1'-9"		18'-3"
R8	"	"	7'-7"	2'-3"		18'-3"
R9	"	c	6'-3"	"		17'-0"
R10	"	"	6'-1"	"		16'-9"
R11	"	"	5'-3"	3'-3"		15'-3"
R12	"	"	4'-7"	3'-3"		16'-3"
R13	"	"	4'-7"	3'-3"		15'-9"
R14	"	b4'-4"	4'-0"	3'-3"		15'-3"
R15	"	"	3'-8"	3'-10"		15'-6"
R16	"	b	3'-2"	4'-1"		15'-3"
R17	"	a2'-7"	2'-7"	4'-3"		13'-0"
R18	"	"	3'-0"	4'-3"		14'-9"
R19	"	"	2'-6"	4'-3"		14'-6"



MARK	SIZE	S	T	U	V	RAD.	TOT. LGTH.
AD1	#4	24'-3"	7'-3"	6"	3'-6"	3'-10"	34'-9"
AD2	"	"	7'-0"	"	"	"	34'-3"
AR1	3/4"ø	"	7'-10"	45°	"	"	36'-0"
AR2	"	"	7'-5"	"	"	"	35'-8"
AR3	"	10'-9"	7'-3"	"	6'-2"	6'-7"	35'-3"

MARK	SIZE	X	Y	TOT. LGTH.
AM1	#4	2'-11"	3'-3"	6'-3"
AN1	"	3'-3"	3'-3"	6'-9"



MARK	SIZE	U	T	TOT. LGTH.
AU33	1/2"ø	3'-0"	3'-0"	9'-3"
AU34	"	"	2'-6"	8'-6"
AU37	"	1'-7"	3'-0"	8'-6"
AU40	"	"	2'-6"	7'-6"
AU41	"	3'-0"	1'-7"	7'-3"
AU42	"	0'-10"	2'-1"	5'-9"
AU43	"	1'-7"	1'-7"	5'-6"
AU44	"	0'-10"	1'-7"	5'-3"
AU45	"	"	2'-1"	5'-9"
U9	"	0'-7"	2'-5"	6'-0"
U10	"	"	2'-0"	5'-0"

MARK	SIZE	X	Y	TOT. LGTH.
AM1	1/2"ø	2'-11"	3'-3"	6'-3"
AN1	"	3'-3"	3'-3"	6'-9"

·BILL·OF·SPLICE·BARS·

(SPLICE WITH REG. BARS CUT FOR TESTS)

Pier or Abut.	Mark	No. Pcs.	Size	Length	Location	Total Length	Weight
	REINFORCING STEEL						
1	C2	10	1"ø	13'-4"	Splice with 1"ø Bars cut for Tests	130'-0"	351 #
1	A2	12	3/4"ø	10'-6"	" " " " " "	126'-0"	338 #
1	U6	3	1/2"ø	8'-0"	" " 3/4"ø " " " "	72'-0"	62 #
1	W2	3	3/8"ø	7'-0"	" " 3/4"ø " " " "	21'-0"	8 #
					TOTAL STEEL		@ 850 #

* Weights for 1930 Stds.
1-3'-0" Sample required for each 16000 Lbs. of 1"ø Bars.
3-3'-0" Samples required for each 16000 Lbs. of 3/4"ø, 1/2"ø and 3/8"ø Bars.

NOTE:- ALL DIMENSIONS SHOWN ON BENDING DIAGRAMS FOR REINFORCING BARS ARE MEASURED ON CENTER LINE OF BAR.

·COMPLETE·SUMMARY·

UNIT	CONCRETE			REINFORCING STEEL				
	CLASS 'D'	CLASS 'E'	CLASS 'F'	1"ø	3/4"ø	5/8"ø	3/8"ø	TOTAL
	Cu. Yds.	Cu. Yds.	Cu. Yds.	Lin. Ft.	Lbs.	Lbs.	Lbs.	Lbs.
ABUT. NO. 1	4.5	261.5	4.0	34.33	7295	3279	5469	626	16569
SPAN "A"	285.5	—	11.8	145.5	22411	10690	6207	2900	42206
PIER NO. 2	18.5	193.0	1.0	7.0	13298	4825	1263	27	19413
SPAN "B"	273.5	—	11.8	145.5	21866	10204	5711	2900	40681
PIER NO. 3	18.5	193.0	1.0	7.0	13298	4825	1263	27	19413
SPAN "C"	273.5	—	11.8	145.5	21866	10204	5711	2900	40681
PIER NO. 4	18.5	193.0	1.0	7.0	13298	4825	1263	27	19413
SPAN "D"	285.5	—	11.8	145.5	22411	10690	6207	2900	42208
ABUT. NO. 5	6.0	308.5	5.0	46.92	8303	4726	6850	971	20850
SPLICE BARS	—	@ 2.5	—	—	35	230	60	5	610
TOTALS	1184.0	1151.5	49.2	686.25	144397	64507	40206	13285	282726

@ Headwalls for Ford Pipes (See Drwg. C-a)
* Headwall for 15" Pipe (See Drwg. C).

MISCELLANEOUS

12 C.I. Gutter Drains (See Drwgs. C2, C4, and Special Provisions)
6"ø C.I. Drain Pipe = 12.0 Lin. Ft. (Spans "A", "B", and "D")
4"ø C.I. Downs at Steel Tubing approx. 3/16" thick = 28.5 Lin. Ft. (Spans "A", "B", and "D")
2"ø W.I. Pipe = 67.0 Lin. Ft. (Abuts. #1 & #5 - Piers #2, #3 & #4)
Filling Material (To be fur. Abuts. 1668 Cu. Yds. +10% = 1835 Cu. Yds.
30 Lin. Ft. 8" Corrugated Metal Pipe (See Drwg. C-a)
1900 Lbs. C.I. Pipe and Fittings (See Drwg. C-a)
180 Lin. Ft. 15" V.C. Tile Pipe (See Drwg. C)
2 - Std. Catch Basins (Curb & Gutter Type) (See Sheet 18)

NOTE:- For itemized list of Rivet Gage Wall materials see Bill of Materials for Abut. #1 - Drwg. C-a.

BENDING DIAGRAMS, SPLICE BARS & SUMMARY
REINFORCED CONCRETE BRIDGE
4 SPANS @ 72' SKEW 15°L. 5'WALKS 44'-0"ROADWAY
OVER WABASH RIVER. ON STATE ROAD: I-O

INDIANA STATE HIGHWAY COMMISSION
WELLS COUNTY

SCALE:-
DECEMBER 15, 1933
RECOMMENDED FOR APPROVAL:-
Ed Handy
Chief Engineer
PROJECT:- I
SECTION:- O
STATION:- 66+10
DRAWING:- C 11 OF 13
STRUCTURE NO. 230
BRIDGE CONTRACT NO. 692

I-O-230

~ BILLS OF MATERIALS ~

BRIDGES OVER 20' SPAN

FED. AID PROJ. NO.	STATE	FED. NO.	STATE PROJ. NO.	SHEET NO.	TOTAL SHTS.
7	IND.	1	1934	15	20

SECTION-O

SPAN "A" - SPAN "D" SAME

REINFORCING STEEL

Bent Straight	Mark	No. Pcs.	Size	Length	Location	Tot. Length	Weight
					Footing "A"		
	Cs	6	1"	34'-9"	Horiz. & Vert. Sp. Wls.	208'-6"	
	Cs	6	"	34'-9"	" " " "	208'-6"	
		100	"	34'-9"	Sp. Wls.	3475'-0"	
		100	"	34'-9"	Arch Ring Extrados	3475'-0"	
		150	"	28'-0"	" " Intrados	4200'-0"	
					Total 1" Bars	8300'-0"	27811#
	Gt	24	1¼"	38'-9"	Longt. Gutters (2 Top-12 Bott.)	930'-0"	
	Wt	100	"	6'-6"	Vert. Sp. Wls. (F.P.) & Transv. Gutters	650'-0"	
		3	"	14'-0"	Sloping Face of Cstfs.	42'-0"	
		3	"	21'-0"	" " " "	63'-0"	
		3	"	20'-0"	" " " "	60'-0"	
		3	"	18'-5"	" " " "	55'-3"	
		3	"	17'-0"	" " " "	51'-0"	
		3	"	16'-0"	" " " "	48'-0"	
		3	"	16'-3"	" " " "	48'-9"	
		3	"	15'-6"	" " " "	47'-6"	
		3	"	15'-3"	" " " "	45'-9"	
		3	"	15'-0"	" " " "	45'-0"	
		3	"	14'-9"	" " " "	44'-3"	
		12	"	14'-6"	" " " "	174'-0"	
		170	"	28'-9"	Transv. Ar. Ring	4547'-6"	
					Total ¾" Bars	7032'-0"	10690#
	Hs	8	½"	38'-0"	Horiz. Sp. Wls. (4F.P.-4R.F.)	304'-0"	
	Hs	8	"	23'-0"	" " " (2 " - 2 ")	92'-0"	
	Hs	8	"	18'-6"	" " " (4 " - 4 ")	148'-0"	
	Hs	8	"	13'-6"	" " " (4 " - 4 ")	108'-0"	
	Hs	4	"	11'-0"	" " " (2 " - 2 ")	44'-0"	
	Hs	4	"	9'-0"	" " " (2 " - 2 ")	72'-0"	
	Hs	8	"	6'-6"	" " " (4 " - 4 ")	52'-0"	
	Hs	8	"	17'-0"	Vert. " " (F.P.)	136'-0"	
		2	"	15'-3"	" " " "	30'-6"	
		2	"	12'-3"	" " " "	24'-6"	
		2	"	11'-3"	" " " "	22'-6"	
		2	"	10'-3"	" " " "	20'-6"	
		2	"	9'-3"	" " " "	18'-6"	
		2	"	8'-3"	" " " "	16'-6"	
		2	"	7'-3"	" " " "	14'-6"	
		2	"	6'-3"	" " " "	12'-6"	
		2	"	5'-3"	" " " "	10'-6"	
		2	"	4'-3"	" " " "	8'-6"	
		2	"	3'-3"	" " " "	6'-6"	
		117	"	6'-0"	Counterforts (F.P.)	702'-0"	
		4	"	19'-6"	" " " (")	78'-0"	
		4	"	18'-0"	" " " (")	72'-0"	
		4	"	16'-6"	" " " (")	66'-0"	
		4	"	15'-0"	" " " (")	60'-0"	
		4	"	13'-6"	" " " (")	54'-0"	
		4	"	12'-0"	" " " (")	48'-0"	
		4	"	10'-6"	" " " (")	42'-0"	
		4	"	9'-0"	" " " (")	36'-0"	
		4	"	7'-6"	" " " (")	30'-0"	
		4	"	6'-0"	" " " (")	24'-0"	
		136	"	5'-6"	Handrails (78 F.P.-58 R.F.)	748'-0"	
		4	"	5'-6"	Counterforts (F.P.)	22'-0"	
		28	"	5'-9"	Stirrups Ar. Ring	161'-0"	
		100	"	3'-3"	" " "	325'-0"	
		100	"	3'-3"	" " "	325'-0"	
		56	"	2'-9"	" " "	154'-0"	
		56	"	2'-9"	" " "	154'-0"	
		84	"	2'-1"	" " "	180'-0"	
		112	"	2'-0"	" " "	224'-0"	
		140	"	2'-0"	" " "	280'-0"	
		580	"	1'-9"	" " "	1015'-0"	
				18'-6"	Horiz. Ties - Cstfs.	18'-6"	
				17'-0"	" " " "	17'-0"	
				15'-6"	" " " "	15'-6"	
				14'-0"	" " " "	14'-0"	
				12'-6"	" " " "	12'-6"	
				11'-0"	" " " "	11'-0"	
				9'-6"	" " " "	9'-6"	
				8'-0"	" " " "	8'-0"	
				6'-6"	" " " "	6'-6"	
				5'-0"	" " " "	5'-0"	
				3'-6"	" " " "	3'-6"	
				2'-0"	" " " "	2'-0"	
				14'-9"	Vert. Ties - Cstfs.	14'-9"	
				13'-3"	" " " "	13'-3"	
				11'-9"	" " " "	11'-9"	
				10'-3"	" " " "	10'-3"	
				8'-9"	" " " "	8'-9"	
				7'-3"	" " " "	7'-3"	
				5'-9"	" " " "	5'-9"	
				4'-3"	" " " "	4'-3"	
				2'-9"	" " " "	2'-9"	
				1'-3"	" " " "	1'-3"	
				6'-0"	Ties (Sea. Bracket)	330'-0"	
				5'-6"	Vert. Ties Regular Brackets	55'-0"	
				5'-3"	" " " "	5'-3"	
				4'-3"	" " Expansion	8'-6"	
				3'-3"	" " "	3'-3"	
				12'-7"	Counterfort Ties	13'-2"	
				10'-6"	" " "	14'-0"	
				8'-6"	" " "	13'-6"	
				6'-6"	" " "	13'-0"	
				7'-0"	" " "	14'-0"	
				6'-0"	" " "	12'-0"	
				5'-0"	" " "	10'-0"	
				4'-0"	" " "	8'-0"	
				3'-0"	" " "	6'-0"	
	Vt	4		15'-3"	Vert. Sp. Wls. (F.P.)	61'-0"	
	Vs	2		13'-3"	" " " (")	26'-6"	
	Vs	2		11'-3"	" " " (")	22'-6"	
	Vs	2		11'-0"	" " " (")	22'-0"	
	Vs	2		10'-3"	" " " (")	20'-6"	
	Vs	2		8'-3"	" " " (")	16'-6"	
	Vs	4		6'-3"	" " " (")	25'-0"	
	Vs	4		5'-3"	" " " (")	21'-0"	
	Vs	6		4'-3"	" " " (")	25'-6"	
	Vs	4		3'-3"	" " " (")	13'-0"	
	Vs	4		2'-3"	" " " (")	9'-0"	
	Vs	6		4'-0"	" " " (")	24'-0"	
	Vs	18		5'-3"	" " " (")	95'-6"	
	Vs	18		4'-3"	" " " (")	77'-0"	
	Vs	36		3'-3"	" " " (")	117'-0"	
	Vs	10		2'-3"	" " " (")	22'-6"	
	Vs	18		1'-3"	" " " (")	23'-6"	
				7'-0"	Ties - Brackets (32 Rec.-4 Exp.)	252'-0"	
				6'-0"	" " " (Exp.)	24'-0"	
					Total ¾" Bars	7691'-0"	23008#
	Wt	140	½"	24'-9"	Longt. Sidewalk Slab (72 Top-68 Bott.)	5465'-0"	6207#
	Wa	76	"	23'-9"	" " " (36 " - 34 ")	1665'-0"	
		22	"	24'-3"	Handrails	1067'-0"	
		144	"	23'-3"	"	511'-0"	
				6'-3"	Transv. Sidewalk Slab & Sidewalk Ledge	225'-0"	
					Total ½" Bars	8933'-0"	23008#
					Total Steel		82210#

SPAN "A" - SPAN "D" SAME (Continued)

~ CONCRETE ~

Class "D" -
Section #1 - Arch Ring (48-0" Width) Betw. Spandrel Const. Jts. Incl. Cstfs. Pcgs. = 84.5 Cu. Yds.
" #2 " " " " " " " = 84.0 " "
" #3 - Arch Ring (0-4.36" Width) Betw. Underside Top Const. Jts. - Incl. Gutter, Spandrel Walls, Brackets & Counterforts. = 50.0 " "
" #4 - (Same Items as Section No. 3.) = 49.5 " "
Sidewalks = 31.1 " "
Total Class "D" = 299.1 " "

Class "F" -
Handrail - (11.8 Cu. Yds.) = 145.5 Lin. Ft.

~ MISCELLANEOUS ~

2-6"ø C.I. Drains at 1'-6"
2-C.I. Gutter Drains (See Drwg. Co and Special Provisions.) = 3.0 Lin. Ft.

SPAN "B" - SPAN "C" SAME

Bent Straight	Mark	No. Pcs.	Size	Length	Location	Tot. Length	Weight
					Footing "B"		
	Cs	12	1"	34'-9"	Horiz. & Vert. Sp. Wls.	417'-0"	
		6	"	30'-3"	" " Sp. Wls.	181'-6"	
		100	"	34'-9"	Arch Ring Extr.	3475'-0"	
		150	"	28'-0"	" " Intr.	4200'-0"	
					Total 1" Bars	8093'-6"	27866#
	Gt	24	1¼"	38'-9"	Longt. Gutters (2 Top-12 Bott.)	930'-0"	
	Wt	80	"	6'-6"	Vert. Sp. Wls. (F.P.)	520'-0"	
		6	"	20'-0"	Sloping Face of Counterforts.	120'-0"	
		6	"	18'-9"	" " " "	112'-6"	
		6	"	16'-0"	" " " "	96'-0"	
		6	"	15'-6"	" " " "	93'-0"	
		6	"	14'-9"	" " " "	88'-6"	
		12	"	14'-6"	" " " "	174'-0"	
		164	"	28'-9"	Transv. Ar. Ring	4714'-0"	
					Total ¾" Bars	6713'-0"	10306#
	Hs	8	½"	38'-0"	Horiz. Sp. Wls (4F.P.-4R.F.)	304'-0"	
	Hs	8	"	18'-6"	" " " (4 " - 4 ")	148'-0"	
	Hs	8	"	13'-6"	" " " (4 " - 4 ")	108'-0"	
	Hs	4	"	9'-0"	" " " (2 " - 2 ")	72'-0"	
	Hs	4	"	11'-0"	" " " (2 " - 2 ")	44'-0"	
	Hs	8	"	6'-6"	" " " (4 " - 4 ")	52'-0"	
	Hs	8	"	17'-0"	Vert. " " (F.P.)	136'-0"	
		2	"	15'-3"	" " " "	30'-6"	
		2	"	12'-3"	" " " "	24'-6"	
		2	"	11'-3"	" " " "	22'-6"	
		2	"	10'-3"	" " " "	20'-6"	
		2	"	9'-3"	" " " "	18'-6"	
		2	"	8'-3"	" " " "	16'-6"	
		2	"	7'-3"	" " " "	14'-6"	
		2	"	6'-3"	" " " "	12'-6"	
		2	"	5'-3"	" " " "	10'-6"	
		2	"	4'-3"	" " " "	8'-6"	
		2	"	3'-3"	" " " "	6'-6"	
		117	"	6'-0"	Counterforts (F.P.)	702'-0"	
		4	"	19'-6"	" " " (")	78'-0"	
		4	"	18'-0"	" " " (")	72'-0"	
		4	"	16'-6"	" " " (")	66'-0"	
		4	"	15'-0"	" " " (")	60'-0"	
		4	"	13'-6"	" " " (")	54'-0"	
		4	"	12'-0"	" " " (")	48'-0"	
		4	"	10'-6"	" " " (")	42'-0"	
		4	"	9'-0"	" " " (")	36'-0"	
		4	"	7'-6"	" " " (")	30'-0"	
		4	"	6'-0"	" " " (")	24'-0"	
		136	"	5'-6"	Handrails (78 F.P.-58 R.F.)	748'-0"	
		4	"	5'-6"	Counterforts (F.P.)	22'-0"	
		28	"	5'-9"	Stirrups Arch Ring	161'-0"	
		100	"	3'-3"	" " "	325'-0"	
		100	"	3'-3"	" " "	325'-0"	
		56	"	2'-9"	" " "	154'-0"	
		56	"	2'-9"	" " "	154'-0"	
		84	"	2'-1"	" " "	180'-0"	
		112	"	2'-0"	" " "	224'-0"	
		140	"	2'-0"	" " "	280'-0"	
		580	"	1'-9"	" " "	1015'-0"	
				18'-6"	Horiz. Ties - Cstfs.	18'-6"	
				17'-0"	" " " "	17'-0"	
				15'-6"	" " " "	15'-6"	
				14'-0"	" " " "	14'-0"	
				12'-6"	" " " "	12'-6"	
				11'-0"	" " " "	11'-0"	
				9'-6"	" " " "	9'-6"	
				8'-0"	" " " "	8'-0"	
				6'-6"	" " " "	6'-6"	
				5'-0"	" " " "	5'-0"	
				3'-6"	" " " "	3'-6"	
				2'-0"	" " " "	2'-0"	
				14'-9"	Vert. Ties - Cstfs.	14'-9"	
				13'-3"	" " " "	13'-3"	
				11'-9"	" " " "	11'-9"	
				10'-3"	" " " "	10'-3"	
				8'-9"	" " " "	8'-9"	
				7'-3"	" " " "	7'-3"	
				5'-9"	" " " "	5'-9"	
				4'-3"	" " " "	4'-3"	
				2'-9"	" " " "	2'-9"	
				1'-3"	" " " "	1'-3"	
				6'-0"	Ties (Sea. Bracket)	330'-0"	
				5'-6"	Vert. Ties Regular Brackets	55'-0"	
				5'-3"	" " " "	5'-3"	
				4'-3"	" " Expansion	8'-6"	
				3'-3"	" " "	3'-3"	
				12'-7"	Counterfort Ties	13'-2"	
				10'-6"	" " "	14'-0"	
				8'-6"	" " "	13'-6"	
				6'-6"	" " "	13'-0"	
				7'-0"	" " "	14'-0"	
				6'-0"	" " "	12'-0"	
				5'-0"	" " "	10'-0"	
				4'-0"	" " "	8'-0"	
				3'-0"	" " "	6'-0"	
	Vs	2		11'-3"	Sp. Wls. (F.P.)	22'-6"	
	Vs	2		11'-0"	" " " (")	22'-0"	
	Vs	2		9'-0"	" " " (")	18'-0"	
	Vs	2		7'-0"	" " " (")	14'-0"	
	Vs	2		5'-0"	" " " (")	10'-0"	
	Vs	2		3'-0"	" " " (")	6'-0"	
	Vt	10		7'-0"	Ties - Brackets (92 Rec.-4 Exp.)	682'-0"	
					(Exp.) each	24'-0"	
					Total ¾" Bars	6464'-0"	21116#
	Wt	140	½"	24'-9"	Longt. Sidewalk Slab (72 Top-68 Bott.)	5465'-0"	
	Wa	76	"	23'-9"	" " " (36 " - 34 ")	1662'-6"	
		44	"	24'-3"	Handrails	1066'-6"	
		22	"	23'-3"	"	511'-0"	
		144	"	6'-3"	Transv. Sidewalk Slab & Sidewalk Ledge	225'-0"	
					Total ½" Bars	8930'-0"	23008#
					Total Steel		45124#

~ CONCRETE ~

Class "D" -
Section #1 - Arch Ring (48-0" Width) Betw. Spandrel Const. Jts. Incl. Cstfs. Pcgs. = 84.5 Cu. Yds.
" #2 " " " " " " " = 84.0 " "
" #3 - Arch Ring (4'-4⅜" Width) Betw. Underside Top Const. Jts. Incl. Gutter, Spandrel Walls, Brackets & Counterforts. = 46.5 " "
" #4 - (Same Items as Section No. 3.) = 46.5 " "
Sidewalks = 31.1 " "
Total Class "D" = 212.6 " "

Class "F" -
Handrail - (11.8 Cu. Yds.) = 146.8 Lin. Ft.

~ MISCELLANEOUS ~

2-6"ø C.I. Drains at 1'-6"
2-C.I. Gutter Drains (See Drwg. Co and Spec. Prov.) = 3.0 Lin. Ft.

ABUTMENT No. 1

REINFORCING STEEL

Bent Straight	Mark	No. Pcs.	Size	Length	Location	Tot. Length	Weight
					Footing "A"		
	ADs	4	1"	35'-7"	Sloping Face - Cstfs. 1-2-1 & 1-2-W.	142'-0"	
		50	"	28'-0"	Arch Ring Extrados & Vert. Abut.	1400'-0"	
		6	"	29'-0"	Sloping Face - Cstfs. 1-2-1 & 1-2-W.	174'-0"	
		50	"	18'-5"	Arch Ring Intrados & Vert. Abut.	925'-0"	
		6	"	29'-5"	Sloping Face - Cstfs. 1-2-E & 1-2-P.	59'-0"	
					Total 1" Bars	2702'-0"	7295#
	ADc	18	½"	18'-3"	Horiz. 12 Gutter & ~ Up Wing walls	328'-6"	
	AAs	4	"	11'-0"	Longt. West Wing Pcg. under Cstfs	44'-0"	
	AAs	4	"	2'-3"	" " " "	9'-0"	
	AAs	34	"	5'-7"	Transv. Abut. Pcg. Incl. Gutter & Vert. Top Wing wall	189'-6"	
	AAt	42	"	6'-9"	" " " "	285'-6"	
	AAt	2	"	36'-0"	Sloping Face Cstfs. 1-2-E & 1-2-W	144'-0"	
	AAs	3	"	33'-5"	" " " "	100'-3"	
	AAs	3	"	26'-9"	" " " "	80'-3"	
	AAs	12	"	16'-6"	Cont. Brkts. - Rear Pls.	198'-0"	
	AAs	4	"	13'-6"	" " - Front "	54'-0"	
	AAs	4	"	12'-6"	" " - Next to Frame Pls.	75'-0"	
	AAt	6	"	36'-0"	Sloping Face Cstfs. A-1-W	216'-0"	
	AAt	6	"	36'-6"	Transv. Abut.	219'-0"	
		6	"	4'-0"	Anchor Bars for Ladder Straps	24'-0"	
					Total ¾" Bars	2187'-9"	5276#
	ADs	23	¾"	17'-9"	Horiz. Wing walls	408'-3"	
	ADd	60	"	13'-0"	" " "	780'-0"	
	ALs	10	"	25'-9"	Vert. Pcs. Through top Transv. Gutter	257'-6"	
	ALs	4	"	10'-6"	Mid. Bent Pl. & Transv. Gutter	42'-0"	
	ALs	2	"	8'-3"	" " " "	16'-6"	
	ADs	2	"	21'-6"	Horiz. - Cstfs.	43'-0"	
	ADs	3	"	21'-0"	" "	63'-0"	
	ADs	2	"	20'-0"	" "	40'-0"	
	AUs	4	"	19'-5"	" "	77'-8"	
	AUs	4	"	18'-6"	" "	74'-0"	
	AUs	4	"	17'-6"	" "	70'-0"	
	AUs	2	"	16'-9"	" "	33'-6"	
	AUs	2	"	16'-0"	Brkts. - Rear Pls.	32'-0"	
	AUs	2	"	16'-0"	Horiz. - Cstfs.	32'-0"	
	AUs	2	"	15'-0"	" "	30'-0"	
	AUs	2	"	14'-0"	" "	28'-0"	
	AUs	2	"	13'-0"	" "	26'-0"	
	AUs	2	"	12'-0"	" "	24'-0"	
	AUs	2	"	11'-0"	" "	22'-0"	
	AUs	2	"	10'-0"	" "	20'-0"	
	AUs	10	"	10'-6"	Vert. End Curtain Walls	105'-0"	
	AUs	2	"	9'-9"	Horiz. - Cstfs.	19'-6"	
	AUs	2	"	9'-3"	" Brkts. - Front Pls.	18'-6"	
	AUs	2	"	8'-9"	Vert. " - Rear "	17'-6"	
	AUs	24	"	8'-3"	" " - " "	198'-0"	
	AUs	2	"	7'-9"	Sloping " " "	15'-6"	
	AUs	2	"	7'-3"	Horiz. - Rear Pls.	14'-6"	
	AUs	2	"	6'-9"	Vert. - Brkts. Front Pls.	13'-6"	
	AUs	2	"	6'-3"	" " Rear "	12'-6"	
	AUs	24	"	6'-3"	Sloping " " "	150'-0"	
	AUs	2	"	5'-9"	" Brkts. - East Wall only	11'-6"	
	AUs	2	"	5'-3"	Vert. " " " "	10'-6"	
	AUs	2	"	5'-0"	" " Front Pl.	10'-0"	
	AUs	2	"	4'-9"	" " East Wall only	9'-6"	
	AVs	22	"	25'-9"	Horiz. - Cstfs. - Sloping Brkts. - Stirrups - Diagonals - Expansion	570'-0"	
	AWs	8	"	6'-9"	4 " - 45° Rear Pl.	54'-0"	
	AWs	4	"	6'-3"	2 " " " "	25'-0"	
	AVs	4	"	6'-3"	Rear Pl. Brkts.	25'-0"	
	AVs	4	"	7'-0"	Front " "	28'-0"	
	AVs	2	"	7'-0"	Bracket (East Wing Only)	14'-0"	
	Er	48	"	5'-3"	Vert. - Handrail and Post	252'-0"	
	Er	4	"	5'-3"	Horiz. - Cap Wall Sidewalk (R.F.)	21'-0"	
	Xs	16	"	11'-0"	" " " (")	176'-0"	
	Xs	4	"	9'-3"	" " " (R.E.)	37'-0"	
	Xd	16	"	6'-3"	" " " (")	100'-0"	
	Xf	80	"	5'-3"	" Endwalls	420'-0"	
	Xs	5	"	20'-3"	Vert. " (as Endwall (Ea. Face))	62'-5"	
	Za	4	"	16'-3"	" Stirrups - Cap Wall (as End Wall - Cstfs.	65'-0"	
		4	"	24'-0"	" " " "	96'-0"	
		1	"	23'-3"	Cstfs. only - 5 Cap Wall Sidewalk	23'-3"	
		6	"	18'-0"	" " - 4 Wingwalls - 2 Cstfs.	108'-0"	
		17	"	16'-0"	" " - Cap Wall Sidewalk, Wing Abut Pcs.	272'-0"	
		4	"	14'-6"	" " - Cstfs. A-1-W	58'-0"	
		2	"	13'-6"	" " - "	27'-0"	
		4	"	11'-6"	" " - Cstfs. A-1-W	46'-0"	
		2	"	11'-0"	" " - "	22'-0"	
		18	"	10'-0"	" " - Cstfs. (Cap Wall Sidewalk, Wing Abut Pcs.	180'-0"	
		12	"	9'-0"	Transv. West Wing Pcs.	108'-0"	
		2	"	8'-6"	Vert. - Cstfs. Abut. W.	17'-0"	
		2	"	8'-0"	" - Cstfs. (Cap Wall Sidewalk	16'-0"	
		74	"	6'-0"	Horiz. - End Curtain Walls	740'-0"	
		41	"	4'-6"	Vert. - Stub bars in Pcs.	246'-6"	
					Total ¾" Bars	6392'-3"	5669#
	Abs	46	½"	17'-9"	Longt. - 36 Rear Sidewalk, 11 West Slabs	816'-6"	
	Abs	16	"	6'-3"	Transv. - West	98'-0"	
	Abs	16	"	6'-3"	" "	98'-0"	
		34	"	6'-3"	" "	212'-6"	
		2	"	7'-3"	Horiz. - Handrail	14'-6"	
		28	"	6'-3"	Transv. - East Sidewalk	175'-0"	
		33	"	6'-3"	Horiz. - H.R. Posts (Rear)	206'-3"	
		3	"	5'-11"	" " (Front)	17'-9"	
					Total ½" Bars	1567'-9"	424#
					Total Steel		13488#

~ CONCRETE ~

Class "E" - Footing
" "E" - Above Frg. - Base of Constr. Jt. About up to Skew back (Quarter Circle) up to Underside Top) Wing Walls, Diaphragm walls, & Cstfs. up to Const. Jt. = 88.5 Cu. Yds.
" "E" - Above Frg. - Base of Constr. Jt. About up to Skew back (Quarter Circle) up to Underside Top) Wing Walls and Counterforts up to Constr. Jt. = 83.0 " "
" "E" - Skew Wing Wall, Cstfs. Gutter, & Brkts. above Constr. Jt. = 34.0 " "
" "E" - Wing Wall, Cstfs. Gutter, & Brkts. above Const. Jt. = 56.0 " "
Total Class "E" = 261.5 " "

Class "D" - Sidewalk Slabs = 4.5 " "
" " - Handrail (4.0 Cu. Yds.) = 34.5 Lin. Ft.

ABUTMENT No. 1 (Continued)

~ MISCELLANEOUS ~

6-6"ø C.I. Drains or 6" Steel Tubing approx. ⅜" thick (6'10"-6'8 1/2") = 40.0 Lin. Ft.
2-2"ø W.I. Pipes x 3'-9" = 6.8 " "

MATERIALS FOR RIVER GAGE WELL (See Special Provisions)

Lumber: 2"x6" (1"x6" Long	Steps in Well - Concrete	18'-0"
2"x4" " " "	Additional Steps Required	1'-0"
Metal Work: - Gate, Ladder & Connections	Inside Well	26'-0"
2 3½"x 3½" Exp. Bolts	For attaching 2"x6" Cypress to Well	2
Rubber Gasket With Primer, Cement Both Sides & Cover - West Sidewalk		1
Total Metal Work		26'-0"
Pipe & Fittings: - 6" Galv. Iron Pipe (2 @ 13'-3")	Air Shafts in well	26'
6"ø Galv. Iron Pipe (90° Ells) Cap, Tee & Nipples - Intakes pipe to well		44'
2" Std. Pipe to Blow Brass Gate Valve	- Well intake pipe	18'
	Total Pipe & Fittings	110'

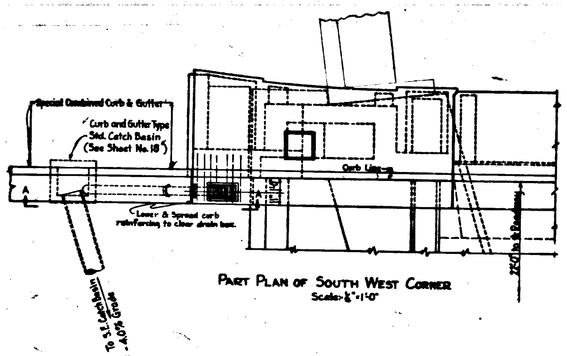
ABUTMENT No. 5

REINFORCING STEEL

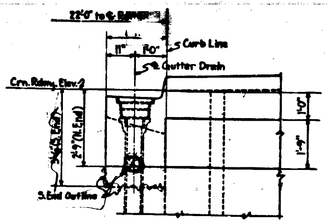
Bent Straight	Mark	No. Pcs.	Size	Length	Location	Tot. Length	Weight
					Footing "A"		
		80	1"	28'-0"	Arch Ring Extrados & Vert. Abut.	1400'-0"	
		50	"	18'-0"	" " Intrados " " "	925'-0"	
	ADs	6	"	29'-0"	Sloping Face - Cstfs.	174'-0"	
		18	"	5'-3"	" " "	95'-0"	
					Total 1" Bars	3494'-0"	8505#
	AGs	18	¾"	25'-9"	Horiz. 12 Gutter - & Top Wall	463'-6"	
	AAs	14	"	11'-0"	Frg. under Cstfs.	154'-0"	
	AAs	4	"	2'-3"	" " "	9'-0"	
	AAs	56	"	5'-3"	Transv. Ws. Frg. Toes.	294'-0"	
	AAs	34	"	6'-9"	" Abut. Frg. Heel	229'-6"	
	AAt	60	"	6'-7"	" Gutter & Vert. Wg. Wall	405'-0"	
	AAs	8	"	35'-6"	Sloping Face Cstfs.	284'-0"	
	AAs	3	"	33'-5"	" " "	99'-9"	
	AAs	3	"	26'-9"	" " "	80'-3"	
	AAs	16	"	12'-6"	Cantil. Brackets (Pilasters)	204'-0"	
	AAt	6	"	36'-0"	" " "	216'-0"	
		5	"	30'-6"	Sloping Fs. - Cstfs. D-10-E.	91'-6"	
		20	"	26'-0"	Transv. Abut.	520'-0"	
					Total ¾" Bars	3109'-5"	4726#
	AHs	16	½"	25'-3"	Horiz. Wings	404'-0"	
	AHs	60	"	20'-6"	" "	1230'-0"	
	AHs	6	"	14'-3"	" "	85'-6"	
	AUs	4	"	25'-9"	Vert. F.F. Was. & Transv. Gutter	309'-6"	
	AUs	4	"	11'-0"	" Mid. Bent Pl. & Transv. Gutter	44'-0"	
	AUs	2	"	8'-3"	" " " "	16'-6"	
	AUs	4	"	27'-6"	Horiz. Cstfs.	110'-0"	
	AUs	4	"	21'-0"	" Cstfs.	84'-0"	
	AUs	4	"	20'-0"	" "	80'-0"	
	AUs	4	"	19'-3"	" "	77'-0"	
	AUs	4	"	18'-6"	" "	74'-0"	
	AUs	8	"	17'-6"	" "	140'-0"	
	AUs	2	"	16'-9"	" "	33'-6"	
	AUs	2	"	16'-0"	" "	32'-0"	
	AUs	2	"	15'-0"	" "	30'-0"	
	AUs	2	"	14'-0"	" "	28'-0"	
	AUs	2	"	13'-0"	" "	26'-0"	
	AUs	2	"	12'-0"	" "	24'-0"	
	AUs	2	"	11'-0"	" "	22'-0"	
	AUs	2	"	10'-0"	" "	20'-0"	
	AUs	2	"	9'-0"	" "	18'-0"	
	AUs	2	"	8'-0"	" "	16'-0"	
	AUs	2	"	7'-0"	" "	14'-0"	
	AUs	2	"	6'-0"	" "	12'-0"	
	AUs	2	"	5'-0"	" "	10'-0"	
	AUs	2	"	4'-0"	" "	8'-0"	
	AUs	6	"	3'-0"	" "	18'-0"	
	AUs	4	"	16'-0"	Vert. - End Curtain Walls	64'-0"	
	AUs	6	"	14'-9"	" " " "	88'-6"	
	AUs	2	"	9'-3"	Horiz. - Cstfs.	18'-6"	
	AUs	2	"	8'-9"	" Brackets (Front Pil.)	17'-6"	
	AUs	2	"	8'-3"	Vert. " " "	16'-6"	
	AUs	2	"	7'-9"	" " (Rear ")	15'-6"	
	AUs	2	"	7'-3"	Sloping " " "	14'-6"	
	AUs	2	"	6'-9"	Horiz. - Cstfs.	13'-6"	
	AUs	2	"	6'-3"	" Brackets (Rear Pil.)	12'-6"	
	AUs	2	"	5'-9"	Vert. " " "	11'-6"	
	AUs	2	"	5'-3"	" " (Front ")	10'-6"	
	AUs	2	"	4'-9"	Sloping " " "	9'-6"	
	AUs	6	"	7'-6"	Horiz. - Cstfs.	45'-0"	
	AUs	4	"	6'-6"	" Brackets (Front Pil.)	26'-0"	
	AUs	4	"	6'-0"	Vert. " " "	24'-0"	
	AUs	4	"	5'-6"	" " (Rear ")	22'-0"	
	AUs	2	"	5'-0"	Sloping " " "	10'-0"	
	AUs	6	"	7'-3"	Horiz. - Cstfs.	43'-6"	
	AUs	4	"	7'-0"	Vert. Bracket (Front Pil.)	28'-0"	
	AUs	4	"	7'-0"	" " (Rear ")	28'-0"	
	AUs	24	"	6'-3"	Horiz. - Front Pil.	150'-0"	
	AUs

BRIDGES OVER 20' SPAN					
DIST. PROJECT NO.	STATE	PROJ. NO.	YEAR	SHEET NO.	SHEETS IN SECTION
7	IND.	1	1934	16	20

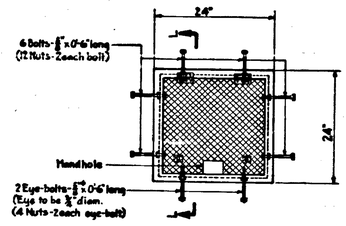
SECTION-"D"



PART PLAN OF SOUTH WEST CORNER
Scale:- ¼"=1'-0"

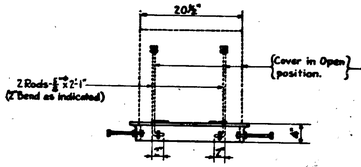


END ELEV. OF DRAIN BLOCKS
Scale:- ½"=1'-0"

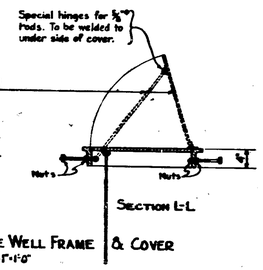


PLAN

NOTE- To be furnished with flush hasps for locking.

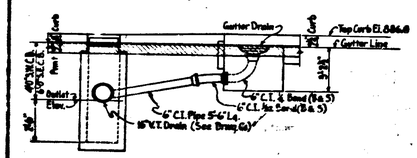


FRONT ELEV.

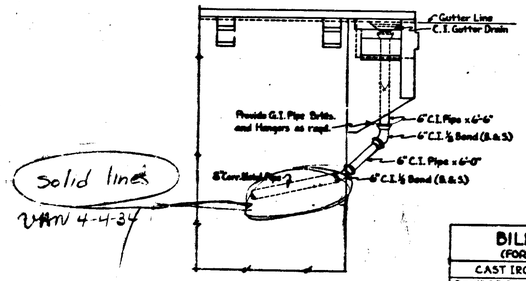


SECTION L-L

DETAIL OF RIVER GAGE WELL FRAME & COVER
Scale:- 1"=1'-0"



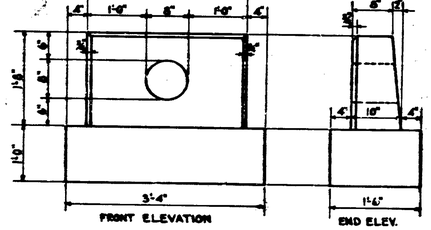
SECTION A-A
SECTION A'A' OPP. HAND



Solid lines
VAN 4-4-34

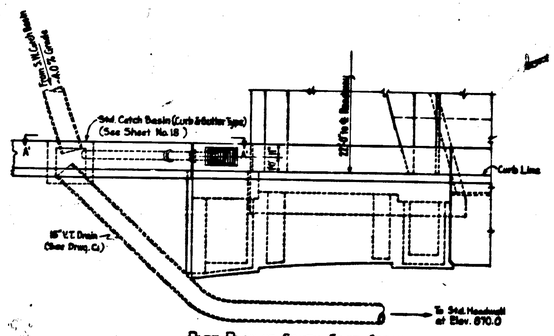
PART ELEVATION B-B
Scale:- ½"=1'-0"

BILL OF MATERIALS (FOR END GUTTER DRAINS)	
CAST IRON PIPE & FITTINGS (A.W.W.A)	
Std. 6" C.I. Pipe Fittings (2 Std. 6" C.I. ⅛ Bends B & S, 4 Std. 6" C.I. ⅛ Bends + B & S, 2 Std. 6" C.I. ⅛ Bends - B & S)	700 lb
Std. 6" C.I. Pipe (2 Pcs. 6'-6" Long, 2 Pcs. 6'-0" Long, 2 Pcs. 5'-6" Long)	1200 lb
Total Weight	1900 lb
4- C.I. Gutter Drains (See Drwg. C4 & Spec. Provi.)	
50 Lin. Ft. 8" Corr. Metal Pipe.	
2- Spec. Headwalls (0.25 Cu. Yds. Class "E" Conc. Each)	

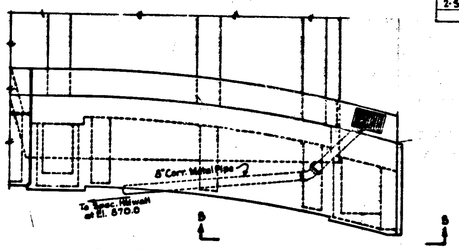


FRONT ELEVATION
END ELEV.

SPECIAL HEADWALL FOR 8" CORR. METAL PIPE
(@ Toe of Slope)
Scale:- 1"=1'-0"



PART PLAN OF SOUTH EAST CORNER
Scale:- ½"=1'-0"



PART PLAN OF NORTH EAST CORNER
NORTH WEST CORNER OPPOSITE HAND
Scale:- ½"=1'-0"

END GUTTER DRAIN, SPECIAL HEADWALL, AND GAGE WELL COVER DETAILS

REINFORCED CONCRETE BRIDGE

4 SPANS @ 72' SKEW 15° L. 5' WALKS 44'-0" ROADWAY
OVER WABASH RIVER ON STATE ROAD:- 1-0

INDIANA STATE HIGHWAY COMMISSION

WELLS COUNTY

SCALE:- AS NOTED DECEMBER 15, 1933

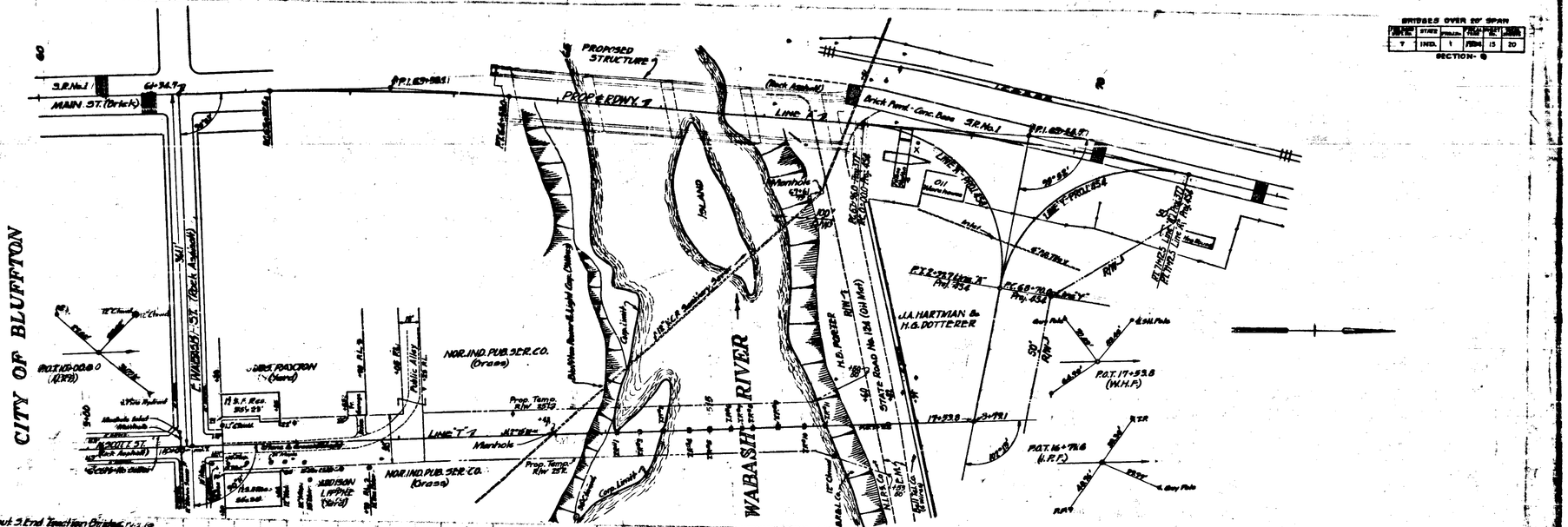
RECOMMENDED FOR APPROVAL:- [signature] ENGINEER OF BRIDGES

PROJECT:- 1 STATION:- 66+10
SECTION:- D

DRAWING:- C9 OF 13 STRUCTURE NO. 230

BRIDGE CONTRACT NO. 692

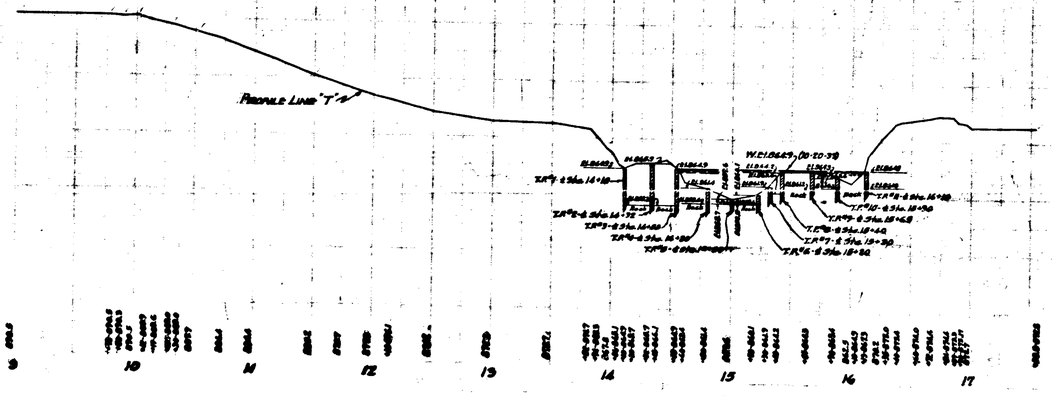
1-0-230



BRIDGES OVER 20' SPAN

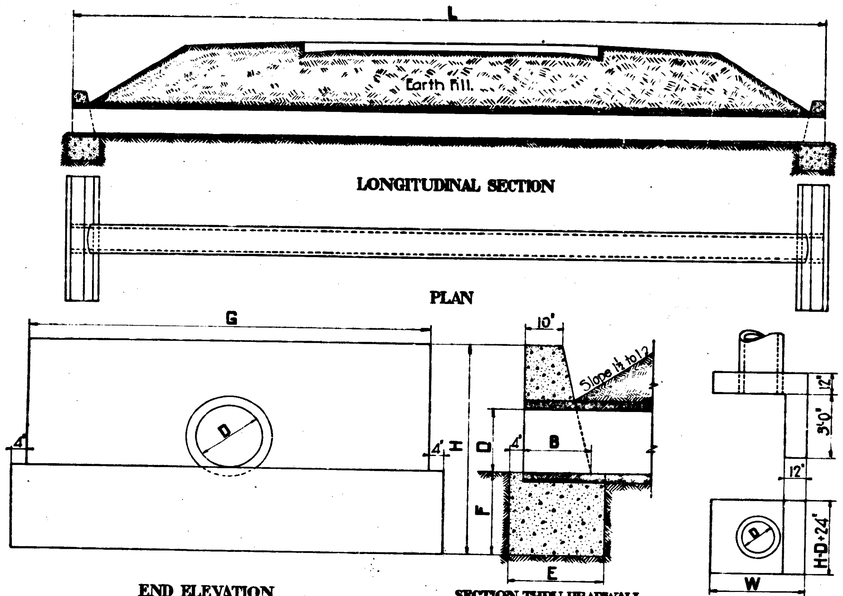
DISTRICT	STATE	ROUTE	SECT	SHEET NO.	SHEETS
7	IND.	1	FED.	15	20

SECTION- S



ROAD PLAN & PROFILE
FOR TEMPORARY BRIDGE
1-0-230
BRIDGE CONTRACT 692

1-0-230



HEADER AT ROAD CROSSINGS

NOTE:- All exposed edges and corners to be chamfered ¾ inch. Pipe not to be used under more than 10 foot fill.

NOTE:- 1. Headwalls to be made parallel to intersecting roads, without footings.

D	F	H	B	E	G	CONCRETE 2 HEADWALLS CU. YDS.
15"	1'-6"	4'-3"	1'-6"	2'-2"	8'-0"	4.0
18"	2'-0"	5'-0"	1'-6"	2'-2"	9'-0"	5.4
24"	2'-0"	5'-8"	1'-8"	2'-4"	10'-0"	7.1
30"	2'-0"	6'-2"	1'-8"	2'-4"	10'-0"	7.6
36"	2'-0"	6'-9"	2'-0"	2'-8"	12'-0"	11.0

D	W	CONCRETE 2 HEADWALLS CU. YDS.
15"	4'-0"	1.76
18"	4'-3"	1.90
24"	4'-9"	2.30
30"	5'-6"	2.63
36"	6'-0"	3.33

PIPE CULVERTS

ESTIMATE OF QUANTITIES FOR CONC.-BIT. CONC.- OR BRICK PVMT.

EXCAVATION CU. YDS.				SPECIAL BORROW CU. YDS.	DRAINAGE TILE & PROTECTION FOR SID. DITCHES LIN. FT.	METAL SHOULDERS LIN. FT.	PAVEMENT SQ. YDS.			CONCRETE CU. YDS.			STEEL		RIP-RAP SQ. YDS.	CONCRETE PAVED SIDE-DITCH LIN. FT.	CATCH BASINS		BITUMINOUS EXPANSION JOINTS PREMOULDED OR POURED LIN. FT.		
CLASS "A"	CLASS "B"	CLASS "C"	PEAT				CON-CRETE	BITUM. OR BRICK BASE	BITUM. OR BRICK SURFACE	HEAD WALL CLASS "A" 2.0	CULVERTS CLASS "A"	CULVERTS CLASS "B"	FOR CULVERTS LBS.	FOR PAVEMENT LBS.			TYPE	GRATES & CURB INLETS WHERE CATCH BASIN IS COVERED BY GRATE	½"	1"	½" PREFORMED
																	2				

INTEGRAL CONCRETE CURBING LIN. FT.	COMBINED CURB & GUTTER LIN. FT.	PAVEMENT REMOVAL SQ. YDS.	STONE OR GRAVEL SURFACE CU. YDS.	MONUMENTS	CONCRETE ℄ MARKERS	GUARD RAIL LIN. FT.	PIPE LIN. FT.										SALVAGED ROAD METAL CU. YDS.	BLASTING LIN. FT. PRELIM.	BLASTING LIN. FT. FINAL	
							15" VITRIFIED													
							100'													

STEEL :- OFFICIAL WEIGHT - ½" = 0.86 LBS. PER LINEAL FOOT. (Deformed)
⅝" - 1.502 LBS. PER LINEAL FOOT. (Smooth)

BRIDGE AND CULVERT DATA

FEDERAL ROAD DISTRICT NO.	STATE	PROJECT	FISCAL YEAR	SHEET NO.	TOTAL SHEETS
7	IND.	1	1934	17	20

SECTION :- O

STRUCTURE NUMBER	LOCATION	DESCRIPTION SIZE	DESCRIPTION KIND	LENGTH "L"	HEIGHT "H"	WINGS "W"	FLOW LINE UP STREAM	FLOW LINE DOWN STREAM	CONCRETE - CU. YDS. HEAD WALLS CLASS "A"	CONCRETE - CU. YDS. CULVERTS CLASS "A"	CONCRETE - CU. YDS. CULVERTS CLASS "B"	STEEL REINFORCING LBS.	REMARKS	PLANS OR SHEET NO.	STRUCTURE NUMBER

1-0-230
BRIDGE CONTRACT #692

NOTE:- WHERE PIPE IS CALLED FOR ON THESE PLANS AND THERE IS NO DESIGNATION OF TYPE, BIDS WILL BE RECEIVED ON ANY OF THE FOLLOWING FOUR TYPES: REINFORCED CONCRETE, TRIPLE STRENGTH VITRIFIED CLAY, CORRUGATED METAL OR CAST IRON AS COVERED BY THE SPECIFICATIONS ON THIS PROJECT. IF THE PROPOSAL INDICATES QUANTITIES OF A PARTICULAR PIPE, BIDS SHALL BE SUBMITTED ON THAT TYPE. NOTE:- FOR ALL VIT. CLAY PIPE WITH LESS THAN 15" DIA. SEE PARAGRAPH 194 STD. 1929 SPECIFICATIONS.

END STR

PROJECT	DESIGNATION
BR-231-2( )	9406690
CONTRACT	BRIDGE FILE
B-25747	1-90-230B

STRUCTURE	TYPE	SPANS	SKEW	OVER	℄ STATION
1-90-230B	REINFORCED CONCRETE ARCH	4 @ 21946	15° LT.	WABASH RIVER	2+014.728
	PREFABRICATED STEEL TRUSS BRIDGE	30275, 46500 & 24325	0°	WABASH RIVER	2+009.525 "C"

INDEX

SHEET NUMBER	SUBJECT
1	TITLE AND INDEX
2-4	TRAFFIC MAINTENANCE DETAILS
5	GENERAL PLAN EXISTING STRUCTURE
6	GENERAL PLAN PROPOSED STRUCTURE
7	GENERAL PLAN
8	GENERAL NOTES AND APPROACH DETAILS
9-16	FLOOR DETAILS
17-19	RAILING DETAILS
20	BRIDGE SUMMARY
21	APPROACH DETAILS
22	SOIL BORINGS
23	LAYOUT
24-25	GENERAL PLAN
26-27	BENTS NO.1 AND NO.4 DETAILS
28-29	PIERS NO.2 AND NO.3 DETAILS
30	BRIDGE SUMMARY
30A	RAILING DETAILS

B-25747

INDIANA DEPARTMENT OF TRANSPORTATION

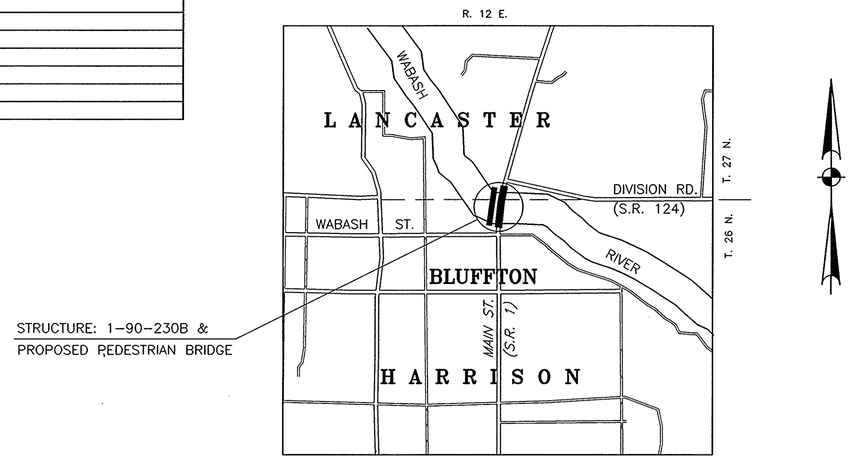
BRIDGE PLANS

FOR SPANS OVER 6.1 METERS
ON STATE ROAD 1

PROJECT NO. BR-231-2( )

DECK RECONSTRUCTION ON STRUCTURE: 1-90-230B, STATE ROAD 1 OVER WABASH RIVER LOCATED IN SECTION 4, T.26 N. R.12 E., LANCASTER TOWNSHIP, WELLS COUNTY, INDIANA AT R.P. 125+11.

PROPOSED PEDESTRIAN STRUCTURE LOCATED 20m WEST OF STATE ROAD 1 OVER WABASH RIVER LOCATED IN SECTION 4, T.26 N. R.12 E., LANCASTER TOWNSHIP, WELLS COUNTY, INDIANA AT R.P. 125+11.



LOCATION IN WELLS COUNTY
Not to Scale

TRAFFIC DATA	
A.A.D.T. (1994)	25300 V.P.D.
A.A.D.T. (2022 PROJECTED)	35700 V.P.D.
D.H.V. (PROJECTED)	V.P.H.
DIRECTIONAL DISTRIBUTION	
TRUCKS	% D.H.V.
	2.5 % A.A.D.T.
DESIGN DATA	
POSTED SPEED	50 K.P.H.
PROJECT DESIGN CRITERIA	3R NON-FREEWAY
FUNCTIONAL CLASSIFICATION	PRINCIPAL ARTERIAL
RURAL/URBAN	URBAN
TERRAIN	LEVEL
ACCESS CONTROL	NONE
E.S.A.L.	7,500,000



PROJECT LOCATION: WELLS COUNTY
FORT WAYNE DISTRICT

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

INDIANA DEPARTMENT OF TRANSPORTATION
STANDARD SPECIFICATIONS DATED 1999
TO BE USED WITH THESE PLANS

REVISIONS	
DATE	SHEET NO.
04-22-04	1, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 25, 26, 27, 28, 29 & 30
	ADDED 15-1 & 18-1
02-10-05	1, DELETED 23 & 24; ADDED 23A & 24A
05-25-05	1, ADDED SHEET 30A

8450 WESTFIELD BLVD., SUITE 300
INDIANAPOLIS, IN. 46240
317 713-4615
FAX 317 713-4616

BF&S
Butler Fairman Seufert
CONSULTING ENGINEERS

509 WEST 84TH DRIVE, SUITE G
MERRILLVILLE, IN. 46410
219 769-2333
FAX 219 769-2377

CERTIFIED BY: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

APPROVED FOR LETTING: 3-3-04
CHIEF DIVISION OF DESIGN DATE

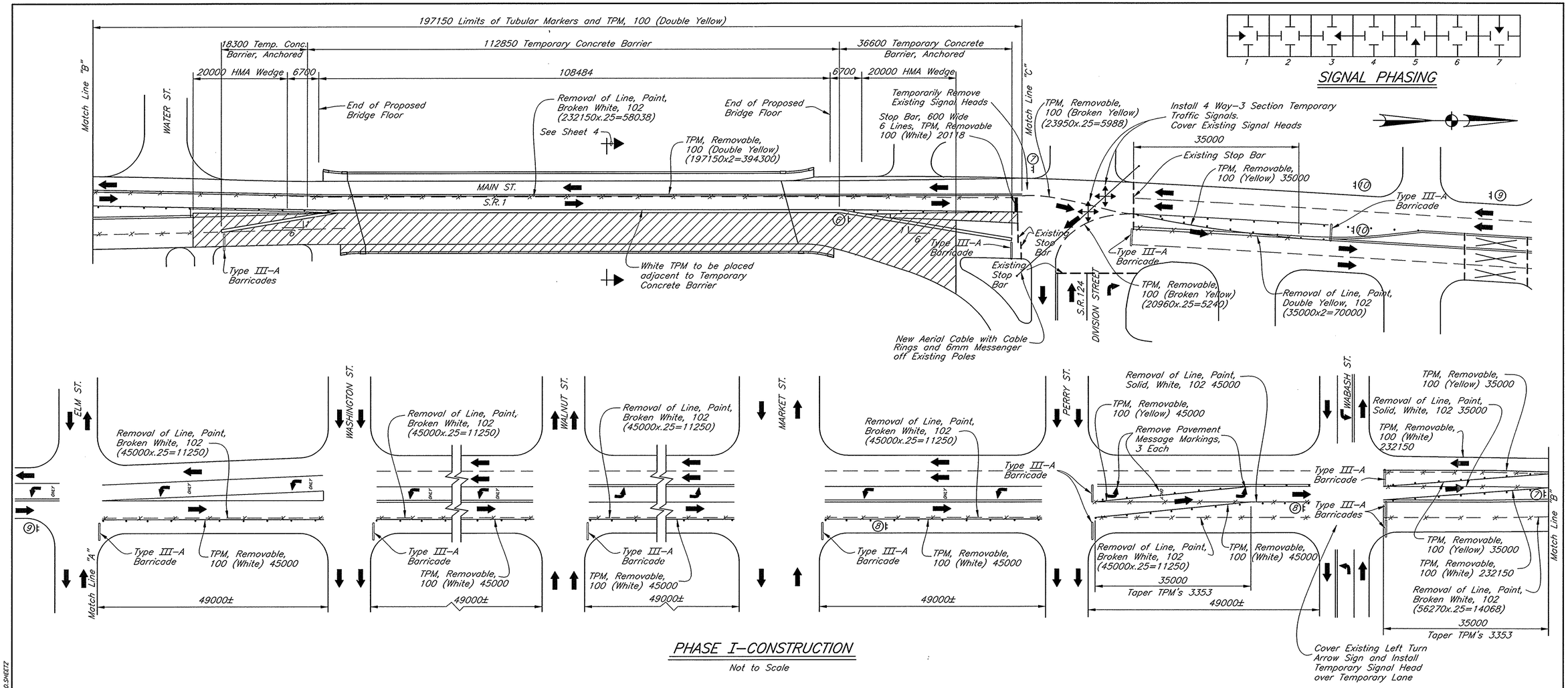
JEFFREY S. NICHOLSON
REGISTERED
No. 19900078
STATE OF INDIANA
PROFESSIONAL ENGINEER

CONTRACT	PROJECT
B-25747	BR-231-2( )

BRIDGE FILE
1-90-230B
DESIGNATION
9406690
SHEET
1 OF 30

BFS NO. 3841.0201

R:\3841.0201\VUTCAD\BRIDGE\0201B101.dwg02-04-04 AT 11:30 D:SHEET1



PHASE I–CONSTRUCTION

Not to Scale

LEGEND

TPM=Temporary Pavement Marking (Type, Color and Width as Noted)

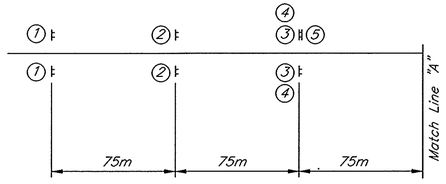
–x–x–x– Indicates Line Removal

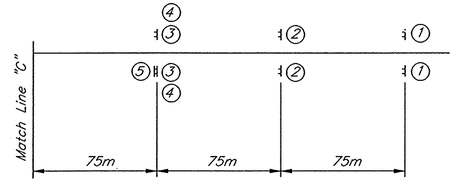
• Indicates Drum with Type "C" Steady Burning Light (Not a Pay Item) (Typ.)

Construction Area

SIGN LEGEND

1. XW20-1 Road Construction Ahead
2. XW20-1 Road Construction 500 FT.
3. XW20-1 Road Construction 250 FT.
4. XW13-1-A Speed Advisory
5. XG20-2 End Construction
6. XW1-4-A(R) Reverse Curve
7. XW6-3-B Two Way Traffic
8. XW1-4-A(L) Reverse Curve
9. XG20-5 Lane Closure Notice Sign
10. R3-7-A(L) Left Lane Must Turn Left





PLAN–CONSTRUCTION SIGNS

Not to Scale

Notes: Signs shall be placed as shown on the Plans or as directed by the Engineer.

All Construction Signs to have Low Intensity Flashing Yellow Light, Type "A".

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

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JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: D. SHEETZ DRAWN: D. SHEETZ
CHECKED: _____ CHECKED: _____

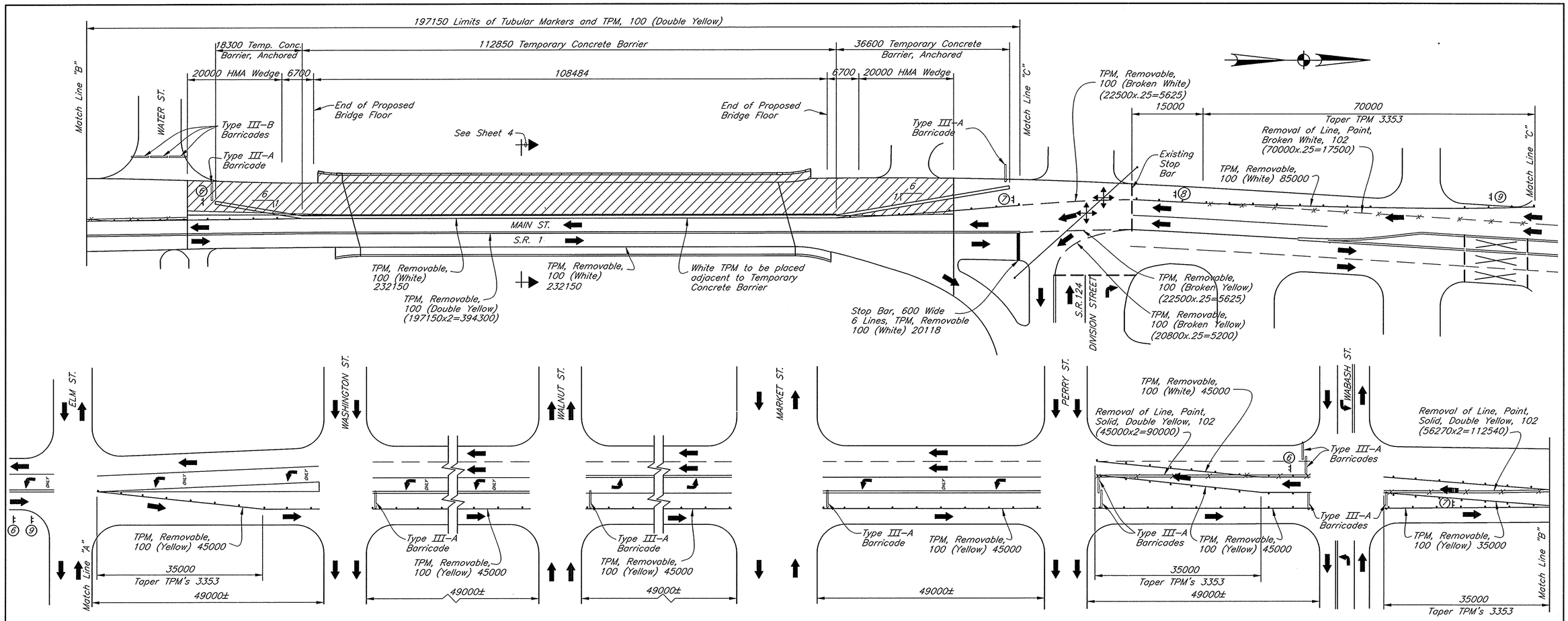
INDIANA DEPARTMENT OF TRANSPORTATION

TRAFFIC MAINTENANCE DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	2 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS NO. 3841.0201

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PHASE II-CONSTRUCTION

Not to Scale

LEGEND

TPM=Temporory Pavement Marking (Type, Color and Width as Noted)

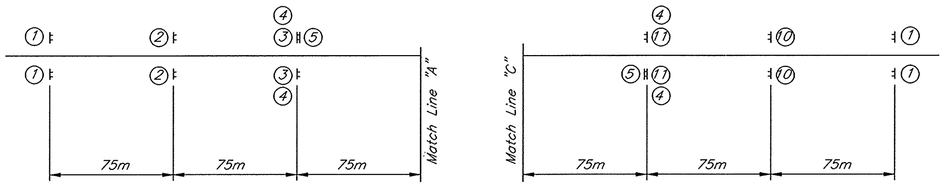
-x-x-x- Indicates Line Removal

• Indicates Drum with Type "C" Steady Burning Light (Not a Pay Item) (Typ.)

Construction Area

SIGN LEGEND

1. XW20-1 Road Construction Ahead
2. XW20-1 Road Construction 500 FT.
3. XW20-1 Road Construction 250 FT.
4. XW13-1-A Speed Advisory
5. XG20-2 End Construction
6. XW1-4-A(R) Reverse Curve
7. XW6-3-B Two Way Traffic
8. XW1-4-A(L) Reverse Curve
9. XG20-5 Lane Closure Notice Sign
10. XW20-5(R) Right Lane Closed Ahead
11. XW4-2-A(R) Merge



PLAN-CONSTRUCTION SIGNS

Not to Scale

Notes: Signs shall be placed as shown on the Plans or as directed by the Engineer.

All Construction Signs to have Low Intensity Flashing Yellow Light, Type "A".

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DESIGN ENGINEER DATE

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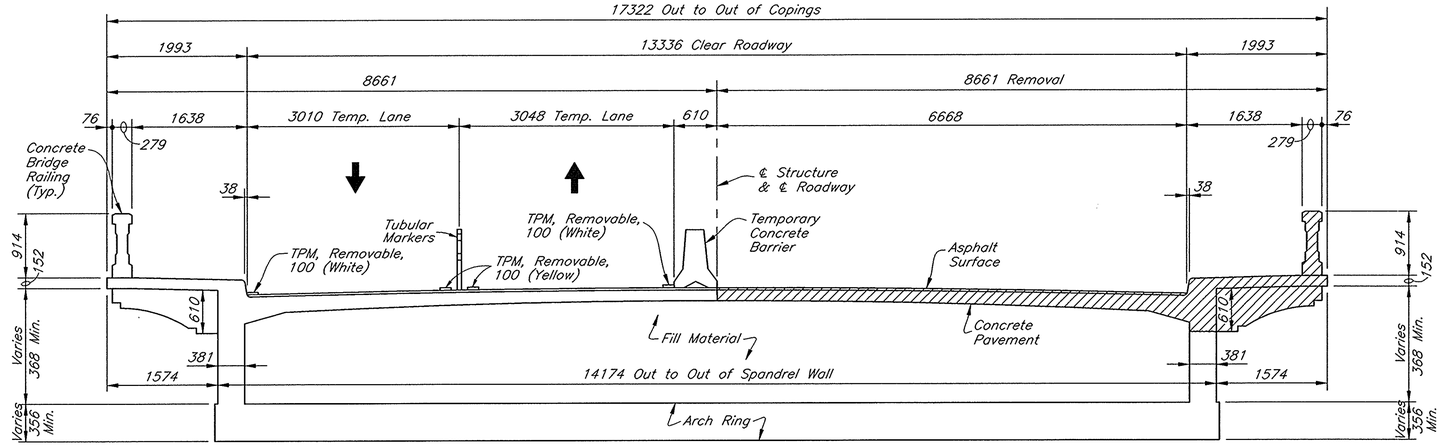
INDIANA DEPARTMENT OF TRANSPORTATION

TRAFFIC MAINTENANCE DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	3 OF 30
CONTRACT	PROJECT
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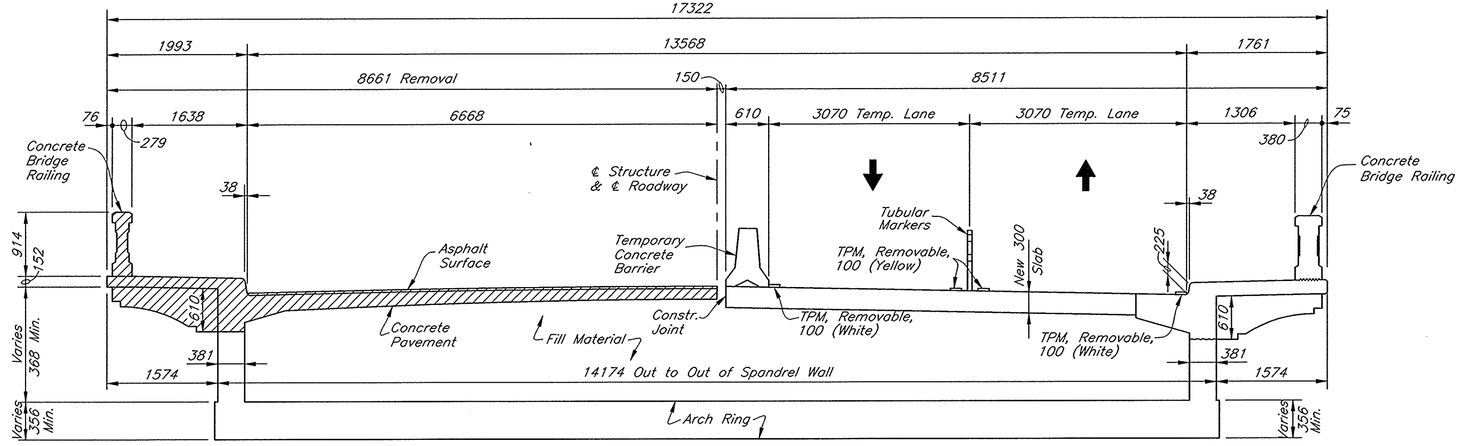
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PHASE I
Scale=1:40

Note: Hatched Areas indicate Portions to be Removed.



PHASE II
Scale=1:40

ESTIMATED QUANTITIES		
PAY ITEM	UNIT	TOTAL
Maintaining Traffic	L.S.	1
Temporary Concrete Barrier	m	113
Construction Signs, Type A	EA.	22
Construction Signs, Type B	EA.	2
Temporary Pavement Marking, Removable, 100mm (White)	m	✳1741
Temporary Pavement Marking, Removable, 100mm (Yellow)	m	⊛1605
Temporary Pavement Marking, 100mm (White)	m	——
Temporary Pavement Marking, 100mm (Yellow)	m	——
Removal of Line, Paint, Solid, 102mm	m	353
Removal of Line, Paint, Broken, 102mm	m	188
Line, Thermoplastic, Solid, White, 100mm	m	404
Line, Thermoplastic, Solid, Yellow, 100mm	m	555
Line, Thermoplastic, Broken, White, 100mm	m	199
Temporary Concrete Barrier, Anchored	m	55
Standard Barricade, Type III-A	EA.	13
Standard Barricade, Type III-B	EA.	3
Pavement Message Markings, Thermoplastic, Word(Only)	EA.	1
Pavement Message Markings, Thermoplastic, Lane Indication Arrow	EA.	2
Temporary Traffic Signals	L.S.	1

✳ Includes 411m for unrestricted Traffic between Phases.
⊛ Includes 411m for unrestricted Traffic between Phases.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

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JEFFREY S. NICHOLSON
REGISTERED
No. 19900078
STATE OF INDIANA
PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: D. SHEETZ DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

INDIANA DEPARTMENT OF TRANSPORTATION

TRAFFIC MAINTENANCE DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	I-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	4 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

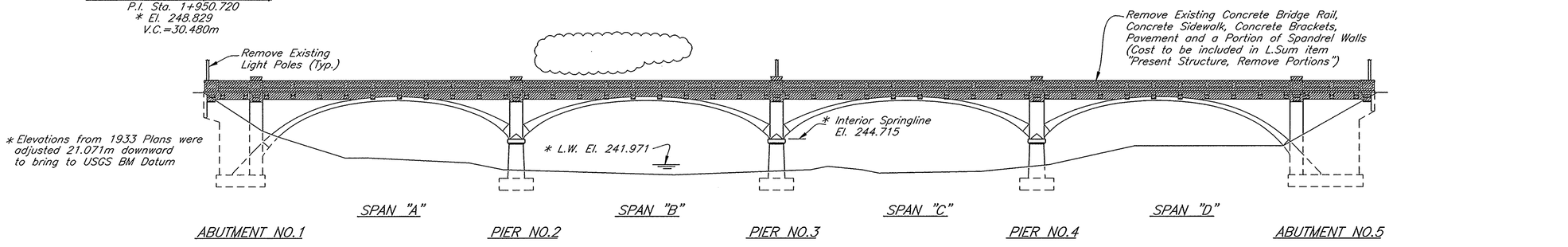
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STRUCTURE IS BUILT ON A 30.480m VERTICAL CURVE AND A 0.00% GRADE

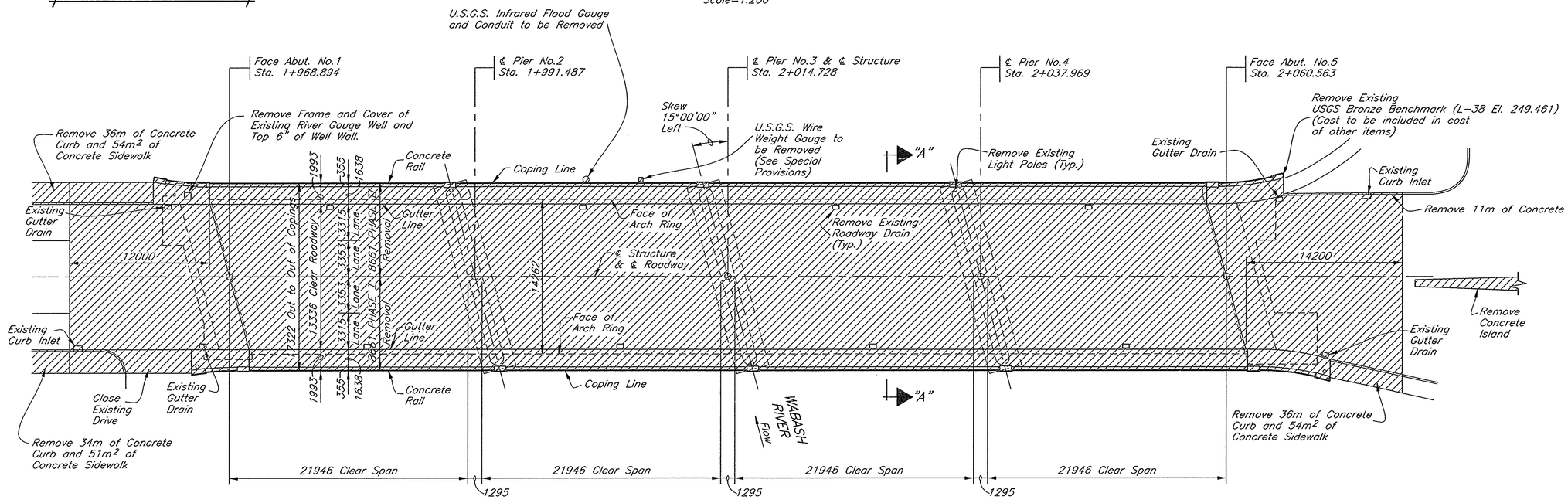
-3.10% 0.00%

ORIGINAL P.G. DATA (1933 Plans)
P.I. Sta. 1+950.720
* El. 248.829
V.C.=30.480m



Note: Hatched Areas indicate Portions to be Removed.

ELEVATION
Scale=1:200



PLAN
Scale=1:200

NOTES
See Sheet 6 for Proposed Structure General Plan.
See Sheet 7 for Section "A-A".
See Sheet 8 for General Notes, Design Data and Material Notes.

REINFORCED CONCRETE ARCH BRIDGE
4 CLEAR SPANS @ 21946, SKEW 15° LEFT, 13336 CLEAR ROADWAY ON STATE ROAD 1 OVER WABASH RIVER

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4-5-04 DO NOT SALVAGE EXISTING FLOWER BOX

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JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: J. NICHOLSON DRAWN: D. SHEETZ
CHECKED: _____ CHECKED: J. NICHOLSON

INDIANA DEPARTMENT OF TRANSPORTATION

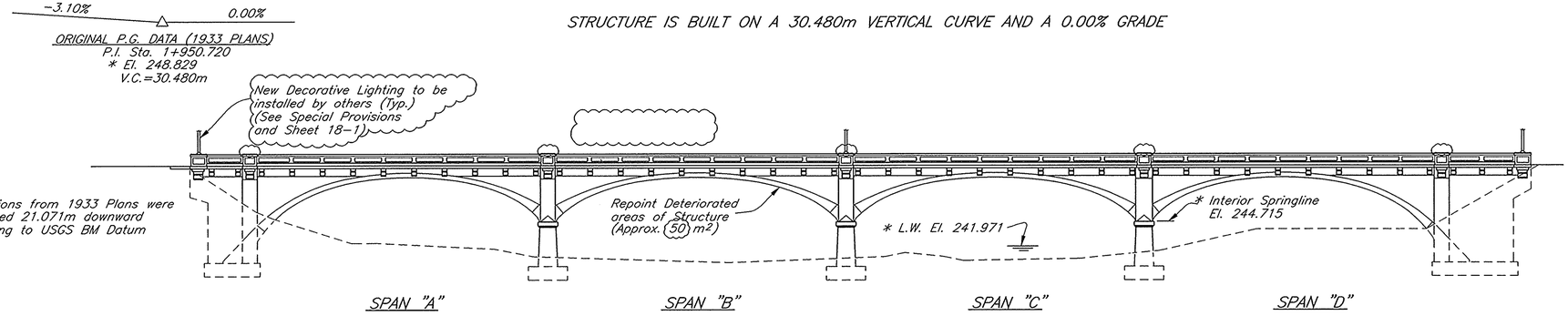
GENERAL PLAN
EXISTING STRUCTURE

HORIZONTAL SCALE	FILE
AS NOTED	I-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	5 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS NO. 3841.0201

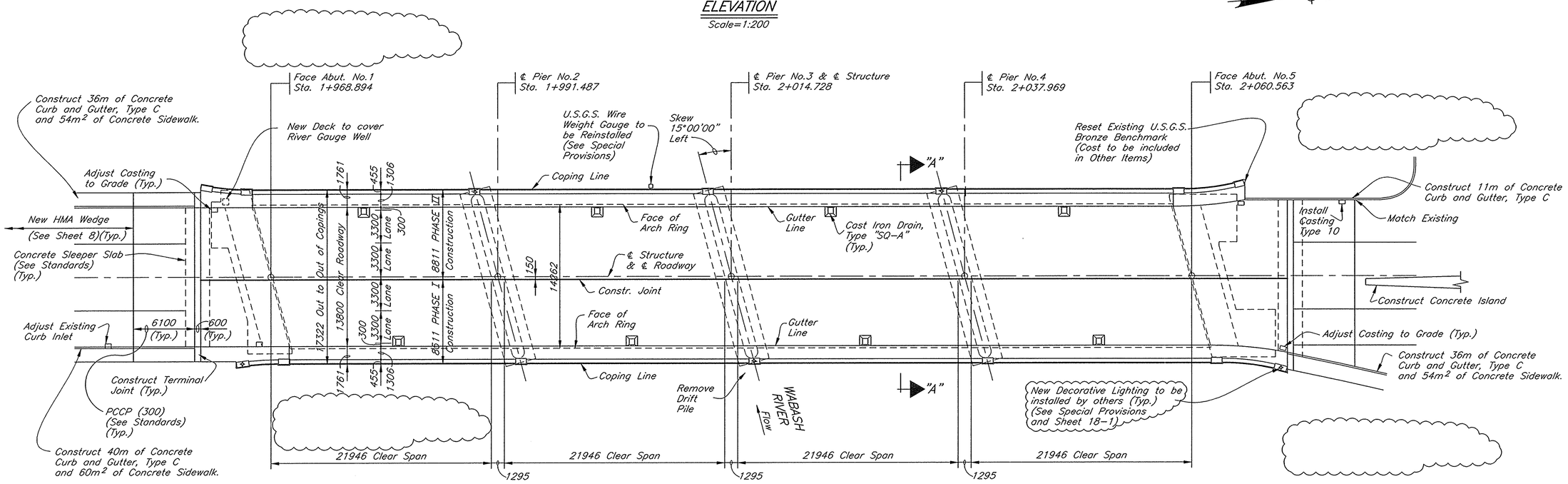
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STRUCTURE IS BUILT ON A 30.480m VERTICAL CURVE AND A 0.00% GRADE



ELEVATION

Scale=1:200



PLAN

Scale=1:200

NOTES

See Sheet 5 for Existing Structure General Plan.
See Sheet 7 for Section "A-A".
See Sheet 8 for General Notes and Design Data.

REINFORCED CONCRETE ARCH BRIDGE

4 CLEAR SPANS @ 21946, SKEW 15° LEFT, 13800 CLEAR ROADWAY ON STATE ROAD 1 OVER WABASH RIVER

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4-6-04 REMOVED FLOWER BOXES, REVISED LIGHTING AND DELETED GUARDRAIL

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RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: J. NICHOLSON DRAWN: D. SHEETZ
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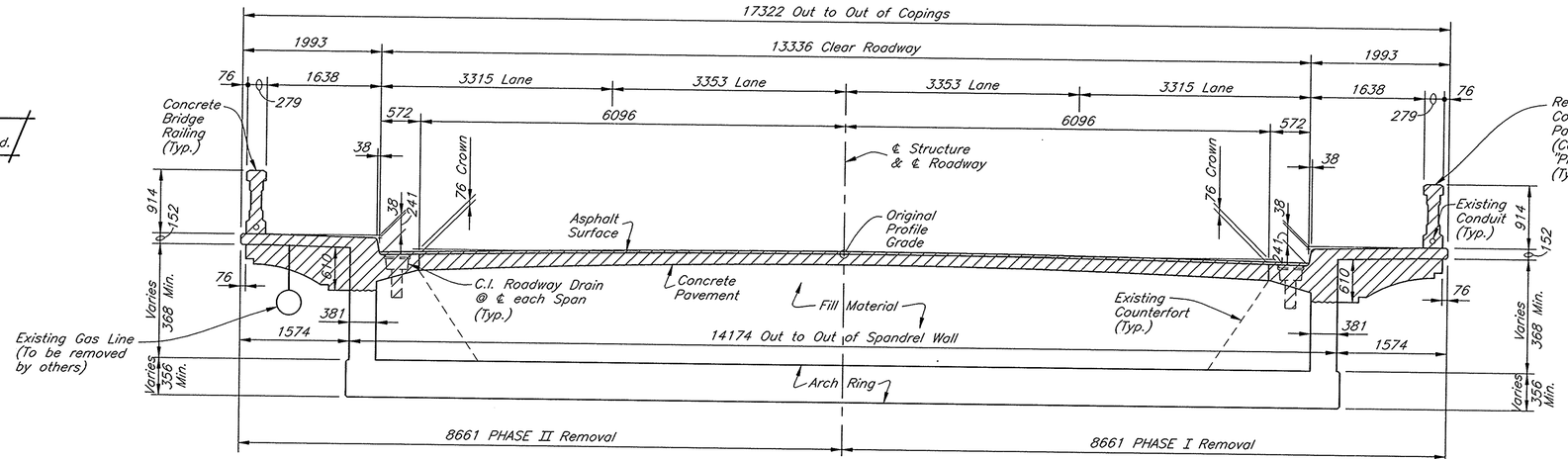
GENERAL PLAN
PROPOSED STRUCTURE

HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	6 OF 30
CONTRACT	PROJECT
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BFS NO. 3841.0201

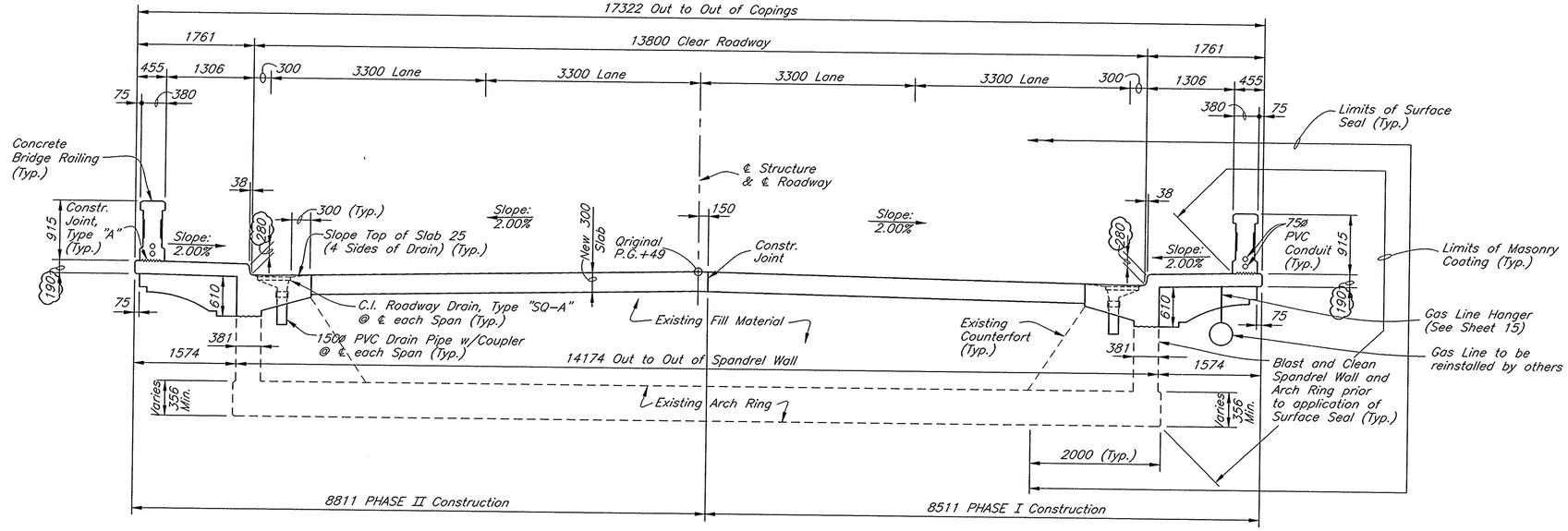
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Note: Hatched Areas indicate Portions to be Removed.



SECTION "A–A" (EXISTING)

Scale=1:40



SECTION "A–A" (PROPOSED)

Scale=1:40

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4–6–04 CURB HEIGHT AND SIDEWALK THICKNESS

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DESIGN ENGINEER DATE

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INDIANA DEPARTMENT OF TRANSPORTATION

GENERAL PLAN

HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	7 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS NO. 3841.0201

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GENERAL NOTES

Plans for the existing structure are on file in the Central Office as Bridge File: 1-0-230 and are available upon request.

Portions of present structure shall be removed.

Where new work is to be fitted to old work, the Contractor shall check all dimensions and conditions in the field and report any errors or discrepancies to the Engineer and assume responsibility for their correctness and the fit of the new part to the old.

The boundaries of removal areas shall be saw cut. All saw cuts shall be a minimum depth of 25mm below the original surface or to the reinforcing steel if cover is less than 25mm.

Concrete shall be Class "C".

Reinforcing steel covering shall be 65mm in top and 25mm in bottom of floor slabs and 50mm in all other areas unless noted.

Epoxy coated reinforcing steel shall be required in various portions of the structure as shown.

Reinforcing steel shall be A.S.T.M. A615M Grade 420.

Chamfer exposed corners of concrete 25 mm unless noted.

All asphalt material required in this contract shall be included in the pay item "HMA for Approaches" except tack coat, which shall be paid for as a separate item.

Surface seal shall be required on various areas of the structure as shown. Estimated quantity = 3726 Sq. Meters.

Masonry coating shall be required on various areas of the structure as shown. Masonry coating shall conform to Federal Color Standard 595a, Color No. 27778 (Antique Ivory). See the Special Provisions. Estimated quantity = 1892 Sq. Meters.

Seal all joints and cracks in the approach pavement with a hot poured joint sealer before placing the HMA wedges. Cost to be included in the cost of other items of the contract.

The calculated clear zone is 3.0 meters.

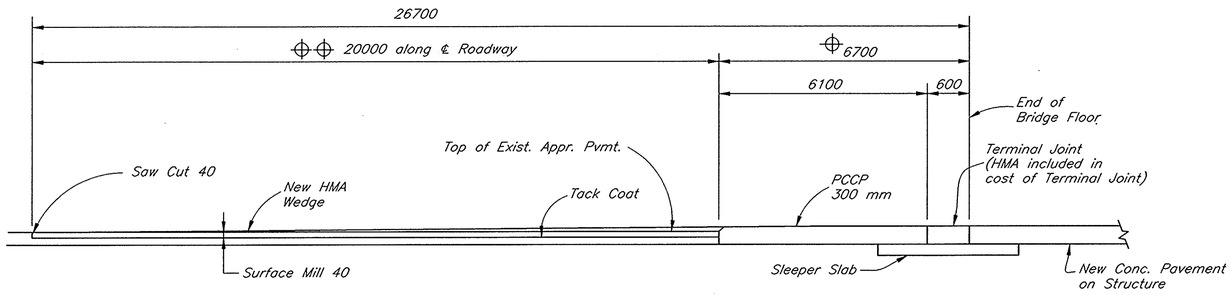
MATERIAL NOTES

HMA Wedge:

90kg/m^2 HMA Type C Surface over Variable Depth
HMA Type C Intermediate.

HMA FOR APPROACHES	
WEDGE	50 Mg

⊕ Top of PCCP shall be a continuation of bridge deck profile.
⊕⊕ Top of Wedge shall be tapered uniformly to meet exist. pavement.



HMA WEDGE APPROACH SECTION

Not to Scale

DESIGN DATA

MATERIAL DESIGN STRENGTHS:

Class "C" Concrete	F'c = 28 MPa
Reinforcing Steel (Grade 420)	Fy = 420 MPa

LIVE LOAD:

4 kPa in accordance with 2002 A.A.S.H.T.O. Specifications.
Load Factor = 2.17

DEAD LOAD:

Actual only. No future wearing surface or metal deck form load assumed.
Load Factor = 1.30

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4-6-04 GENERAL NOTES

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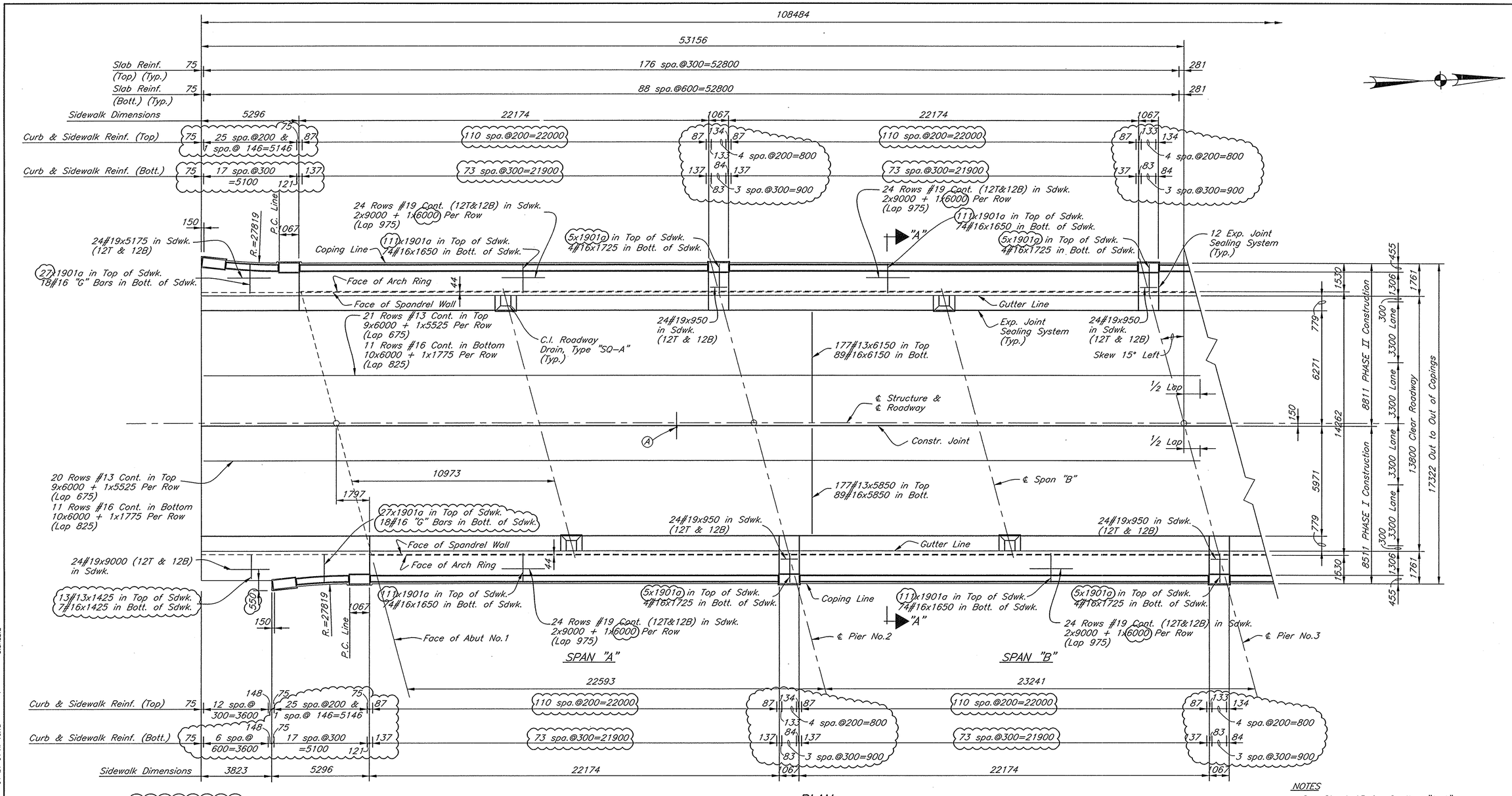
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GENERAL NOTES AND APPROACH DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	8 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

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"G" Bars

⊗ #16 in Bottom (Lap 825) Cut in Field from 7#16x6000

⊗ Cost of Field Cutting shall be included in cost of other items. Number of Bars shown includes 20% for Laps and Waste.

Ⓐ 177-Threaded Tie Bar Assemblies (#13x725 ea.way) (Epoxy Coated) in Top
89-Threaded Tie Bar Assemblies (#16x875 ea.way) (Epoxy Coated) in Bottom

PLAN
SHOWING SIDEWALK AND DECK REINFORCEMENT
SPANS "A" AND "B"

Scale=1:100

NOTES

See Sheet 15 for Section "A-A".
See Sheet 16 for Bar Bending Details and Bills of Materials.
See Sheets 17-19 for Railing Details.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4-5-04: CURB AND SIDEWALK REINFORCING

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DESIGN ENGINEER DATE

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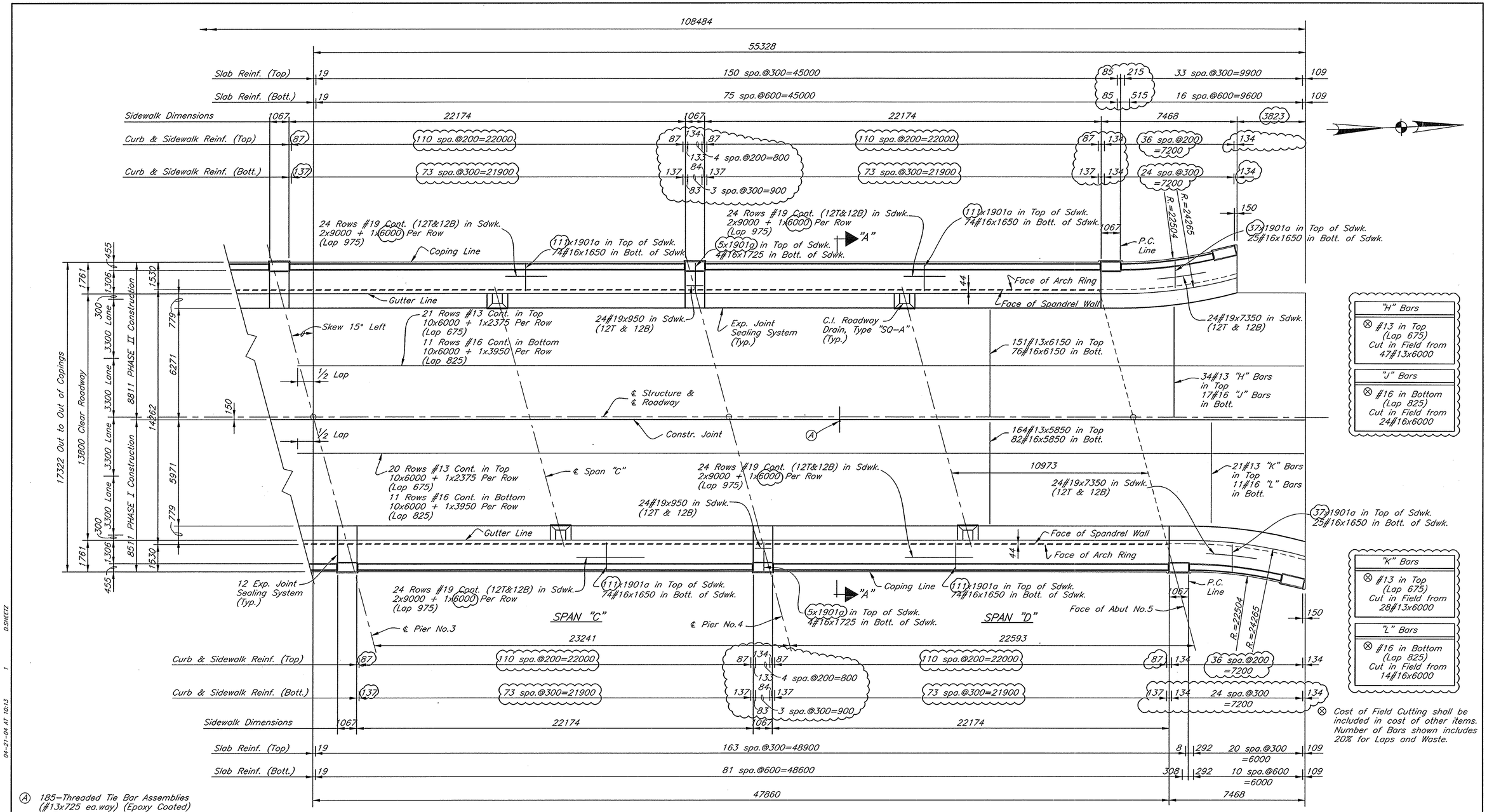
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FLOOR DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	9 OF 30
CONTRACT	PROJECT
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DES No. 3841.0201

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Ⓐ 185-Threaded Tie Bar Assemblies (#13x725 ea.way) (Epoxy Coated) in Top
93-Threaded Tie Bar Assemblies (#16x875 ea.way) (Epoxy Coated) in Bottom

⊗ Cost of Field Cutting shall be included in cost of other items. Number of Bars shown includes 20% for Laps and Waste.

PLAN
SHOWING SIDEWALK AND DECK REINFORCEMENT
SPANS "C" AND "D"

Scale=1:100

NOTES
See Sheet 15 for Section "A-A".
See Sheet 16 for Bar Bending Details and Bills of Materials.
See Sheets 17-19 for Railing Details.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4-5-04 CURB AND SIDEWALK REINFORCING

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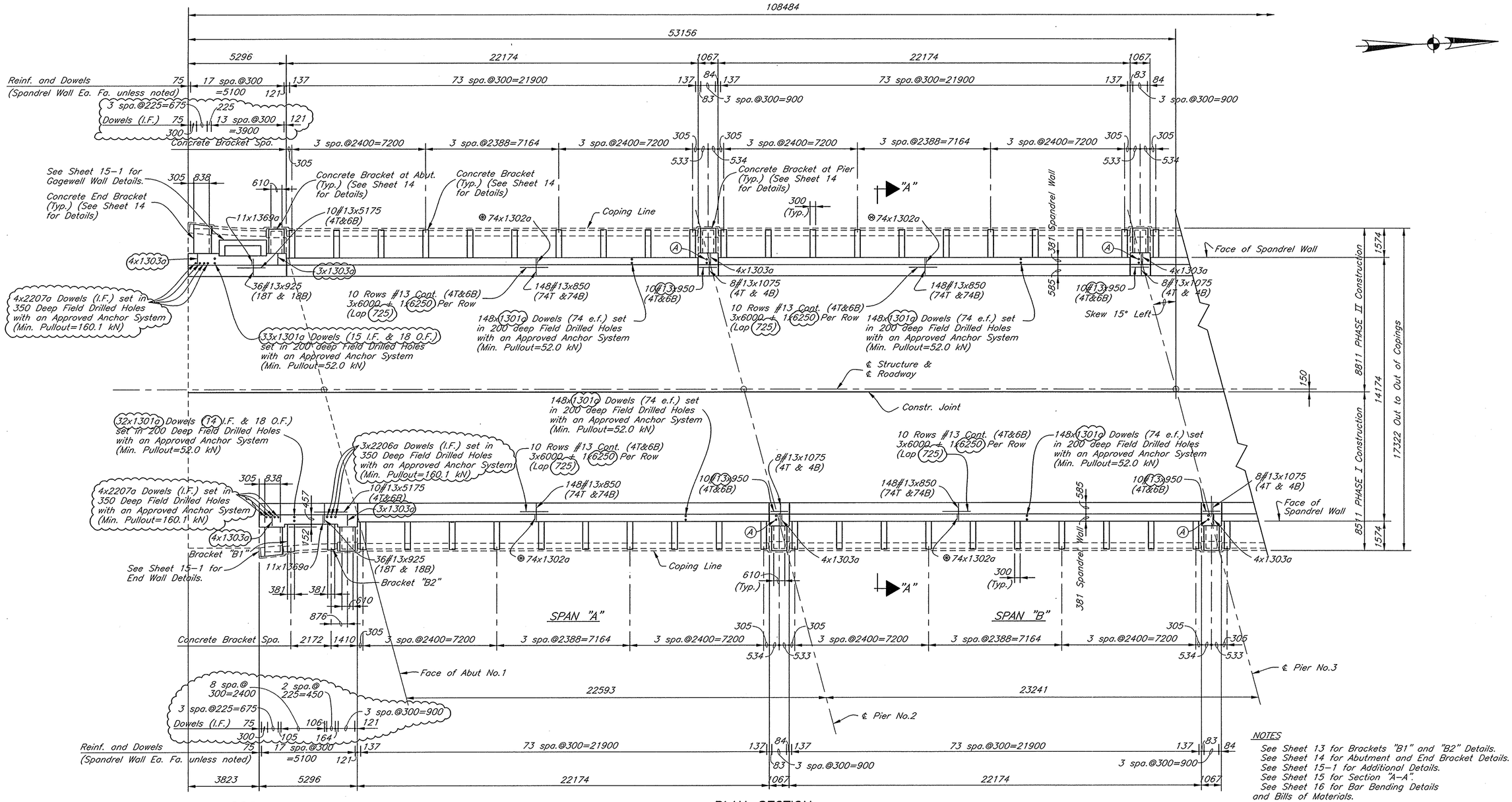
INDIANA DEPARTMENT OF TRANSPORTATION

FLOOR DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	10 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

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℗ Field Cut Bottom Legs of 1302a Bars at top of Arch Ring to provide required cover. Cost to be included in cost of Reinforcing Steel.

Ⓐ 8x1301a Dowels (4 e.f.) set in 200 deep Field Drilled Holes with an Approved Anchor System (Min. Pullout=52.0 kN)

PLAN-SECTION
SHOWING GUTTER AND SPANDREL WALL REINFORCMENT
SPANS "A" AND "B"

Scale=1:100

NOTES

See Sheet 13 for Brackets "B1" and "B2" Details.
See Sheet 14 for Abutment and End Bracket Details.
See Sheet 15-1 for Additional Details.
See Sheet 15 for Section "A-A".
See Sheet 16 for Bar Bending Details and Bills of Materials.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4-5-04 SPANDREL WALL REINFORCING

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DESIGNED: D. SHEETZ DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

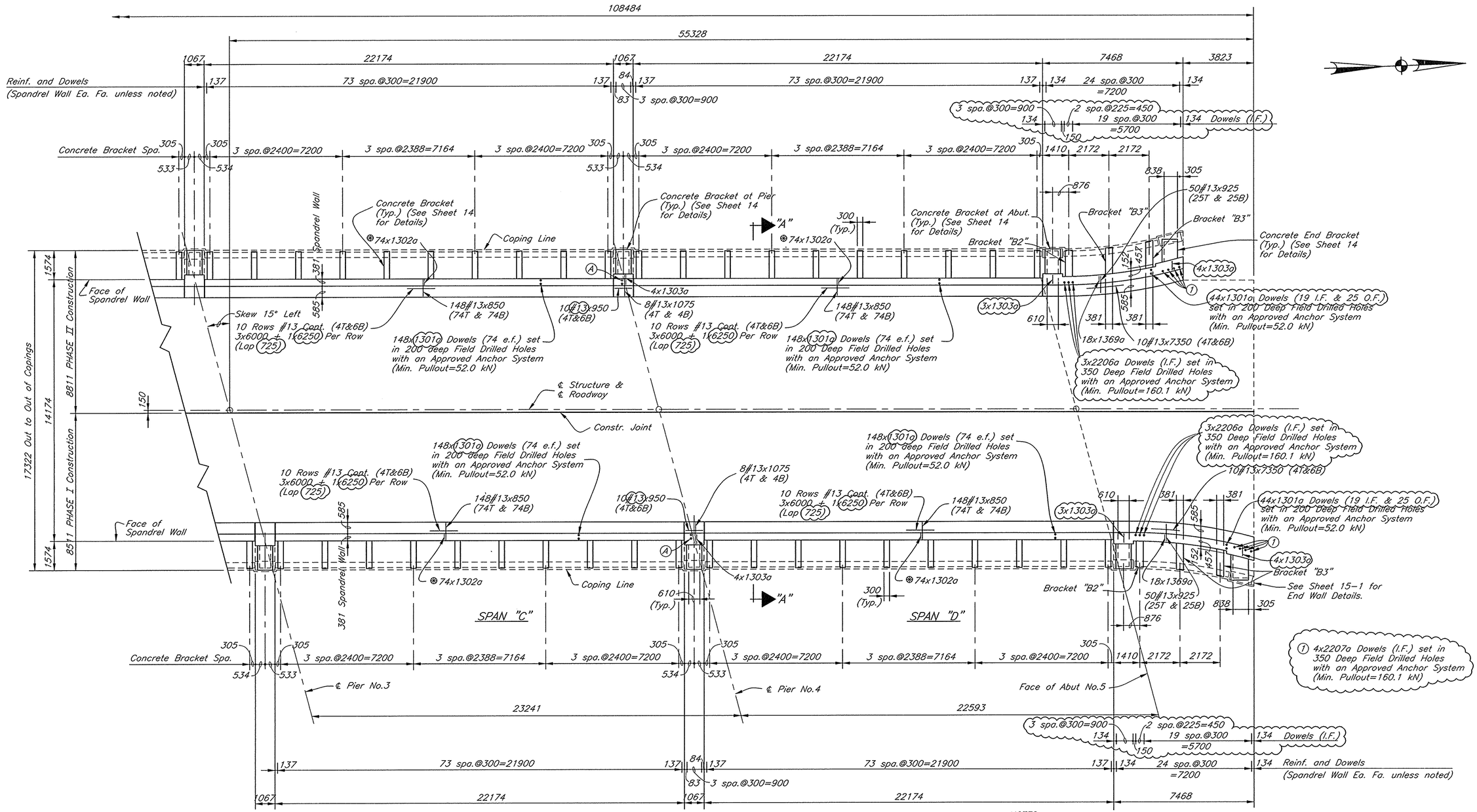
INDIANA DEPARTMENT OF TRANSPORTATION

FLOOR DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	I-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	11 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

DFS No. 3841.02011

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⊗ Field Cut Bottom Legs of 1302a Bars at top of Arch Ring to provide required cover. Cost to be included in cost of Reinforcing Steel.

Ⓐ 8x1301g Dowels (4 e.f.) set in 200 deep Field Drilled Holes with an Approved Anchor System (Min. Pullout=52.0 kN)

PLAN–SECTION
SHOWING GUTTER AND SPANDREL WALL REINFORCMENT
SPANS "C" AND "D"

Scale=1:100

NOTES
See Sheet 13 for Brackets "B2" and "B3" Details.
See Sheet 14 for Abutment and End Bracket Details.
See Sheet 15 for Section "A–A".
See Sheet 15–1 for Additional Details.
See Sheet 16 for Bar Bending Details and Bills of Materials.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4–5–04 SPANDREL WALL REINFORCING

BF&S Butler Fairman Scufert CONSULTING ENGINEERS
8450 WESTFIELD BLVD., SUITE 300 INDIANAPOLIS, IN. 46240 317 713-4615 FAX 317 713-4616
509 WEST 84TH DRIVE, SUITE G MERRILLVILLE, IN. 46410 219 769-2333 FAX 219 769-2377
JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: D. SHEETZ DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

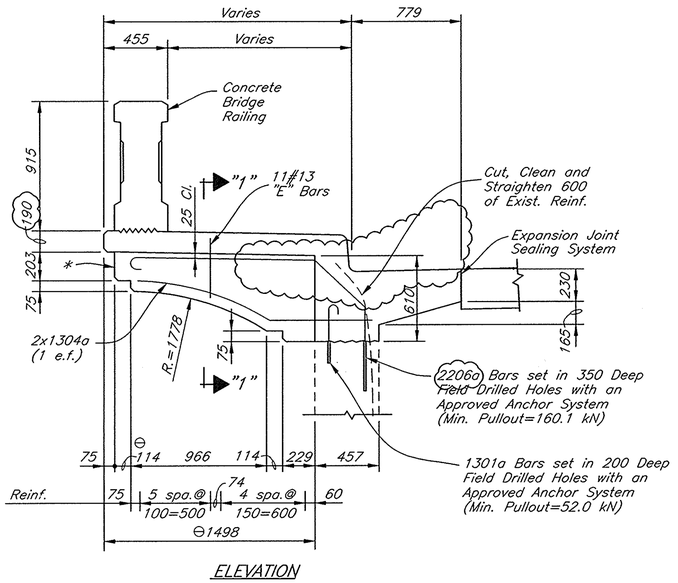
INDIANA DEPARTMENT OF TRANSPORTATION

FLOOR DETAILS

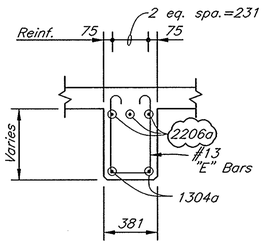
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AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	12 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS NO. 3841.0201

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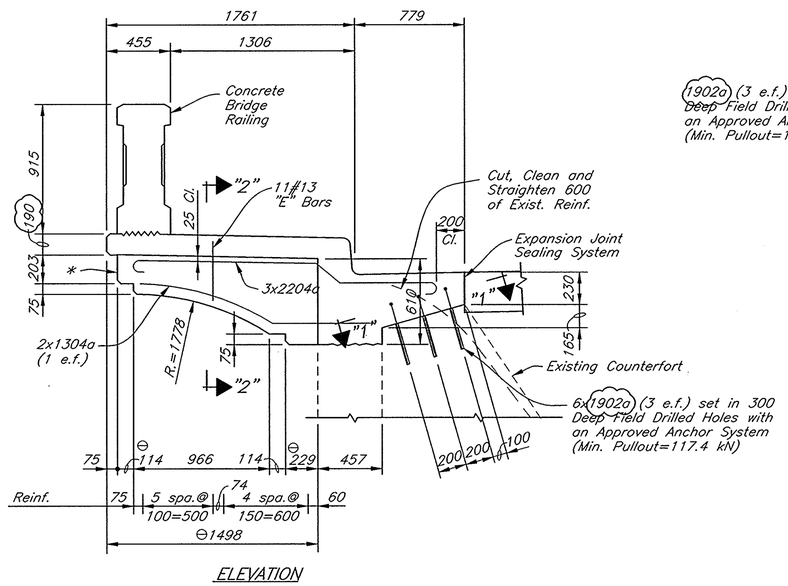
ELEVATION



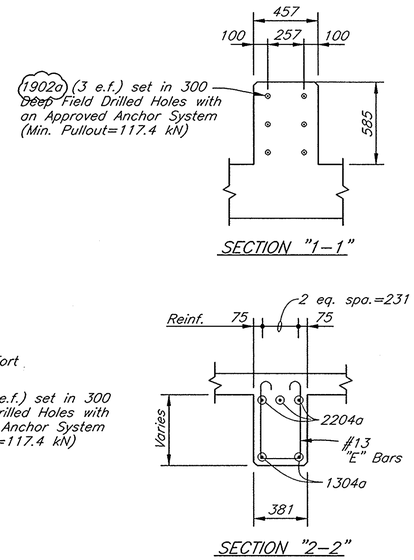
SECTION "1–1"

CONCRETE BRACKET "B2"

Scale=1:20



ELEVATION



SECTION "1–1"

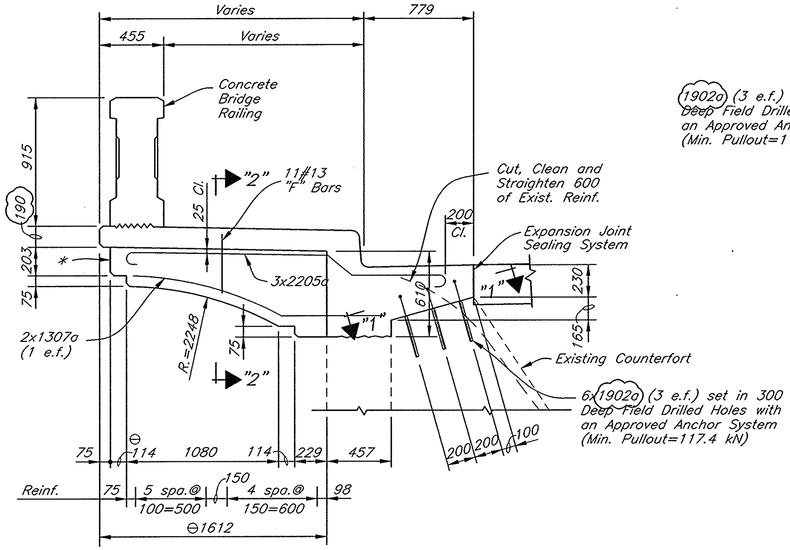
SECTION "2–2"

CONCRETE BRACKET "B3"

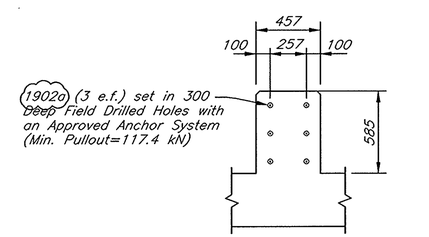
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* Face of Bracket follows curvature of Sidewalk

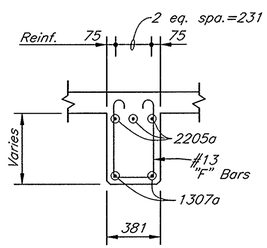
⊖ Dimensions along ℄ Bracket



ELEVATION



SECTION "1–1"



SECTION "2–2"

CONCRETE BRACKET "B1"

Scale=1:20

NOTE
See Sheet 16 for Bar Bending Details and Bills of Materials.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4–6–04 SIDEWALK THICKNESS AND REINFORCING

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Butler Fairman Seufert CONSULTING ENGINEERS
8450 WESTFIELD BLVD., SUITE 300 INDIANAPOLIS, IN. 46240 317 713-4615 FAX 317 713-4616
509 WEST 84TH DRIVE, SUITE G MERRILLVILLE, IN. 46410 219 769-2333 FAX 219 769-2377

JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: DESIGN ENGINEER 2-17-04 DATE

DESIGNED: D. SHEETZ DRAWN: D. SHEETZ

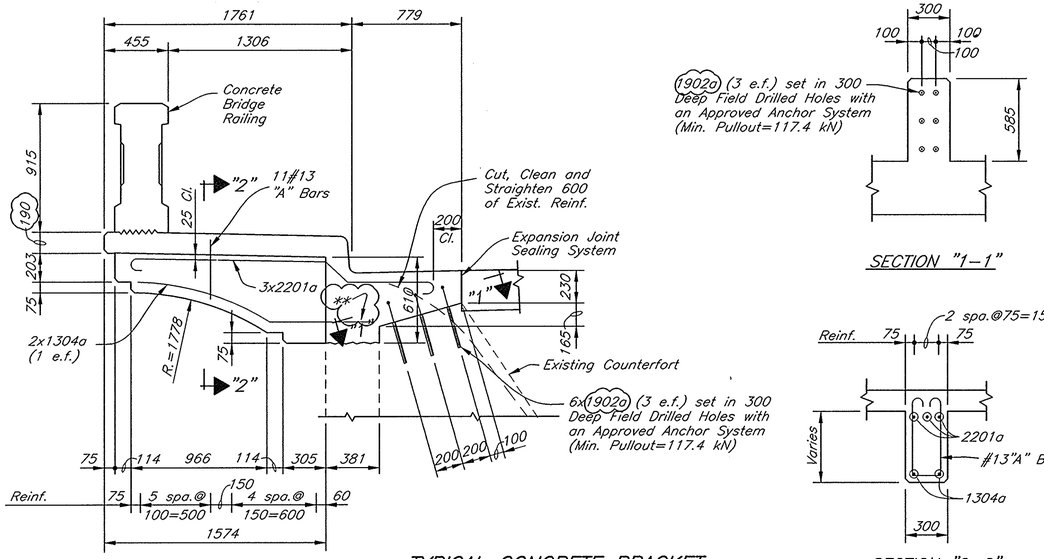
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

INDIANA DEPARTMENT OF TRANSPORTATION

FLOOR DETAILS

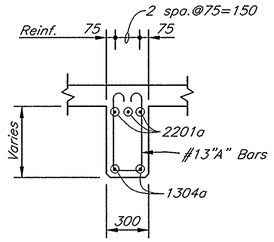
HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	13 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS no. 3841.0201



ELEVATION

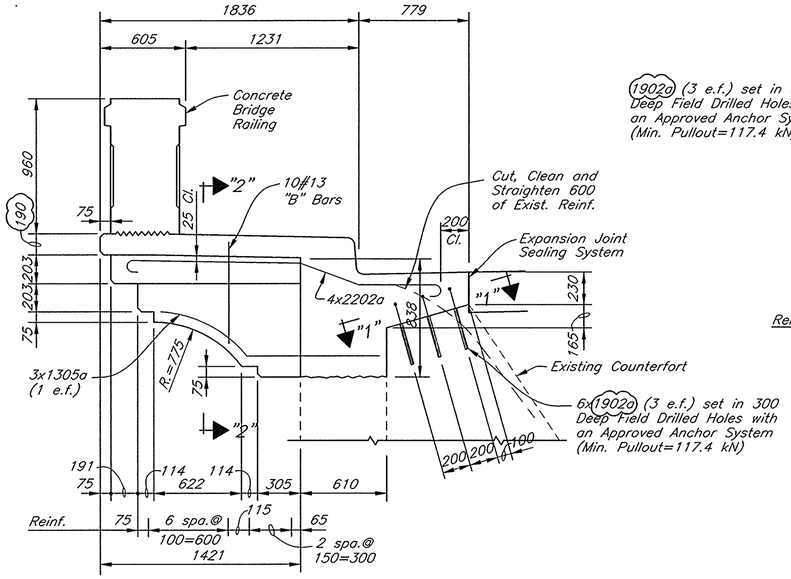
SECTION "1–1"



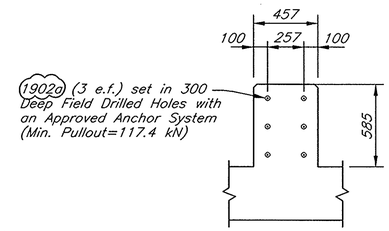
SECTION "2–2"

TYPICAL CONCRETE BRACKET IN SPANS "A" THRU "D"

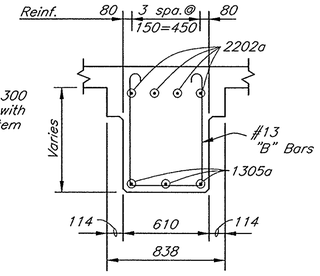
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ELEVATION



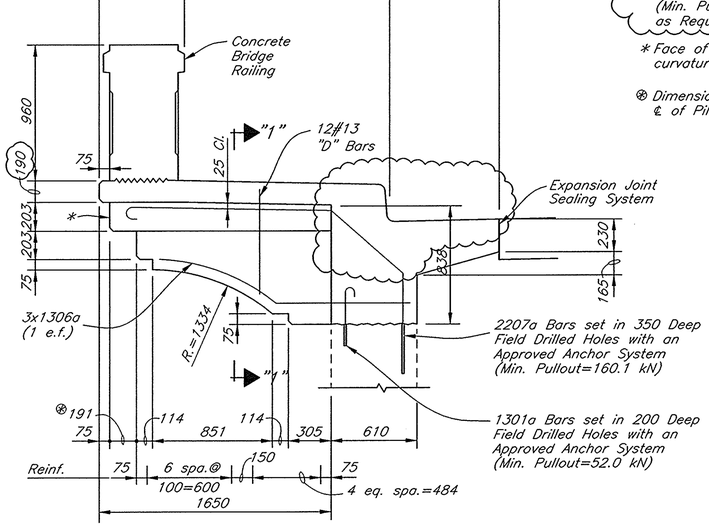
SECTION "1–1"



SECTION "2–2"

TYPICAL CONCRETE BRACKET AT ABUTMENT AND PIER PILASTERS

Scale=1:20

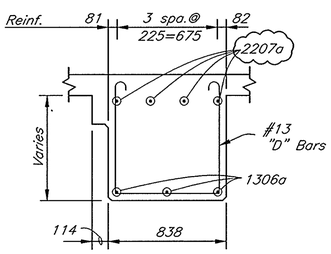


ELEVATION

** Where Bracket overlops Arch Ring, Place Bar in 300 Deep Field Drilled Hole with an Approved Anchor System (Min. Pullout=52.0 kN). Field Cut Bar as Required.

* Face of Bracket follows curvature of Sidewalk

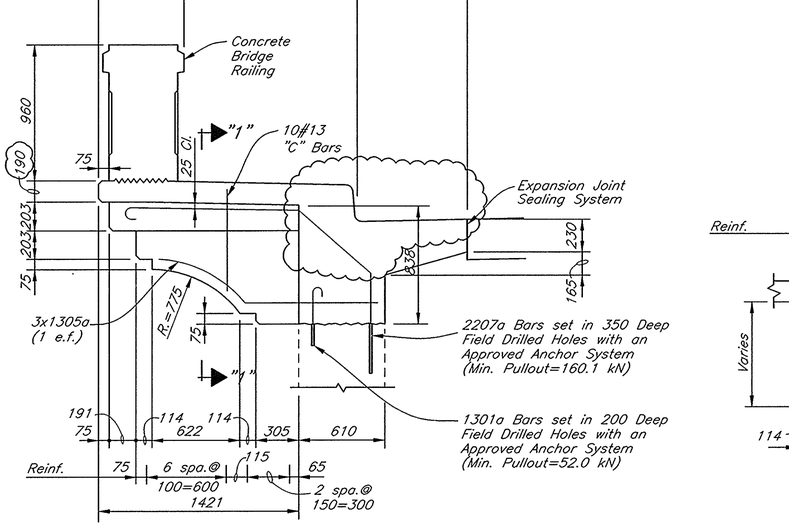
⊛ Dimensions along ℄ of Pilaster



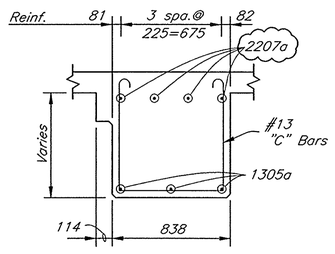
SECTION "1–1"

CONCRETE END BRACKET AT ABUTMENT NO.1

Scale=1:20



ELEVATION



SECTION "1–1"

CONCRETE END BRACKET AT ABUTMENT NO.5

Scale=1:20

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

NOTE
See Sheet 16 for Bar Bending Details and Bills of Materials.

REVISED 4-6-04 SIDEWALK THICKNESS AND REINFORCING

BF&S Butler Fairman Seufert CONSULTING ENGINEERS
8450 WESTFIELD BLVD., SUITE 300 INDIANAPOLIS, IN. 46240 317 713-4615 FAX 317 713-4616
509 WEST 84TH DRIVE, SUITE G MERRILLVILLE, IN. 46410 219 769-2333 FAX 219 769-2377
JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: D. SHEETZ DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

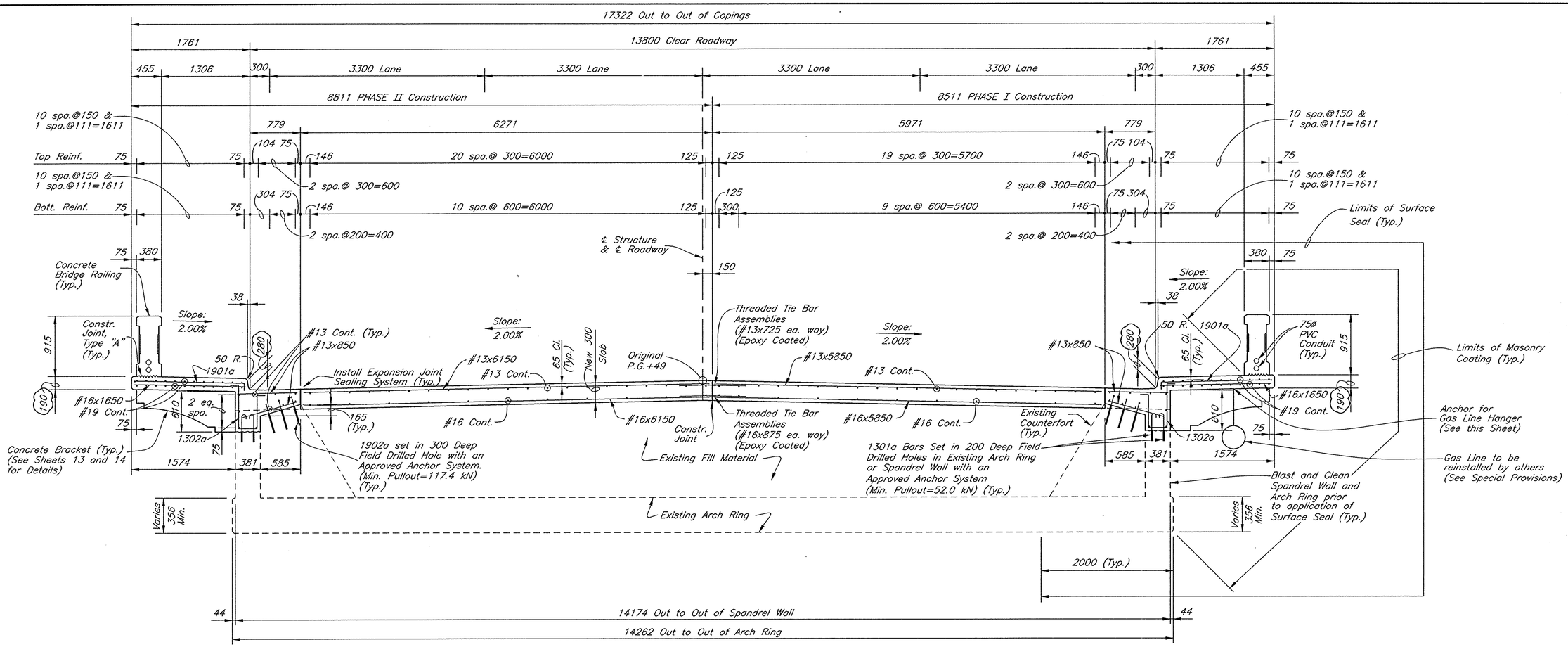
INDIANA DEPARTMENT OF TRANSPORTATION

FLOOR DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	14 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

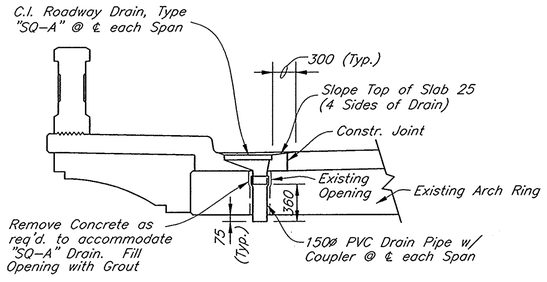
BFS NO. 3841.0201

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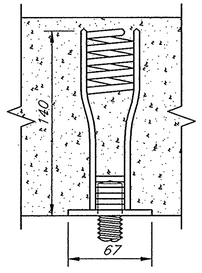
SECTION "A–A"

Scale=1:30



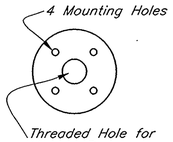
SECTION AT ROADWAY DRAIN

Scale=1:30

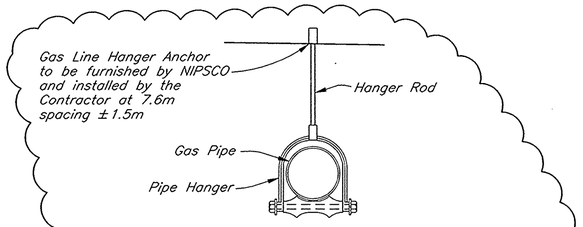


GAS LINE ANCHOR

Not to Scale



BOTTOM VIEW



Note: Gas Pipe, Pipe Hangers, Hanger Rods and Miscellaneous Hardware to be furnished and installed by NIPSCO.

GAS PIPE HANGER DETAIL

Not to Scale

NOTES

See Sheet 16 for Bar Bending Details and Bills of Materials.
See Sheet 18–1 for Conduit and Lighting Details.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4–6–04 CURB HEIGHT, SIDEWALK THICKNESS AND PIPE HANGER DETAIL

Butler Fairman Scufert CONSULTING ENGINEERS
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509 WEST 84TH DRIVE, SUITE G MERRILLVILLE, IN. 46410 219 769-2333 FAX 219 769-2377
JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: D. SHEETZ DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

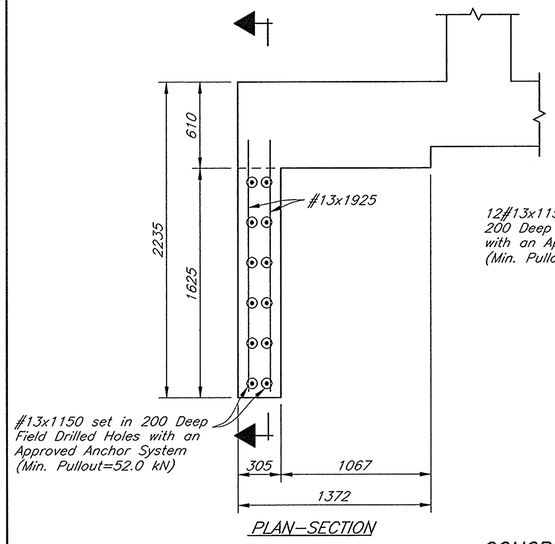
INDIANA DEPARTMENT OF TRANSPORTATION

FLOOR DETAILS

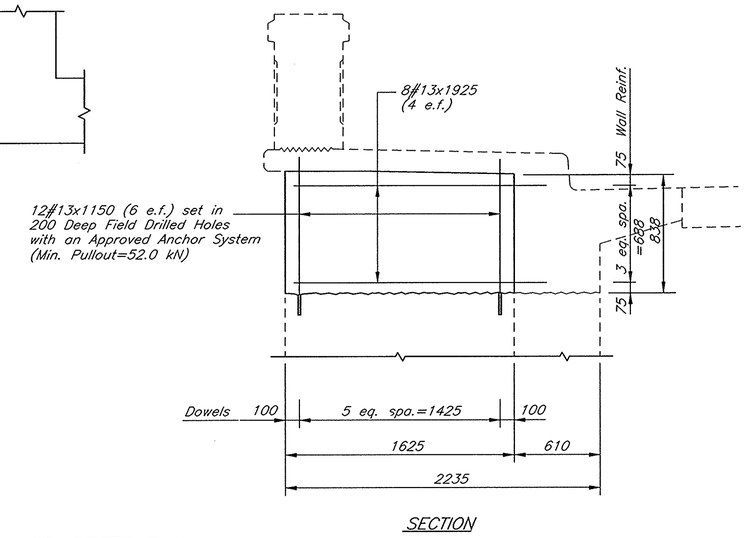
HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	15 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS NO. 3841.0201

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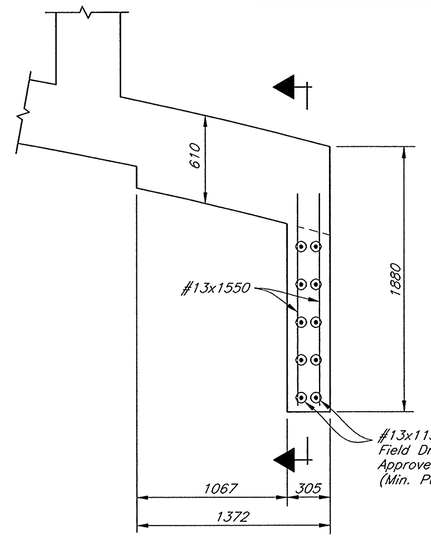
PLAN–SECTION



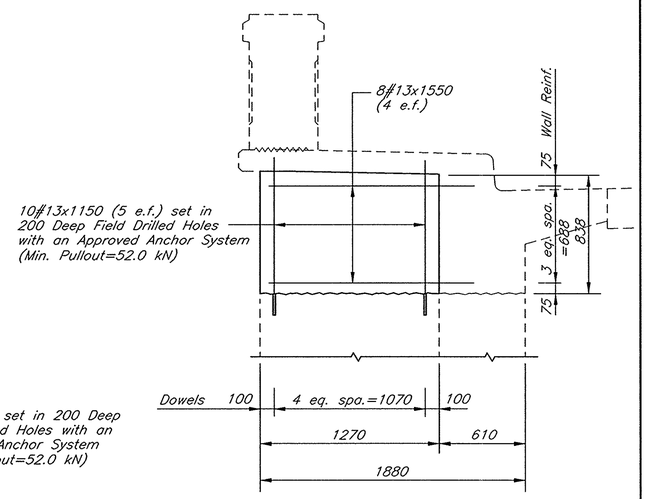
SECTION

CONCRETE END WALL
SOUTHEAST WALL (SHOWN)
SOUTHWEST WALL (OPP. HAND)

Scale=1:20



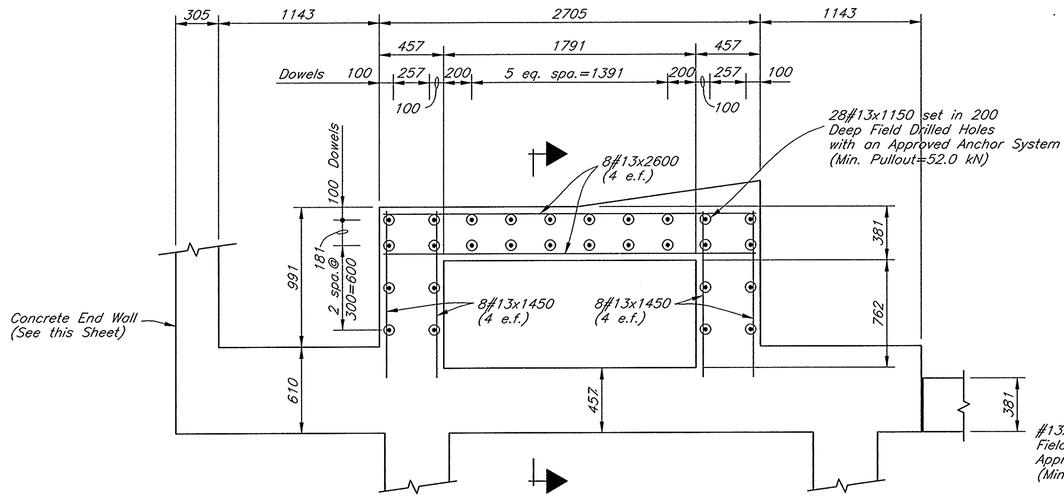
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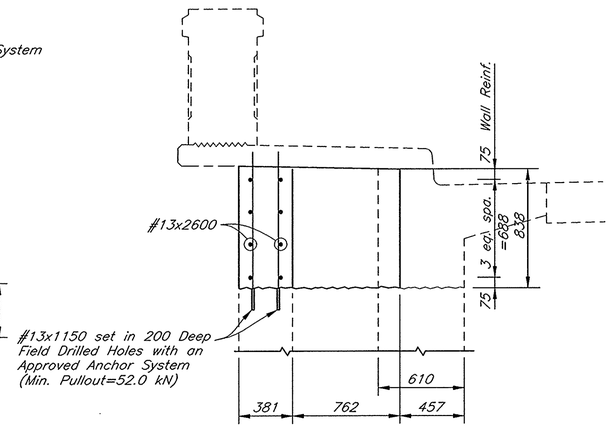
SECTION

CONCRETE END WALL
NORTHEAST WALL (SHOWN)
NORTHWEST WALL (OPP. HAND)

Scale=1:20



PLAN–SECTION



SECTION

CONCRETE GAGEWELL WALL AT SOUTHWEST CORNER

Scale=1:20

NOTE
See Sheet 16 for Bar Bending Details and Bills of Materials.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4-6-04 ADDED THIS SHEET

Butler Fairman Scufert CONSULTING ENGINEERS
8450 WESTFIELD BLVD., SUITE 300 INDIANAPOLIS, IN 46240 317 713-4615 FAX 317 713-4616
509 WEST 84TH DRIVE, SUITE C MERRILLVILLE, IN 46410 219 769-2333 FAX 219 769-2377

JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: D. SHEETZ DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

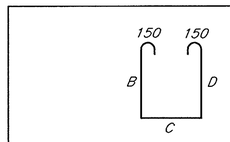
INDIANA DEPARTMENT OF TRANSPORTATION

FLOOR DETAILS

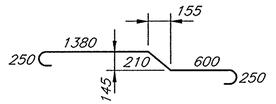
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AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	15-1 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS NO. 3841.0201

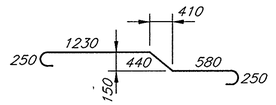
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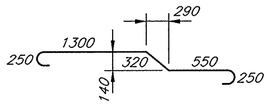
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1311a	1200	350	200	350
1312a	1260	380	200	380
1313a	1300	400	200	400
1314a	1400	450	200	450
1315a	1540	520	200	520
1316a	1680	590	200	590
1317a	1820	660	200	660
1318a	1710	450	510	450
1319a	1850	520	510	520
1320a	1870	530	510	530
1321a	1930	560	510	560
1322a	1990	590	510	590
1323a	2110	650	510	650
1324a	2250	720	510	720
1325a	2450	820	510	820
1326a	2590	890	510	890
1327a	1930	450	730	450
1328a	2070	520	730	520
1329a	2090	530	730	530
1330a	2150	560	730	560
1331a	2210	590	730	590
1332a	2330	650	730	650
1333a	2470	720	730	720
1334a	2670	820	730	820
1335a	2810	890	730	890
1336a	1930	450	730	450
1337a	2090	530	730	530
1338a	2110	540	730	540
1339a	2130	550	730	550
1340a	2170	570	730	570
1341a	2230	600	730	600
1342a	2310	640	730	640
1343a	2450	710	730	710
1344a	2610	790	730	790
1345a	2670	820	730	820
1346a	2810	890	730	890
1347a	1220	320	280	320
1348a	1240	330	280	330
1349a	1260	340	280	340
1350a	1280	350	280	350
1351a	1340	380	280	380
1352a	1380	400	280	400
1353a	1440	430	280	430
1354a	1560	490	280	490
1355a	1700	560	280	560
1356a	1740	580	280	580
1357a	1900	660	280	660
1358a	1220	320	280	320
1359a	1240	330	280	330
1360a	1260	340	280	340
1361a	1280	350	280	350
1362a	1320	370	280	370
1363a	1360	390	280	390
1364a	1440	430	280	430
1365a	1540	480	280	480
1366a	1680	550	280	550
1367a	1760	590	280	590
1368a	1900	660	280	660



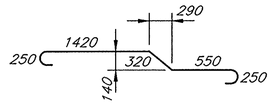
2201a x 2690



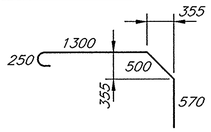
2202a x 2750



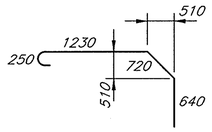
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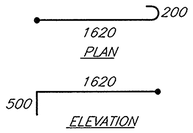
2205a x 2790



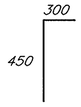
2206a x 2620



2207a x 2840



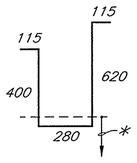
1901a x 2320



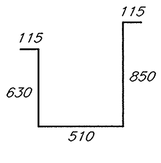
1902a x 750



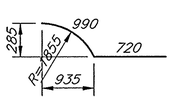
1301a x 830



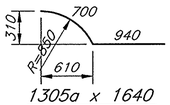
1302a x 1530



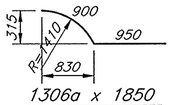
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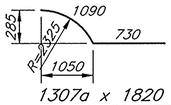
1304a x 1710



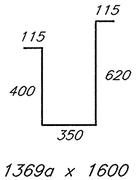
1305a x 1640



1306a x 1850



1307a x 1820



1369a x 1600

BAR BENDING DETAILS

Not to Scale

* Remove this Portion of Bar where necessary to Provide required Cover.

BILL OF MATERIALS FLOOR PHASE I

REINFORCING STEEL

Mark or Size	No. of Bars	Length (mm)	Mass (kg)
2201a	120	2690	
2202a	20	2750	
2204a	6	2670	
2205a	3	2790	
2206a	6	2620	
2207a	8	2840	
Total #22 (Epoxy Coated)			1340
1901a	523	2320	
1902a	288	750	
#19	216	9000	
#19	24	7350	
#19	96	6000	
#19	72	950	
Total #19 (Epoxy Coated)			9374
#16	241	6000	
#16	171	5850	
#16	11	3950	
#16	11	1775	
#16	12	1725	
#16	321	1650	
#16	7	1425	
Total #16 (Epoxy Coated)			4764
1301a	692	830	
1302a	296	1530	
1303a	26	2220	
1304a	88	1710	
1305a	18	1640	
1306a	3	1850	
1307a	2	1820	
1369a	29	1600	
#13	10	7350	
#13	40	6250	
#13	528	6000	
#13	341	5850	
#13	20	5525	
#13	10	5175	
#13	20	2375	
#13	8	1925	
#13	8	1550	
#13	13	1425	
#13	40x11	"A" Bars	
#13	5x10	"B" Bars	
#13	1x10	"C" Bars	
#13	1x12	"D" Bars	
#13	4x11	"E" Bars	
#13	1x11	"F" Bars	
#13	22	1150	
#13	24	1075	
#13	30	950	
#13	86	925	
#13	592	850	
Total #13 (Epoxy Coated)			8533
Total Steel (Epoxy Coated)			24011

CONCRETE

Class "C" in Superstructure	285.8 m³

MISCELLANEOUS

Surface Seal	1552 m²
Field Drilled Holes in Concrete	1032 Each
12 Expansion Joint Sealing System	129 m
4-C.I. Roodway Drains, Type "SQ-A"@87.1 kg Each	348.4 kg
Drain Extensions	4 Each
Threaded Tie Bar Assemblies (Epoxy Coated) (#16x875 ea. way) (in Bott.)	182 Each
Threaded Tie Bar Assemblies (Epoxy Coated) (#13x725 ea. way) (in Top)	362 Each
Masonry Coating	648 m²

BILL OF MATERIALS FLOOR PHASE II

REINFORCING STEEL

Mark or Size	No. of Bars	Length (mm)	Mass (kg)
2201a	120	2690	
2202a	20	2750	
2204a	6	2670	
2206a	3	2620	
2207a	8	2840	
Total #22 (Epoxy Coated)			1291
1901a	523	2320	
1902a	282	750	
#19	192	9000	
#19	24	7350	
#19	96	6000	
#19	24	5175	
#19	72	950	
Total #19 (Epoxy Coated)			9159
#16	165	6150	
#16	251	6000	
#16	11	3950	
#16	11	1775	
#16	12	1725	
#16	321	1650	
Total #16 (Epoxy Coated)			4864
1301a	693	830	
1302a	296	1530	
1303a	26	2220	
1304a	86	1710	
1305a	18	1640	
1306a	3	1850	
1369a	29	1600	
#13	10	7350	
#13	40	6250	
#13	328	6150	
#13	566	6000	
#13	21	5525	
#13	10	5175	
#13	40x11	"A" Bars	
#13	5x10	"B" Bars	
#13	1x10	"C" Bars	
#13	1x12	"D" Bars	
#13	3x11	"E" Bars	
#13	8	2600	
#13	21	2375	
#13	8	1925	
#13	8	1550	
#13	16	1450	
#13	50	1150	
#13	24	1075	
#13	30	950	
#13	86	925	
#13	592	850	
Total #13 (Epoxy Coated)			8809
Total Steel (Epoxy Coated)			24123

BILL OF MATERIALS FLOOR PHASE II (CONTINUED)

CONCRETE

Class "C" in Superstructure	295.2 m³

MISCELLANEOUS

Surface Seal	1578 m²
Field Drilled Holes in Concrete	1058 Each
12 Expansion Joint Sealing System	125 m
4-C.I. Roodway Drains, Type "SQ-A"@87.1 kg Each	348.4 kg
Drain Extensions	4 Each
Masonry Coating	648 m²

"A" Bars 11#13	"B" Bars 10#13	"C" Bars 10#13	"D" Bars 12#13	"E" Bars 11#13	"F" Bars 11#13
1x1308a	1x1318a	1x1327a	1x1336a	1x1347a	1x1358a
1x1309a	1x1319a	1x1328a	1x1337a	1x1348a	1x1359a
1x1310a	1x1320a	1x1329a	1x1338a	1x1349a	1x1360a
1x1311a	1x1321a	1x1330a	1x1339a	1x1350a	1x1361a
1x1312a	1x1322a	1x1331a	1x1340a	1x1351a	1x1362a
1x1313a	1x1323a	1x1332a	1x1341a	1x1352a	1x1363a
1x1314a	1x1324a	1x1333a	1x1342a	1x1353a	1x1364a
1x1315a	1x1325a	1x1334a	1x1343a	1x1354a	1x1365a
1x1316a	2x1326a	2x1335a	1x1344a	1x1355a	1x1366a
2x1317a			1x1345a	1x1356a	1x1367a
			2x1346a	1x1357a	1x1368a
15.50 m	21.34 m	23.54 m	28.32 m	16.06 m	16.00 m

⊕ A.S.T.M. A615M, Grade 420

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

Butler Fairman Scufert CONSULTING ENGINEERS
8450 WESTFIELD BLVD., SUITE 300 INDIANAPOLIS, IN 46240 317 713-4615 FAX 317 713-4616
509 WEST 84TH DRIVE, SUITE G MERRILLVILLE, IN 46410 219 769-2333 FAX 219 769-2377

JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: D. SHEETZ DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

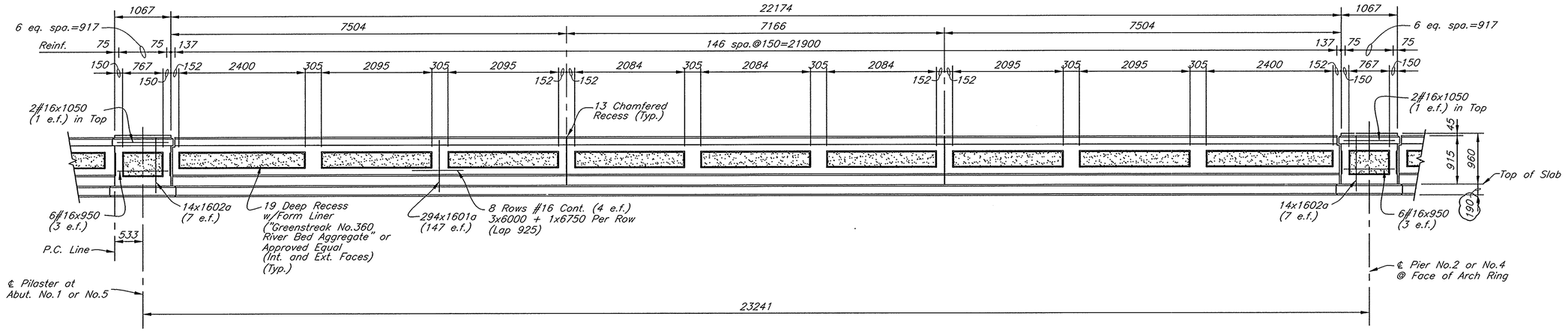
INDIANA DEPARTMENT OF TRANSPORTATION

FLOOR DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	16 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

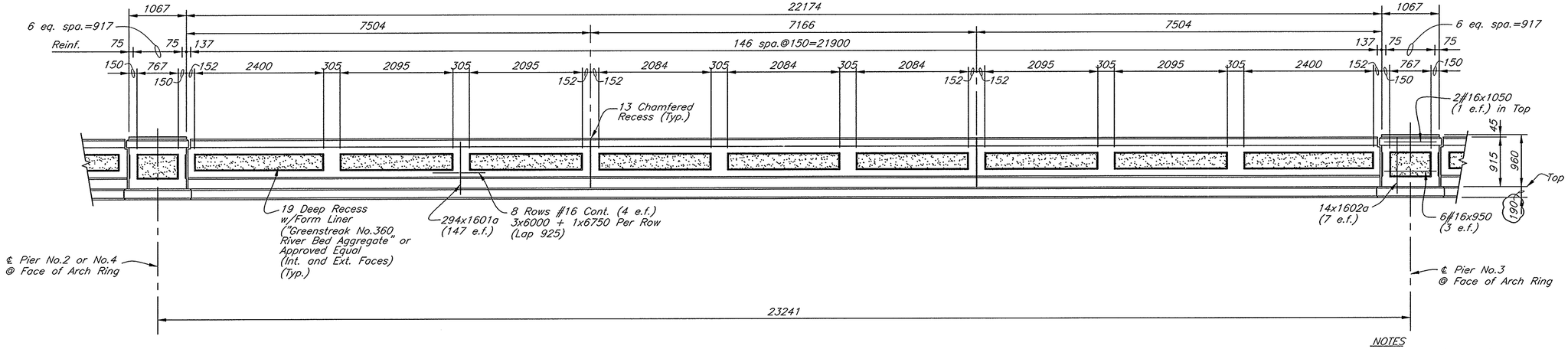
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ELEVATION
SPAN "A" (SHOWN)
SPAN "D" (OPP. HAND)

Scale=1:40



ELEVATION
SPAN "B" (SHOWN)
SPAN "C" (OPP. HAND)

Scale=1:40

NOTES

East Rails (Phase I) are shown, West Rails (Phase II) are same.

See Sheets 18 and 19 for Sections and Additional Details.

See Sheet 19 for Bar Bending Details and Bills of Materials.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4-6-04 SIDEWALK THICKNESS

Butler Fairman Seufert
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JEFFREY S. NICHOLSON
REGISTERED
No. 19900078
STATE OF INDIANA
PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: D. SHEETZ DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

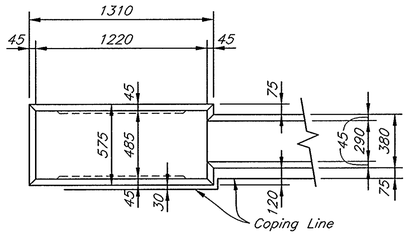
INDIANA
DEPARTMENT OF TRANSPORTATION

RAILING DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	I-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	17 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

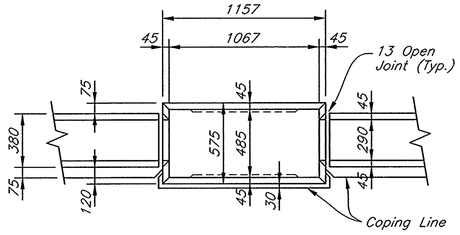
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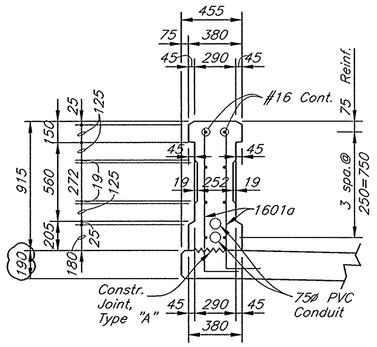


PLAN

Note: Curvature of Wing not shown for Clarity.

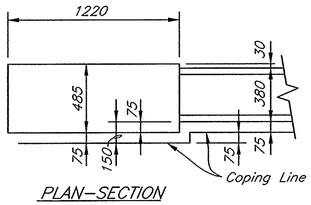


PLAN

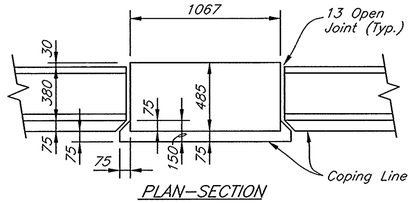


TYPICAL RAIL SECTION

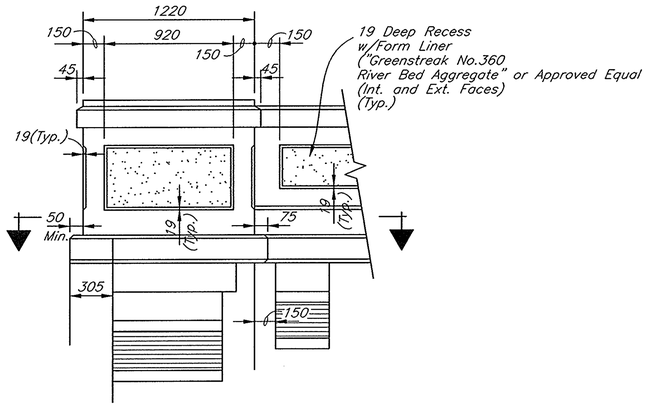
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PLAN–SECTION



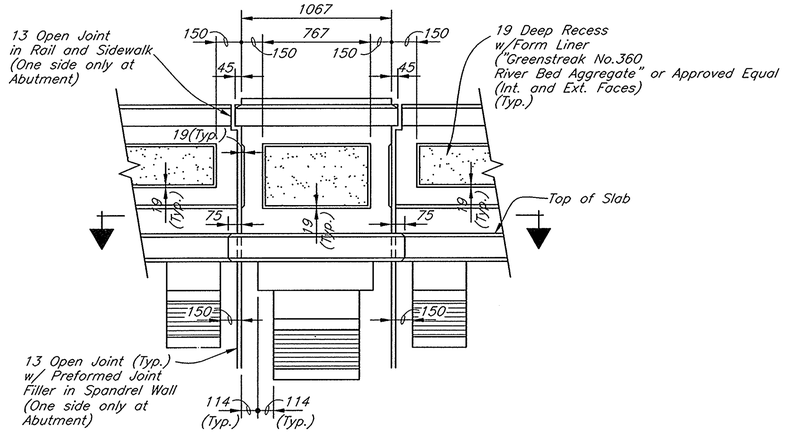
PLAN–SECTION



ELEVATION

TYPICAL PILASTER DETAILS AT ENDS OF WINGS

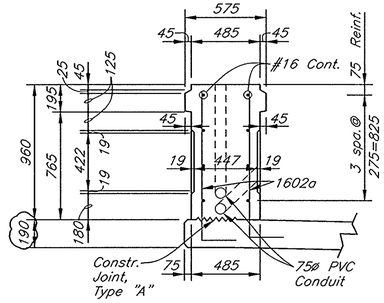
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ELEVATION

TYPICAL PILASTER DETAILS AT ABUTMENTS AND PIERS

Scale=1:20



TYPICAL PILASTER SECTION

Scale=1:20

NOTES

See Sheets 13 and 14 for Bracket Details.
See Sheet 18–1 for Conduit and Lighting Details.
See Sheet 19 for Bar Bending Details and Bills of Materials.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4–6–04 SIDEWALK THICKNESS

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JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: D. SHEETZ DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

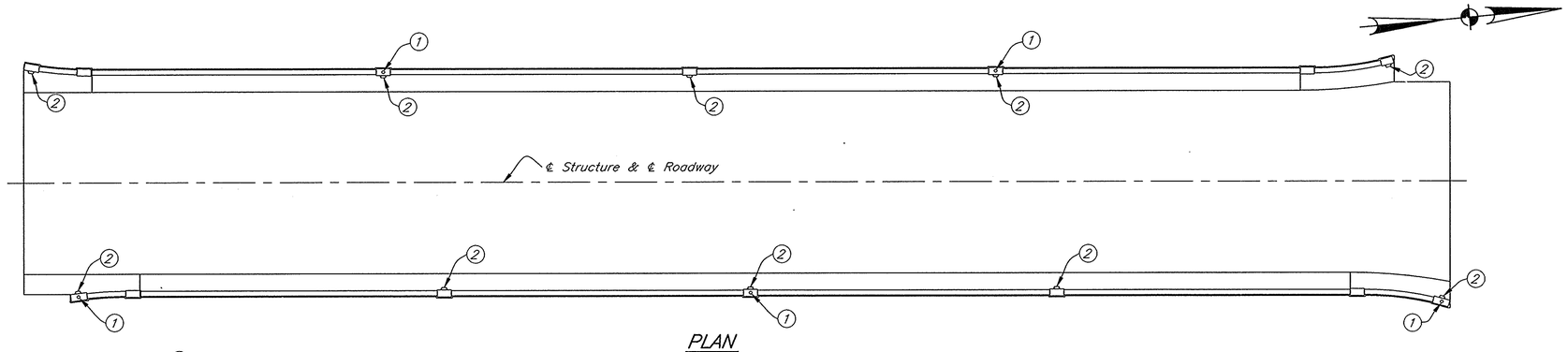
INDIANA DEPARTMENT OF TRANSPORTATION

RAILING DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	18 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS no. 3841.0201

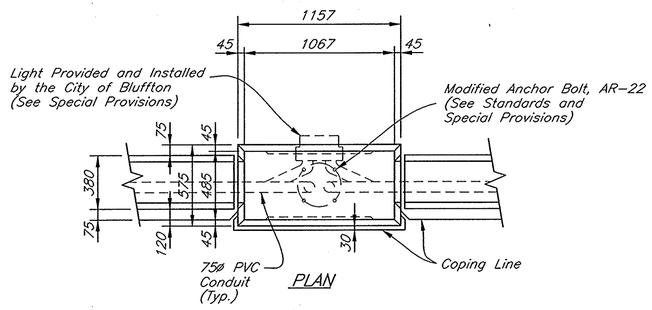
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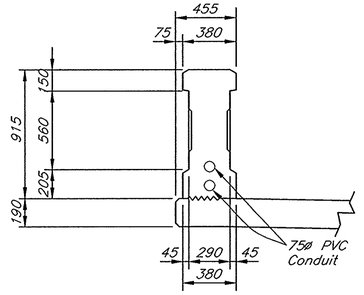
PLAN
Scale=1:200

(1) Light Standard Provided and Installed by the City of Bluffton. Location of Light Standards to be coordinated with the City of Bluffton. (See Special Provisions)

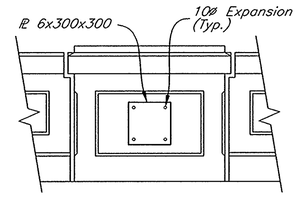
(2) Side Rail Light Provided and Installed by the City of Bluffton. Lighting Cavity Dimensions and locations to be provided by the City of Bluffton and Approved by the Engineer. (See Special Provisions)



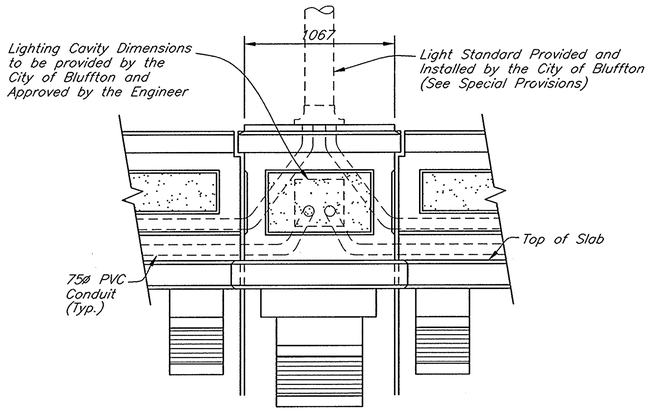
PLAN



TYPICAL RAIL SECTION
Scale=1:20

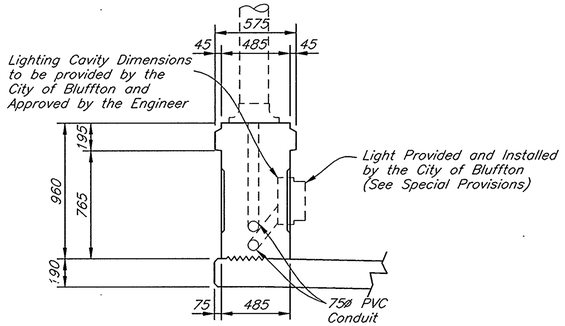


SIDE RAIL LIGHT
TEMPORARY STEEL COVER
Scale=1:20



ELEVATION

TYPICAL PILASTER CONDUIT DETAILS
Scale=1:20



TYPICAL PILASTER SECTION
Scale=1:20

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4-6-04 ADDED THIS SHEET

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RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: D. SHEETZ DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

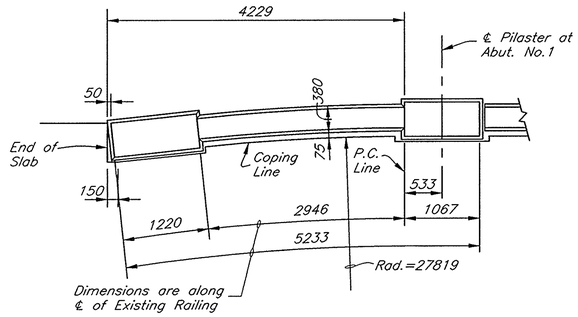
INDIANA DEPARTMENT OF TRANSPORTATION

RAILING DETAILS

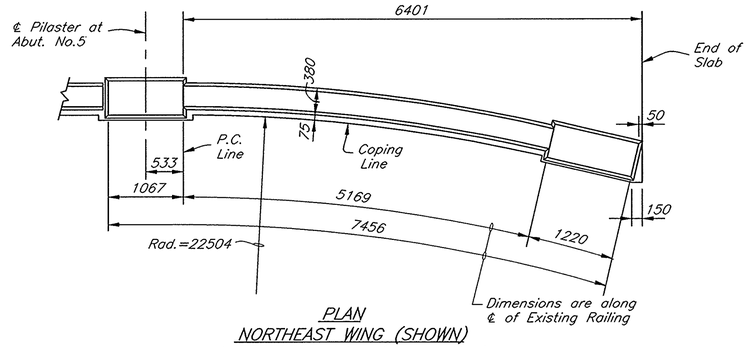
HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
18-1	OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS NO. 3841.0201

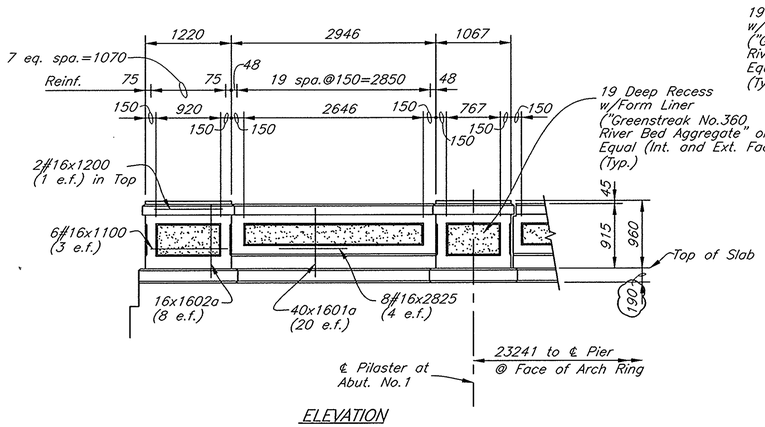
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PLAN
SOUTHEAST WING (SHOWN)
SOUTHWEST WING (OPP. HAND)



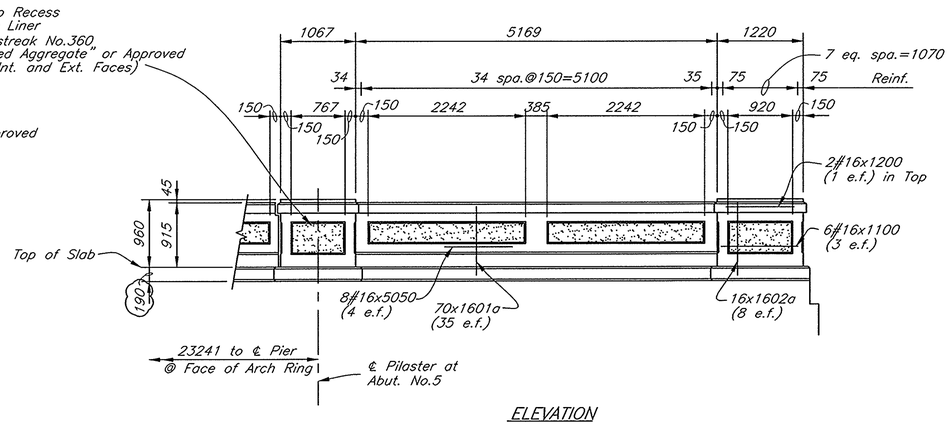
PLAN
NORTHEAST WING (SHOWN)
NORTHWEST WING (OPP. HAND)



ELEVATION

ABUTMENT NO.1

Scale=1:40



ELEVATION

ABUTMENT NO.5

Scale=1:40

BILL OF MATERIALS
CONCRETE RAILING
PHASE I
(PHASE II SAME UNLESS NOTED)

REINFORCING STEEL

Mark or Size	No. of Bars	Length (mm)	Mass (kg)
1601a	1286	(1250)	
1602a	102	(1300)	
#16	32	6750	
#16	96	6000	
#16	8	5050	
#16	8	2825	
#16	4	1200	
#16	12	1100	
#16	10	1050	
#16	30	950	
⊕ Total Steel (Epoxy Coated)			(4116)

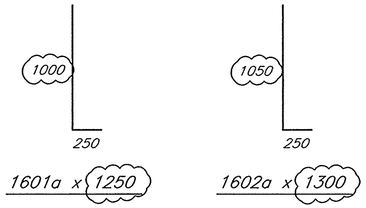
CONCRETE

Class "C" in Railing	31.9 m^3

MISCELLANEOUS

Surface Seal	298 m^2
Masonry Coating	298 m^2
75ø PVC Conduit	216 m
Anchor Rods	
Phase I	12 Each
Phase II	8 Each

⊕ A.S.T.M. A615M, Grade 420



BAR BENDING DETAILS

Not to Scale

NOTES
East Rails (Phase I) are shown, West Rails (Phase II) are same.
See Sheets 18 and 19 for Sections and Additional Details.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4-6-04 SIDEWALK THICKNESS, BAR BENDS AND BILL OF MATERIALS

Butler Fairman Scufert
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509 WEST 84TH DRIVE, SUITE G
MERRILLVILLE, IN. 46410
219 769-2333
FAX 219 769-2377

JEFFREY S. NICHOLSON
REGISTERED
No. 19900078
STATE OF INDIANA
PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: D. SHEETZ DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

INDIANA DEPARTMENT OF TRANSPORTATION

RAILING DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	19 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS NO. 3841.0201

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STRUCTURE QUANTITIES

ITEM	CONCRETE			COMP. AGGR. BASE TYPE "O"	REINF. CONC. BRIDGE APPR. 250 mm	CONC. RAILING CLASS C	REINF. STEEL (PLAIN)	REINF. STEEL (EPOXY COATED)	EST. WEIGHT STR. STEEL	STEEL ENCASED CONC. PILES	STEEL H-PILES	PILE TIPS	PIPE, END BENT DRAIN, 150mm	CAST IRON DRAIN TYPE "SQ-A"	DRAIN EXTENSIONS	75ø PVC CONDUIT	THREADED TIE BAR ASSEMBLIES (EPOXY COATED)	EXP. JOINT SEALING SYSTEM	STRUCTURAL MEMBERS	EST. AREA SURFACE SEAL	EST. AREA MASONRY COATING	FIELD DRILLED HOLES IN CONCRETE	ANCHOR RODS	BRIDGE DECK OVERLAY	ADDITIONAL BRIDGE DECK OVERLAY	SURFACE MILLING	
	CLASS C IN SUPERSTR.	CLASS A IN SUBSTR.	CLASS B IN FOOTING																CONCRETE I-BEAM							
	m^3	m^3	m^3	Mg	m^2	m^3	kg	kg	kg	m	m	EACH	m	kg	EACH	m	EACH	m	m	m^2	m^2	EACH	EACH	m^2	m^3	m^2	
PHASE I																										
SUPERSTRUCTURE																										
Spans "A" thru "D"	285.8							24011						348.4	4		544	129		1552	648	1032				
RAILING																										
Spans "A" thru "D"						31.9		4116								216				298	298		12			
SUB-TOTALS	285.8					31.9		28127						348.4	4	216	544	129		1850	946	1032	12			
PHASE II																										
SUPERSTRUCTURE																										
Spans "A" thru "D"	295.2							24123						348.4	4			125		1578	648	1058				
RAILING																										
Spans "A" thru "D"						31.9		4116								216				298	298		8			
SUB-TOTALS	295.2					31.9		28239						348.4	4	216		125		1876	946	1058	8			
TOTALS	581.0					63.8	⊕	⊕56366						696.8	8	432	544	254		3726	1892	2090	20			

⊕ A.S.T.M. A615M, Grade 420

REVISED 4-20-04 QUANTITIES

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FAX 219 769-2377

JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: J. NICHOLSON DRAWN: D. SHEETZ
CHECKED: D. SHEETZ CHECKED: J. NICHOLSON

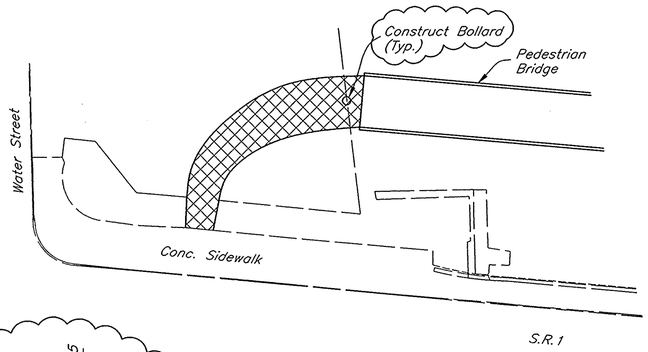
INDIANA DEPARTMENT OF TRANSPORTATION

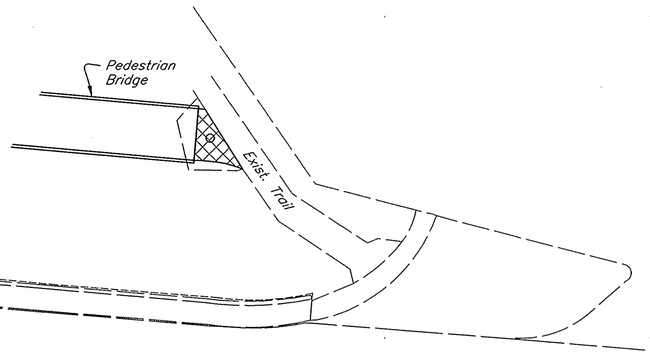
BRIDGE SUMMARY

HORIZONTAL SCALE	FILE
NONE	1-90-230B
VERTICAL SCALE	DESIGNATION
NONE	9406690
SURVEY BOOK	SHEET
	20 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BRS no. 3841.0201

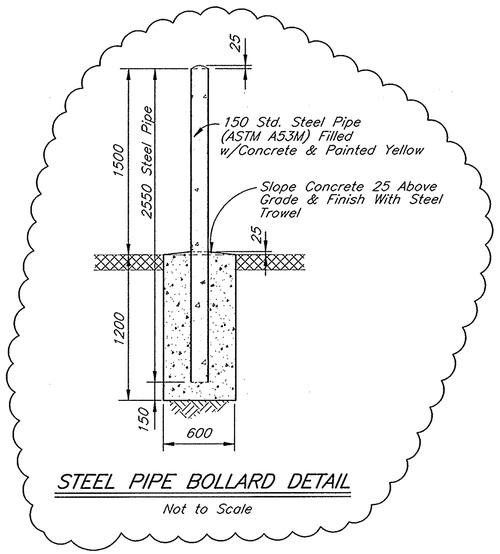
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Indicates 21 Mg of 360 kg/m² HMA for Approaches, Type C

PLAN
Scale=1:200



STEEL PIPE BOLLARD DETAIL
Not to Scale

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4-6-04 CONSTRUCT BOLLARD

Butler Fairman Scufert
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JEFFREY S. NICHOLSON
REGISTERED
No. 19900078
STATE OF INDIANA
PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: D. SHEETZ DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

INDIANA DEPARTMENT OF TRANSPORTATION

APPROACH DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	21 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

SHEET NO. 3841.02\01

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LOG OF TEST BORING

Project: Pedestrian Bridge
Location: Wells County, Indiana
Client: INDOT
4105 West 99th Street Carmel, Indiana 46032
317-875-7040/317-870-0314(Fax)

Boring No. TB-1
Surface Elevation 249.0 m
Proj. # BR-231-2( )
AW Proj. # 03IN0656
Sheet 1 of 1

Str. No. N/A Station 1+953 Offset 0 Line "C"
Datum USGS Weather Sunny Temperature 80 F Inspector T. Coffey

SAMPLE No.	Type	Rec (%)	Blow Count	Depth	DESCRIPTION/CLASSIFICATION and REMARKS	qu (tsf)	δ (pcf)	W (%)	LL (%)	PL (%)	PI (%)
					EL. 249.0						
1SS	SS	0	9-9-7		Asphalt (5") and Crushed Stone (Fill) (Visual)						
					EL. 248.1						
2SS	SS	60	4-8-8								
3SS	SS	100	12-8-9								
4SS	SS	50	3-7-7		Brown, Moist, Stiff to Very Stiff, Clay Loam A-4 Test 02			18.8			
					EL. 244.7						
5SS	SS	90	5-9-13	5	Brown, Moist, Very Stiff to Hard, Clay A-6 (7) Test 01 pH = 7.42 Driving on Boulder at 6.1 m			22.3	31.0	18.3	12.7
6SS	SS	75	3-13 50/4								
7SS	SS	0	50/3								
					EL. 241.7						
					Gray, Highly Weathered Limestone (Visual)						
					EL. 240.2						
				10	Auger Refusal on Apparent Bedrock at 8.8 meters						

WATER LEVEL OBSERVATIONS

While Drilling Q 6.1 m Upon Completion of Drilling ▼ 5.5 m
Time After Drilling 24 Hrs.
Depth to Water ▽ 5.2 m ▼ ▽ ▼
Depth to Cave in

The stratification lines represent the approximate boundary between soil types and the transition may be gradual.

GENERAL NOTES

Start 7/16/03 End 7/16/03 Rig D-50
Driller Steve Editor D. Harness
Remarks Boring backfilled and patched with asphalt after 24 hour groundwater readings. Bag sample taken.

LOG OF TEST BORING

Project: Pedestrian Bridge
Location: Wells County, Indiana
Client: INDOT
4105 West 99th Street Carmel, Indiana 46032
317-875-7040/317-870-0314(Fax)

Boring No. TB-2
Surface Elevation 243.5 m
Proj. # BR-231-2( )
AW Proj. # 03IN0656
Sheet 1 of 1

Str. No. N/A Station 1+986.3 Offset 0 Line "C"
Datum USGS Weather Sunny Temperature 75 F Inspector T. Coffey

SAMPLE No.	Type	Rec (%)	Blow Count	Depth	DESCRIPTION/CLASSIFICATION and REMARKS	qu (tsf)	δ (pcf)	W (%)	LL (%)	PL (%)	PI (%)
					EL. 243.5						
					Water (Visual) Water elevation 243.5m						
					Ground Surface elevation 242.3 m						
					Brown, Wet, Very Soft, Silty Clay (Muck) (Visual)						
					EL. 241.1						
1SS	SS	100	2-7-19		Brown and Gray, Wet, Dense, Gravel with Sand (Bedrock Fragments) (Visual)						
					EL. 239.7						
2SS	SS	0	50/1								
1RC	RC	98			Auger Refusal at 3.8 m Rock Core Initiated at 3.8 m REC. = 98% RQD. = 88% Dark Gray, SSghtly Vuggy, Argillaceous Limestone (Visual)						
					EL. 238.2						
				10	Boring Terminated at 5.3 meters						

WATER LEVEL OBSERVATIONS

While Drilling Q N/A Upon Completion of Drilling ▼ N/A
Time After Drilling 24 Hrs.
Depth to Water ▽ N/A ▼ ▽ ▼
Depth to Cave in

The stratification lines represent the approximate boundary between soil types and the transition may be gradual.

GENERAL NOTES

Start 7/29/03 End 7/29/03 Rig Skid
Driller Winkler Editor D. Harness
Remarks Lull and barge required to access and perform boring.

LOG OF TEST BORING

Project: Pedestrian Bridge
Location: Wells County, Indiana
Client: INDOT
4105 West 99th Street Carmel, Indiana 46032
317-875-7040/317-870-0314(Fax)

Boring No. TB-3
Surface Elevation 243.7 m
Proj. # BR-231-2( )
AW Proj. # 03IN0656
Sheet 1 of 1

Str. No. N/A Station 2+032.8 Offset 0 Line "C"
Datum USGS Weather Sunny Temperature 80 F Inspector T. Coffey

SAMPLE No.	Type	Rec (%)	Blow Count	Depth	DESCRIPTION/CLASSIFICATION and REMARKS	qu (tsf)	δ (pcf)	W (%)	LL (%)	PL (%)	PI (%)
					EL. 243.7						
1SS	SS	90	2-4-5		Topsoil (Visual) Brown, Moist, Medium Stiff to Very Stiff, Clay Loam A-4 Test 02						
					EL. 242.2						
2SS	SS	10	6-9-17		Gray and Brown, Wet, Dense, Gravel with Sand (Bedrock Fragments) (Visual)						
					EL. 241.1						
3SS	SS	60	15-15-25		Auger Refusal at 2.6 m Rock Core Initiated at 2.6 m REC. = 88% RQD. = 30% Dark Gray, Slightly Vuggy, Argillaceous Limestone (Visual) Weak at bedding planes (2-3")						
					EL. 239.6						
1RC	RC	88			REC. = 100% RQD. = 27% Dark Gray, Slightly Vuggy, Argillaceous Limestone (Visual) Weak at bedding planes (2-3")						
2RC	RC	100		5							
					EL. 238.1						
				10	Boring Terminated at 5.6 meters						

WATER LEVEL OBSERVATIONS

While Drilling Q 2.1 m Upon Completion of Drilling ▼ 1.4 m
Time After Drilling 24 Hrs
Depth to Water ▽ 1.2 m ▼ ▽ ▼
Depth to Cave in

The stratification lines represent the approximate boundary between soil types and the transition may be gradual.

GENERAL NOTES

Start 7/31/03 End 7/31/03 Rig Skid
Driller Winkler Editor D. Harness
Remarks

LOG OF TEST BORING

Project: Pedestrian Bridge
Location: Wells County, Indiana
Client: INDOT
4105 West 99th Street Carmel, Indiana 46032
317-875-7040/317-870-0314(Fax)

Boring No. TB-4
Surface Elevation 248.7 m
Proj. # BR-231-2( )
AW Proj. # 03IN0656
Sheet 1 of 1

Str. No. N/A Station 2+060 Offset 4.6m Rt. Line "C"
Datum USGS Weather Sunny Temperature 80 F Inspector T. Coffey

SAMPLE No.	Type	Rec (%)	Blow Count	Depth	DESCRIPTION/CLASSIFICATION and REMARKS	qu (tsf)	δ (pcf)	W (%)	LL (%)	PL (%)	PI (%)
					EL. 248.7						
					Topsoil (Visual)						
1SS	SS	20	3-5-6								
2SS	SS	30	6-4-9					13.8			
3SS	SS	30	4-5-6								
4SS	SS	60	4-5-8		Brown, Moist, Stiff to Very Stiff, Clay Loam A-4 (3) Test 02 pH = 6.82			24.0			
5SS	SS	100	4-8-11	5				19.5	28.9	19.4	9.5
					EL. 243.1						
6SS	SS	75	3-17-14		Brown, Wet, Dense, Sand and Gravel (Visual)						
					EL. 241.4						
7SS	SS	90	18-24-26		Gray, Highly Weathered Limestone (Visual)						
					EL. 240.5						
				10	Auger Refusal on Apparent Bedrock at 8.2 meters						

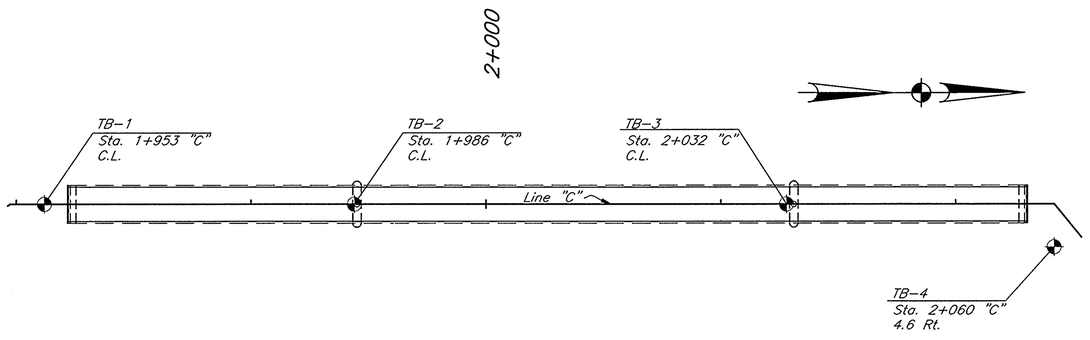
WATER LEVEL OBSERVATIONS

While Drilling Q 7.6 m Upon Completion of Drilling ▼ 3.8 m
Time After Drilling 24 Hrs.
Depth to Water ▽ 4.6 m ▼ ▽ ▼
Depth to Cave in

The stratification lines represent the approximate boundary between soil types and the transition may be gradual.

GENERAL NOTES

Start 7/16/03 End 7/16/03 Rig D-50
Driller Steve Editor D. Harness
Remarks Boring backfilled after 24 hour groundwater readings.



PLAN
Not to Scale

DESIGN LOADING FOR GEOTECHNICAL TESTING				
SUBSTRUCTURE	BENT NO.1	PIER NO.2	PIER NO.3	BENT NO.4
DESIGN LOAD	355 kN	——	——	355 kN
FACTOR OF SAFETY	2.5	——	——	2.5
FACTORED DESIGN LOAD	890 kN	——	——	890 kN
FRICTION IN SCOUR ZONE	N/A	——	——	N/A
DOWN DRAG FRICTION	N/A	——	——	N/A
ULTIMATE LOAD	890 kN	——	——	890 kN
TESTING METHOD	by Formula Std. Spec. 701			

STANDARD PENETRATION TEST
Driving 50.8 O.D. Sampler 450 with 63.5 Kg Hammer falling 760. Count made at 150 intervals. First 150 for seating sampler. "N" is the Sum of the Hammer Blows of the Second and Third 150 intervals of an 450 Drive.

Note:
The entire Geotechnical Report developed by Alt & Witzig Engineering, Inc. will be made available to the Contractor upon request.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

BF&S
Butler Fairman Seufert
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MERRILLVILLE, IN. 46410
219 769-2333
FAX 219 769-2377

JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: ____ DRAWN: L. OSBORNE
CHECKED: ____ CHECKED: J. NICHOLSON

INDIANA DEPARTMENT OF TRANSPORTATION

SOIL BORINGS

HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	22 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS NO. 3841.0201

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All Topo is Referenced from Line "A" unless otherwise noted

EQN:
P.O.T. Sta. 1+950.327 "A", 20.472 Lt.
= P.O.T. Sta. 1+950.000 "C"

℄ Brg. Bent No.1
Sta.
El. 249.250
*31.760 m

℄ Pier No.2
Sta.
El. 248.407
*44.928 m

℄ Pier No.3
Sta.
El. 248.407
*44.293 m

℄ Brg. Bent No.4
Sta.
El. 248.717

EQN:
P.O.T. Sta. 2+059.782 "A", 20.235 Lt.
= P.O.T. Sta. 2+060.000 "C"

Sta. 2+120.326 "A", 2.169m Lt.
Mag Nail Set

EQN:
P.I. Sta. 2+119.793 "A"
= P.I. Sta. 69+54.7 "A" Proj. 377

2+000

Manhole

Proposed Pedestrian Bridge

Line "C"

Line "A"

P.T. Sta. 1+967.311 "A"

P.O.T. Sta. 1+970.115 "A"

P.I. Sta. 1+932.240 "A"

P.O.T. Sta. 2+060.683 "A"

P.C. Sta. 2+071.421 "A"

B.M. #1 Elev. 249.26m

Light under bridge

Existing Bridge

2-Sty Br. Bldg.

2-Sty Fr. Hse.

WATER STREET (Bit.Mat.)

STATE ROAD #1 (Bit.Mat.)

Comm. Dr. (Conc.)

(Bit.Mat.) Remove Old Abutment and Wingwall Adjacent to Existing SR1 Structure

(Grass)

(Conc.)

(C-Stn.)

Sanitary Sewer

CURVE DATA

P.I. Sta. 2+119.793 "A"
Δ= 10°00'00" Rt.
R= 552.894m
T= 48.372m
L= 96.498m
E= 2.112m

CURVE DATA

P.I. Sta. 1+932.240"A"
Δ= 6°20'00" Rt.
R= 635.209m
T= 35.143m
L= 70.214m
E= 0.971m

*From Field Measurements by Primco

REVISED 2-10-05: REVISED SPAN LENGTHS, ELEVATIONS AND GRADES

PVI STA =
PVI ELEV = 249.250

PVI STA =
PVI ELEV = 248.407

PVI STA =
PVI ELEV = 248.407

PVI STA =
PVI ELEV = 248.717

2.65%

Proposed Grade along Line "C"

0.00%

+1.28%

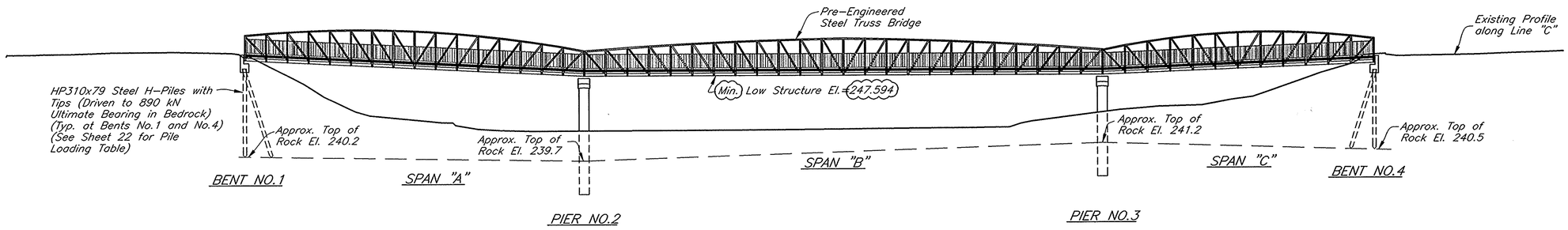
Drop
El. 247.132

(Min.) Low Chord El. 247.594

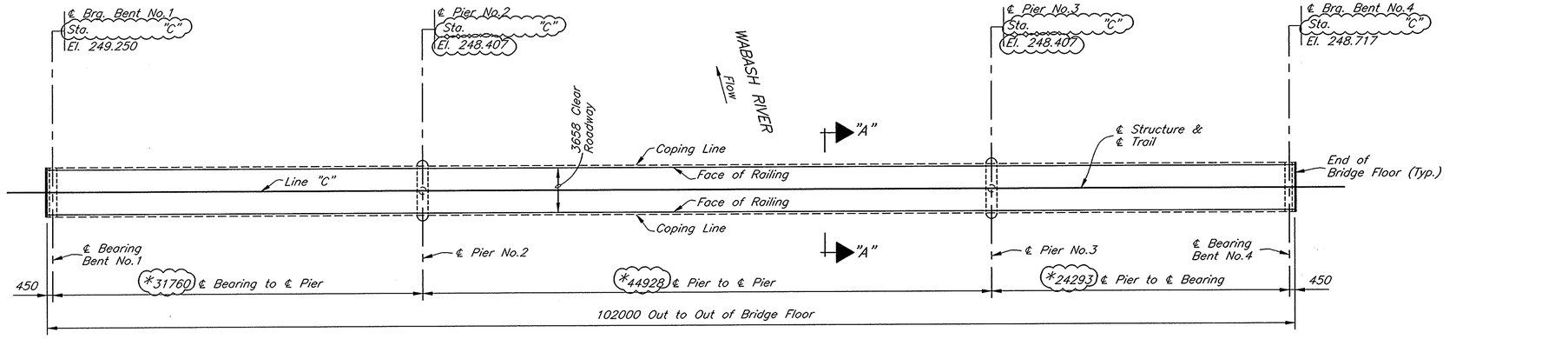
Existing Grade along Line "C"

| | | |

STRUCTURE TO BE BUILT ON A (-2.65%) GRADE, A 0.0% GRADE, AND A (+1.28%) GRADE



ELEVATION
(Not to Scale)



*From Field Measurements by Primco

PLAN
(Not to Scale)

NOTE
See Sheet 25 for Section "A-A".

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 2-10-05: REVISED SPAN LENGTHS, ELEVATIONS AND GRADES

Butler Fairman Scufert
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JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: J. SERA DRAWN: J. EVANS
CHECKED: J. NICHOLSON CHECKED: J. SERA

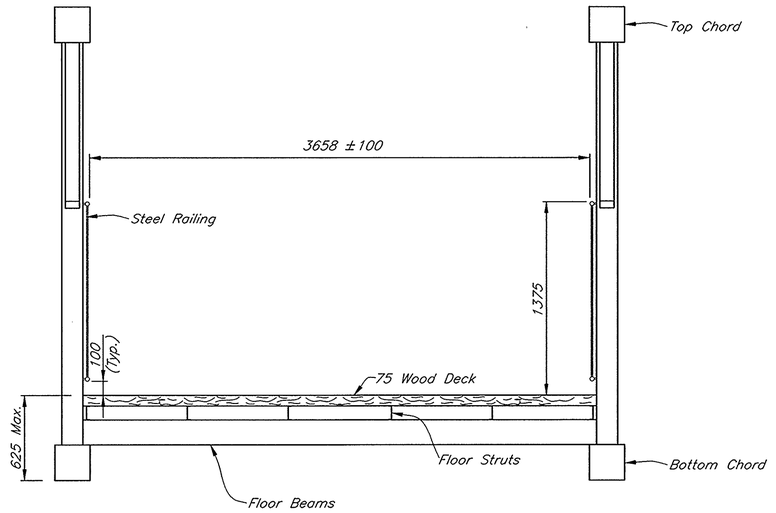
INDIANA DEPARTMENT OF TRANSPORTATION

GENERAL PLAN

HORIZONTAL SCALE	FILE
AS NOTED	
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	24-A OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS No. 3841.0201

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SECTION "A–A"

Scale=1:20

GENERAL NOTES

No present structure at proposed bridge site.

See Standards for details of piles.

Tolerance in position of pile head maximum 50 mm horizontally.

Determine pile lengths by Article 701 of the Specifications.

Caissons shall extend into solid rock as noted.

Steel H–Piles with tips shall be driven to the Ultimate Bearing in rock.

Steel H–Pile Tips shall be Dougherty DFP Hard Bite H–776 or approved equal. See the Special Provisions.

Reinforcing Steel shall not be ordered until rock is uncovered.

Reinforcing steel covering shall be 50 mm.

Reinforcing steel shall be A.S.T.M. A615M, Grade 420.

Concrete shall be Class A in end bents and piers.

Continuous concrete pours shall be made between construction joints shown.

Chamfer exposed corners of concrete 25 mm unless noted.

All asphalt material required in this contract shall be included in the pay item "HMA for Approaches".

Excavation required for placement of Coarse Aggregate at the bridge end bents beyond the limits of Foundation Excavation Unclassified shall not be paid for directly but shall be included in the cost of other items.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4–6–04 GENERAL NOTES

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JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: J. SERA DRAWN: L. OSBORNE
CHECKED: J. NICHOLSON CHECKED: J. SERA

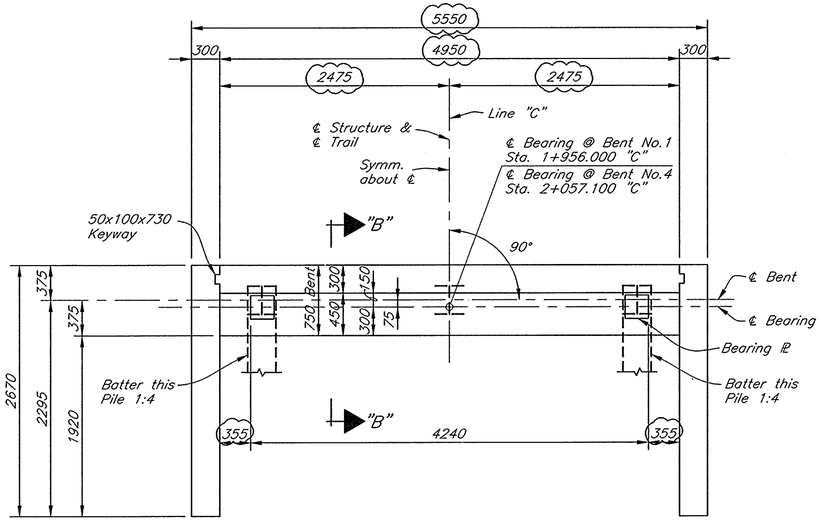
INDIANA DEPARTMENT OF TRANSPORTATION

GENERAL PLAN

HORIZONTAL SCALE	FILE
AS NOTED	
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	25 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS NO. 3841.0201

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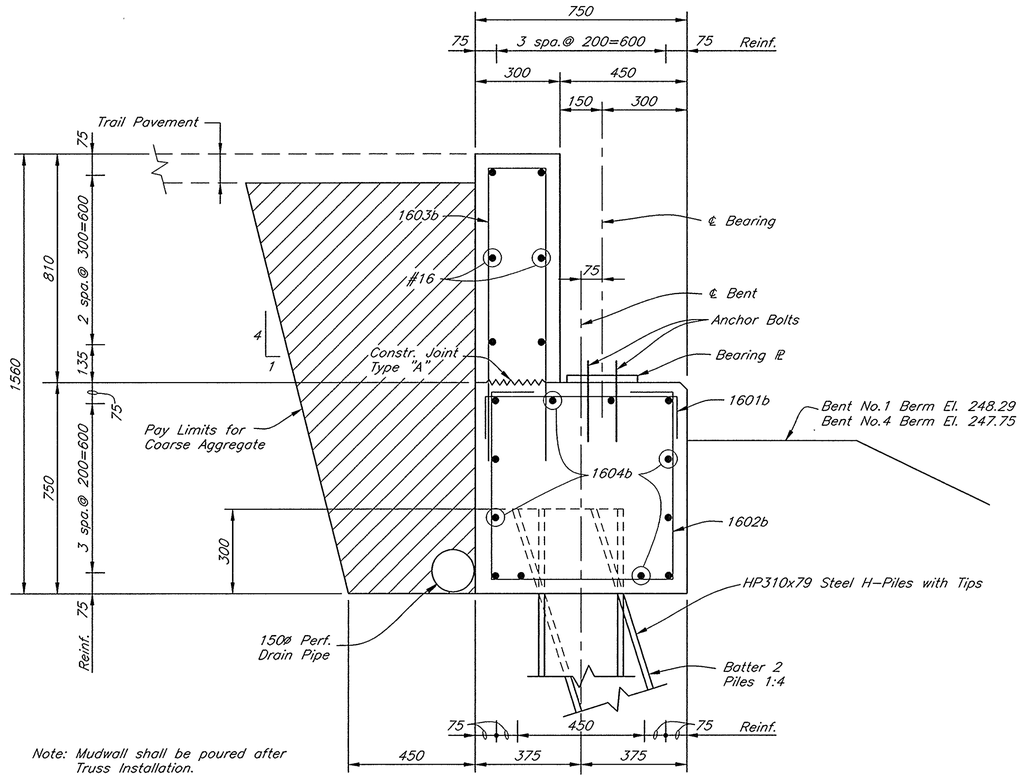
PLAN

Scale=1:30

NORTH-BENT NO.1

SOUTH-BENT NO.4

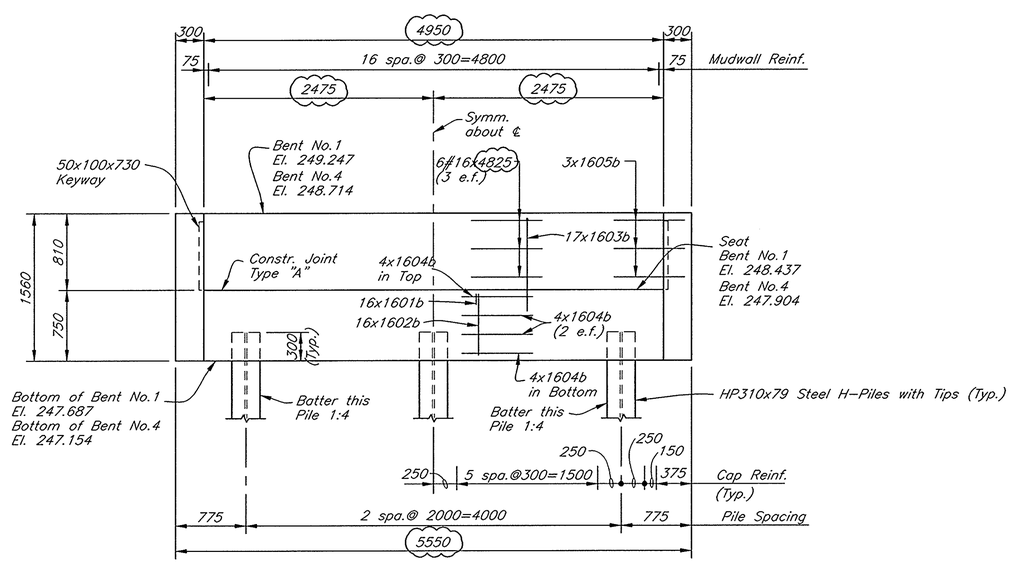
Note:
Seat elevations, bent dimensions and bearing assemblies are dependent upon truss configuration. Bent elevations and dimensions and bearing assemblies shall be designed by the Manufacturer of the Pre-engineered steel truss bridge. (See Special Provisions)



Note: Mudwall shall be poured after Truss Installation.

SECTION "B-B"

Scale=1:10



ELEVATION

Scale=1:30

NOTE

See Sheet 27 for Wingwall Details, Partial Plan-Section, Bar Bending Details and Bill of Materials.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4-6-04 BENT DIMENSIONS

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JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: J. ROZEBOOM DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

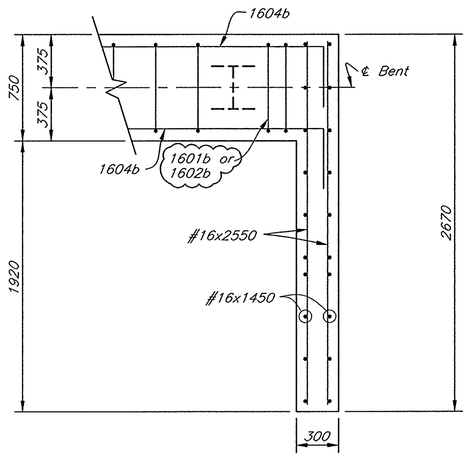
INDIANA DEPARTMENT OF TRANSPORTATION

BENTS NO.1 AND NO.4 DETAILS

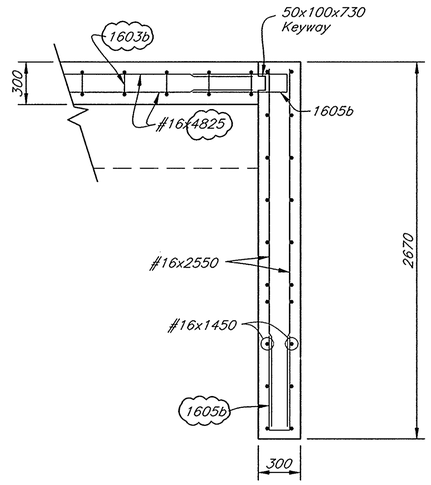
HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	26 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS No. 3841.0201

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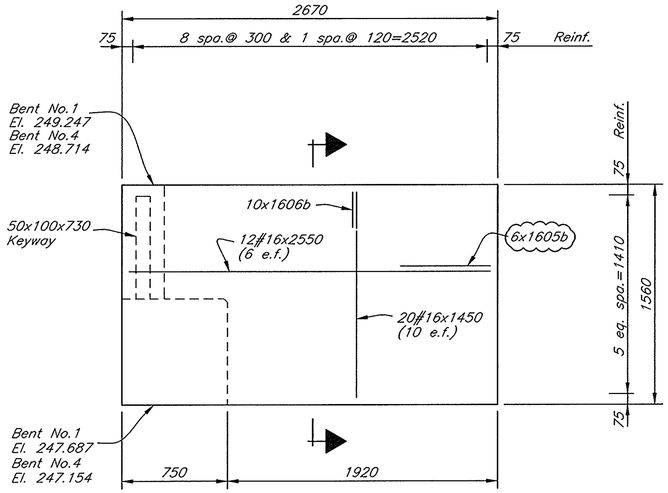


PLAN–SECTION
BELOW CONSTR. JOINT
Scale=1:20

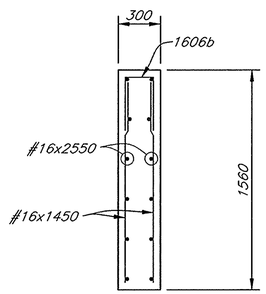


PLAN–SECTION
ABOVE CONSTR. JOINT
Scale=1:20

Note:
Seat elevations, bent dimensions and bearing assemblies are dependent upon truss configuration. Bent elevations and dimensions and bearing assemblies shall be designed by the Manufacturer of the Pre–engineered steel truss bridge. (See Special Provisions)



ELEVATION



SECTION

WINGWALL DETAILS
Scale=1:20

BILL OF MATERIALS
BENT NO.1
(BENT NO.4 SAME)

REINFORCING STEEL

Mark or Size	No. of Bars	Length (mm)	Weight (kg.)
1601b	16	960	
1602b	16	2260	
1603b	17	2480	
1604b	12	(5910)	
1605b	(18)	2220	
1606b	20	1320	
#16	6	(4825)	
#16	24	2550	
#16	40	1450	
⊕ Total Steel (Plain)			(588)

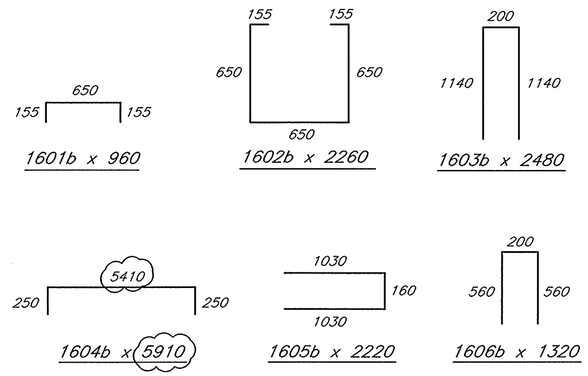
CONCRETE

Class "A" in Substructure	
Mudwall	1.2 m^3
Seat and Wings	5.3 m^3
Total	6.5 m^3

MISCELLANEOUS

HP310x79 Steel H–Piles	
Bent No.1 3 @ 7.8m Each	23.4 m
Bent No.4 3 @ 7.0m Each	21.0 m
Steel H–Pile Tips	3 Each

⊕ A.S.T.M. A615M, Grade 420



BAR BENDING DETAILS
Not to Scale

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4–6–04 REINFORCING AND BILL OF MATERIALS.

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JEFFREY S. NICHOLSON
REGISTERED
No. 19900078
STATE OF INDIANA
PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: _____ 2–17–04
DESIGN ENGINEER DATE

DESIGNED: J. ROZEBOOM DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

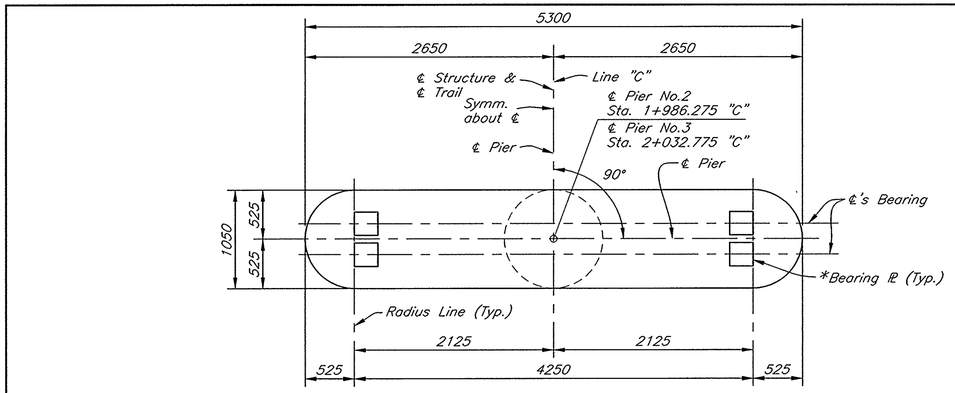
INDIANA
DEPARTMENT OF TRANSPORTATION

BENTS NO.1 AND NO.4 DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	1–90–230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	27 OF 30
CONTRACT	PROJECT
B–25747	BR–231–2( )

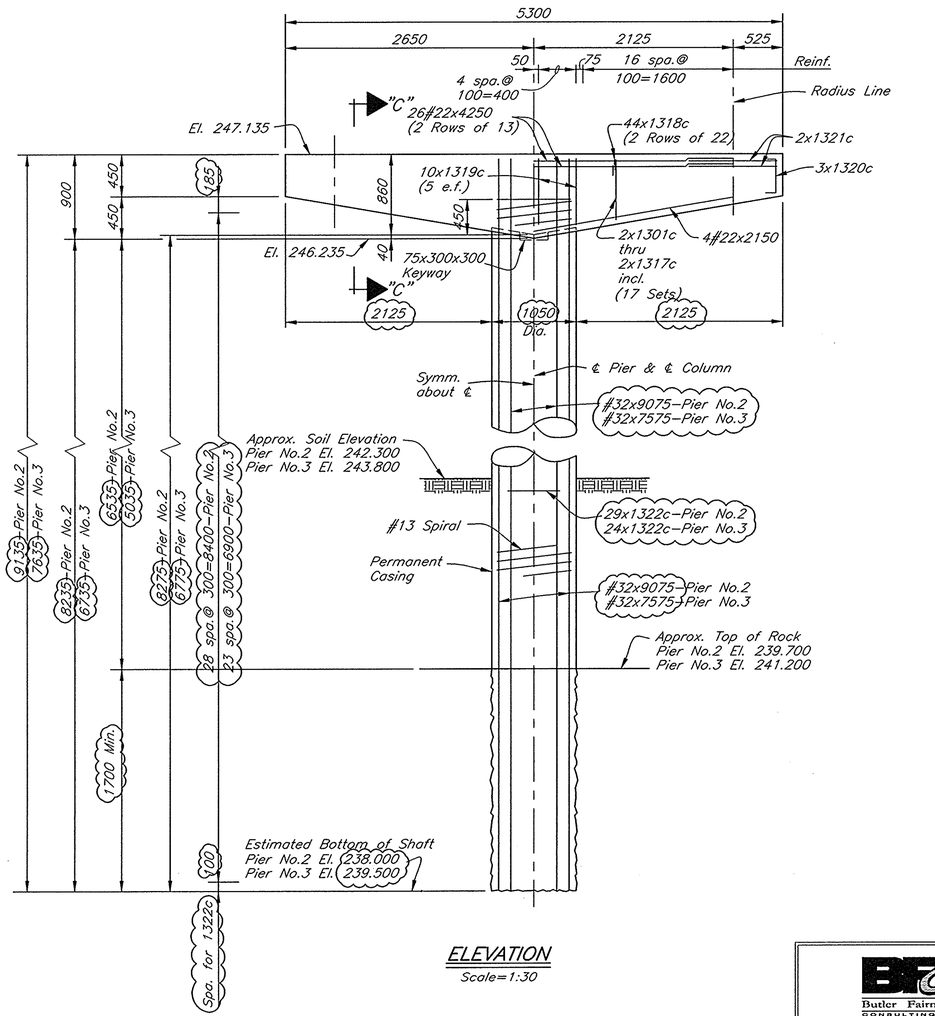
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PLAN

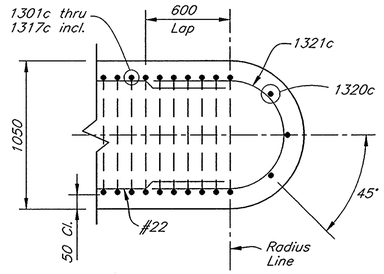
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ELEVATION

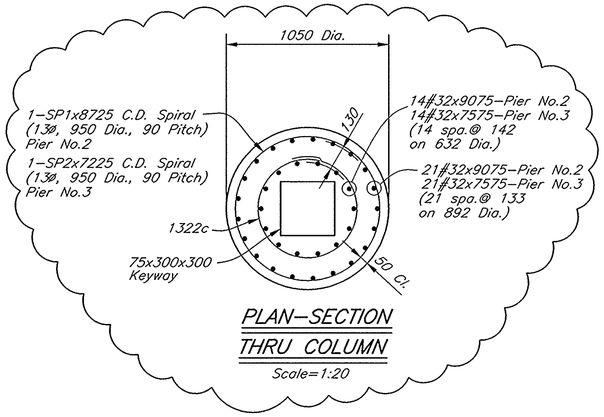
Scale=1:30

Note:
Pier elevations, pier dimensions and bearing assemblies are dependent upon truss configuration. Pier elevations and dimensions and bearing assemblies shall be designed by the Manufacturer of the Pre-engineered steel truss bridge. (See Special Provisions)



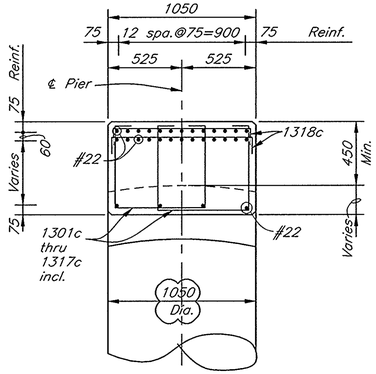
PLAN-SECTION
CAP NOSE

Scale=1:20



PLAN-SECTION
THRU COLUMN

Scale=1:20



SECTION "C-C"

Scale=1:20

NOTE
See Sheet 29 Bar Bending Details and Bill of Materials.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4-9-04 COLUMN HEIGHT AND REINFORCING

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No. 19900078
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PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: J. ROZEBOOM DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

INDIANA DEPARTMENT OF TRANSPORTATION

PIERS NO.2 AND NO.3 DETAILS

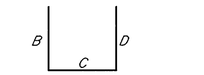
HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	28 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS No. 3841.0201

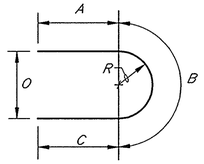
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BAR BENDING SCHEDULE

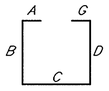
Mark	Length	Type	A	B	C	D	E	G	K	O	R
1301c	2280	S6	115	700	650	700		115			
1302c	2260	↓	↓	690	↓	690		↓			
1303c	2220			670		670					
1304c	2180			650		650					
1305c	2160			640		640					
1306c	2120			620		620					
1307c	2080			600		600					
1308c	2060			590		590					
1309c	2020			570		570					
1310c	1980			550		550					
1311c	1960			540		540					
1312c	1920			520		520					
1313c	1880			500		500					
1314c	1840			480		480					
1315c	1820			470		470					
1316c	1780	↓	↓	450	↓	450		↓			
1317c	1740	S6	115	430	650	430		115			
1318c	1180	17		115	350	115					
1319c	815	17		700	115	—					
1320c	580	17		115	350	115					
1321c	2650	10	600	1450	600					920	460
1322c	2640	T3							475	690	



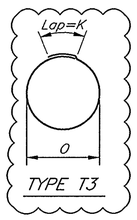
TYPE 17



TYPE 10



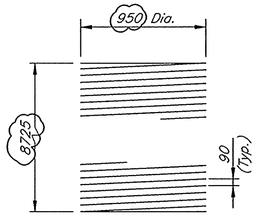
TYPE S6



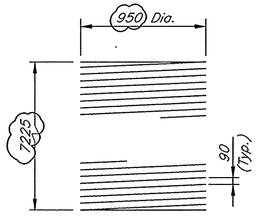
TYPE T3

BAR BENDING DETAILS

Not to Scale



13.0 ø x SP1
SPIRAL



13.0 ø x SP2
SPIRAL

SPIRAL NOTES

All Spirals shall be 950 Diameter Out to Out with a 90 Pitch and be made from 13.0 ø Cold Drawn Wire.
Each Spiral shall have 1 1/2 extra turns Top and Bottom.
Each Spiral shall be supported with 4 Spacers.
Cost of above to be included in Cost of Spirals.
Spirals shall be Spliced by lapping 1 1/2 turns.
No Additional Pay will be made for Splices.
Weights shown include 1 1/2 extra turns Top and Bottom.
Weights shown do not include Spacers or Splices.

BILL OF MATERIALS
PIER NO.2

REINFORCING STEEL

Mark or Size	No. of Bars	Length (mm)	Mass (kg)
#32	35	9075	
Total #32 (Plain)			2034
#22	26	4250	
#22	8	2150	
Total #22 (Plain)			388
1301c	4	2280	
1302c	4	2260	
1303c	4	2220	
1304c	4	2180	
1305c	4	2160	
1306c	4	2120	
1307c	4	2080	
1308c	4	2060	
1309c	4	2020	
1310c	4	1980	
1311c	4	1960	
1312c	4	1920	
1313c	4	1880	
1314c	4	1840	
1315c	4	1820	
1316c	4	1780	
1317c	4	1740	
1318c	88	1180	
1319c	20	815	
1320c	6	580	
1321c	4	2650	
1322c	29	2640	
Total #13 (Plain)			346
SP1	1	8725	
Total 13.0 ø Spirals			305
⊕ Total Steel (Plain)			3073

CONCRETE

Class "A" in Substructure	
Cap	3.7 m^3
Column	7.2 m^3
Total	10.9 m^3

BILL OF MATERIALS
PIER NO.3

REINFORCING STEEL

Mark or Size	No. of Bars	Length (mm)	Mass (kg)
#32	35	7575	
Total #32 (Plain)			1698
#22	26	4250	
#22	8	2150	
Total #22 (Plain)			388
1301c	4	2280	
1302c	4	2260	
1303c	4	2220	
1304c	4	2180	
1305c	4	2160	
1306c	4	2120	
1307c	4	2080	
1308c	4	2060	
1309c	4	2020	
1310c	4	1980	
1311c	4	1960	
1312c	4	1920	
1313c	4	1880	
1314c	4	1840	
1315c	4	1820	
1316c	4	1780	
1317c	4	1740	
1318c	88	1180	
1319c	20	815	
1320c	6	580	
1321c	4	2650	
1322c	24	2640	
Total #13 (Plain)			346
SP2	1	7225	
Total 13.0 ø Spirals			254
⊕ Total Steel (Plain)			2686

CONCRETE

Class "A" in Substructure	
Cap	3.7 m^3
Column	5.9 m^3
Total	9.6 m^3

⊕ A.S.T.M. A615M, Grade 420

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

REVISED 4-20-04 BAR BENDING AND BILLS OF MATERIALS

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JEFFREY S. NICHOLSON
REGISTERED
No. 19900078
STATE OF INDIANA
PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: J. ROZEBOOM DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

INDIANA DEPARTMENT OF TRANSPORTATION

PIERS NO.2 AND NO.3 DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	29 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS NO. 3841.0201

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STRUCTURE QUANTITIES

ITEM	CONCRETE			COMP. AGGR. BASE TYPE "O"	REINF. CONC. BRIDGE APPR. 250 mm	CONC. RAILING CLASS C	REINF. STEEL (PLAIN)	REINF. STEEL (EPOXY COATED)	EST. WEIGHT STR. STEEL	STEEL ENCASED CONC. PILES	STEEL H-PILES	PILE TIPS	PIPE, END BENT DRAIN, 150mm	CAST IRON DRAIN TYPE "SO-A"	PVC PIPE 150 mm	THRIE BEAM BRIDGE RAIL	THREADED TIE BAR ASSEMBLIES (EPOXY COATED)	EXP. JOINT TYPE SS	STRUCTURAL MEMBERS	EST. AREA SURFACE SEAL	EST. AREA MASONRY COATING	FIELD DRILLED HOLES IN CONCRETE	BARRIER DELINEATORS	BRIDGE DECK OVERLAY	ADDITIONAL BRIDGE DECK OVERLAY	SURFACE MILLING	CONCRETE BRIDGE RAILING TRANSITION TYPE TBC
	CLASS C IN SUPERSTR.	CLASS A IN SUBSTR.	CLASS B IN FOOTING																CONCRETE I-BEAM								
	m^3	m^3	m^3	Mg	m^2	m^3	kg	kg	kg	m	m	EACH	m	kg	m	m	EACH	m	m	m^2	m^2	EACH	EACH	m^2	m^3	m^2	EACH
SUBSTRUCTURE																											
Bent No. 1		6.5					588				23.4	3															
Pier No.2		10.9					3073																				
Pier No.3		9.6					2686																				
Bent No.4		6.5					588				21.0	3															
TOTALS		33.5					⊕ 6935	⊕			44.4	6															

⊕ A.S.T.M. A615M, Grade 420

REVISED 4-20-04 QUANTITIES

Butler Fairman & Seufert
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MERRILLVILLE, IN. 46410
219 769-2339
FAX 219 769-2377

JEFFREY S. NICHOLSON REGISTERED No. 19900078 STATE OF INDIANA PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 2-17-04
DESIGN ENGINEER DATE

DESIGNED: J. NICHOLSON DRAWN: D. SHEETZ
CHECKED: D. SHEETZ CHECKED: J. NICHOLSON

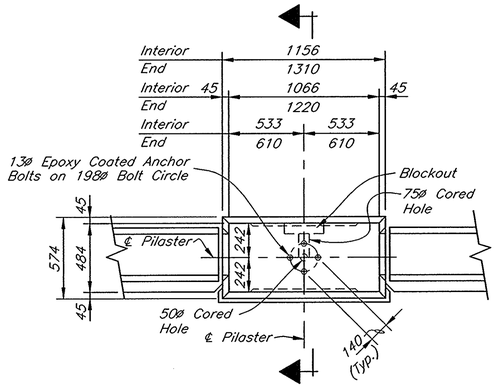
INDIANA DEPARTMENT OF TRANSPORTATION

BRIDGE SUMMARY

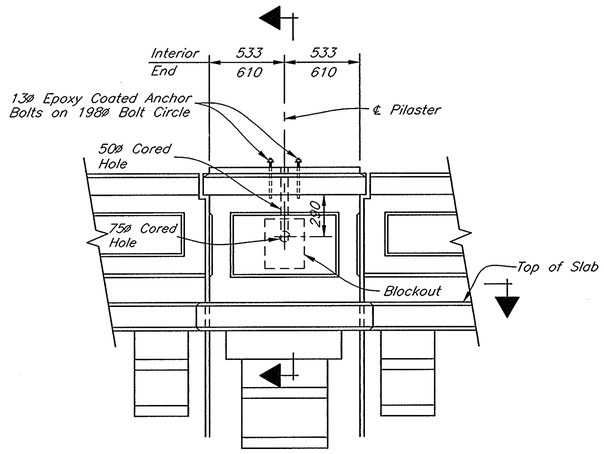
HORIZONTAL SCALE	FILE
NONE	1-90-230B
VERTICAL SCALE	DESIGNATION
NONE	9406690
SURVEY BOOK	SHEET
	30 OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS No. 3841.0201

R:\3841.021\01\AUTOCAD\BRIDGE\TRAILBRIDGE\0201BR02.dwg 04-21-04 AT 10:32 1 D.SHEETZ



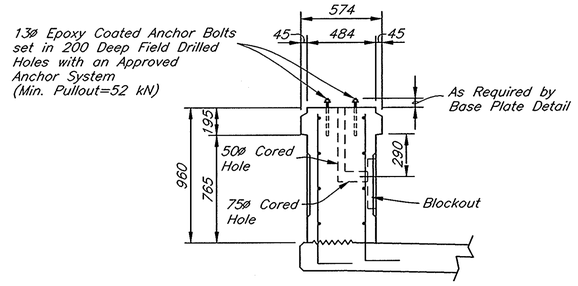
PLAN



ELEVATION

TYPICAL PILASTER DETAILS

Scale=1:20



TYPICAL PILASTER SECTION

Scale=1:20

City of Bluffton performed work.
Not part of Contract B-25747.

All Dimensions in Millimeters (mm) and all Stations and Elevations in Meters (m) unless otherwise noted

Butler Fairman Seufert
CONSULTING ENGINEERS
8450 WESTFIELD BLVD., SUITE 300
INDIANAPOLIS, IN. 46240
317 713-4615
FAX 317 713-4616
509 WEST 84TH DRIVE, SUITE G
MERRILLVILLE, IN. 46410
219 769-2333
FAX 219 769-2377

JEFFREY S. NICHOLSON
REGISTERED
No. 19900078
STATE OF INDIANA
PROFESSIONAL ENGINEER

RECOMMENDED FOR APPROVAL: Jeffrey S. Nicholson 5-18-05
DESIGN ENGINEER DATE

DESIGNED: D. SHEETZ DRAWN: D. SHEETZ
CHECKED: J. NICHOLSON CHECKED: J. NICHOLSON

INDIANA
DEPARTMENT OF TRANSPORTATION

RAILING DETAILS

HORIZONTAL SCALE	FILE
AS NOTED	1-90-230B
VERTICAL SCALE	DESIGNATION
AS NOTED	9406690
SURVEY BOOK	SHEET
	30A OF 30
CONTRACT	PROJECT
B-25747	BR-231-2( )

BFS NO. 3841.0201

R:\3841.021011\AUTOCAD\LBRIDGE\0201B778.dwg 05-18-05 AT 14:25 1 D.SHEETZ